

# Personal Agency and Political Efficacy across Urban-Rural

Friday 9<sup>th</sup> October, 2026 11:57

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We use World Values Survey 7 (2017-2023) across about 50 countries to test urban-rural differences in personal agency and political efficacy. We find that there isn't more personal agency in urban v rural as in "city air makes you free hypothesis"—there is mostly no difference. But political efficacy is significantly higher in urban v rural in rich countries as in "left behind ruralities" hypothesis. And as Toennies hypothesised those doing well financially are "active and alive" in cities or as we find here have greater personal agency (but not political efficacy). Effect sizes are considerable at about ???. As in any correlational study results may not be causal.

WORLD VALUES SURVEY, FREEDOM, AGENCY, AUTONOMY, POLITICAL EFFICACY, URBANISM, URBAN, RURAL, URBAN-RURAL, CITIES

City life and Gesellschaft down the common people to decay and death. Ferdinand Toennies

In the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive. Ferdinand Toennies

"City air makes you free" hypothesis states that urban is more free than rural (Yamagishi et al. 2012).<sup>1</sup> Hence, it'd be already an unexpected finding worth reporting that people don't have more freedom/agency in cities, because typically the expectation is that they do.

Freedom/agency is an end-in-itself, a desirable outcome in just about any respect, except perhaps responsibility. In the Middle Ages there was little freedom, but also little responsibility—for instance the poor and the slaves were not responsible for their lot—their situation was God given. Now the poor have to worry about being poor and about it being their fault—everyone is free after all and their achievements and experiences are of their own making.<sup>2</sup>

We measure freedom as (perceived) personal agency, which we discuss first, and then we turn to our other main variable of interest, (perceived) political efficacy.

## 1 Perceived personal agency

What is "perceived personal agency" or "freedom" for short?<sup>3</sup> To be free is to be able to choose, to control and direct one's own life. Freedom is having control—a person is free if she is her own master. It could be labeled "psychological freedom:" "Freedom means

<sup>1</sup>"Stadt Luft macht frei" (Park et al. [1925] 1984, p. 12). The saying originated in the Middle Ages, and it meant freedom from feudalism in cities, non-feudal islands in a sea of feudalism (Harvey 2012). Feudalism has long been gone, but we may actually be living under feudalism again, so called techno-feudalism (Varoufakis 2024, Morozov 2022). Cloud fiefdoms, rent, and digital serfdom through platforms like Google and Amazon "enslave" both urbanites and ruralities, but still perhaps city may be more free in a sense that there is more choice and opportunity say with local traditional on the ground commerce. While modern cities aren't clearly more free from techno-feudalism, as Middle Ages cities used to be more free from feudalism, cities are still often seen in the literature and popular culture as more free than ruralities.

<sup>2</sup>For instance, people in the US feel relatively free and so they blame the poor for their lot, while Poles feel less free and also blame the poor less.

<sup>3</sup>Here we focus on personal as opposed to societal freedom and follow Okulicz-Kozaryn (2014) in this paragraph.

a degree of harmony between basic motives and overt behavior” (Bay 1970, p.83 cited in Okulicz-Kozaryn 2014). It is “capability to choose,” “awareness of alternatives” and “courage to choose” (Veenhoven 2000).<sup>4</sup> Clearly, there are more choices in city v ruralites, so at least in a sense of choice there is more freedom. But then too much choice may also backfire and even lead to a paralysis (Schwartz 2004).

Freedom/agency is a broad concept and could be sub-divided into dimensions—Hitlin and Long (2009) reviews the concept in sociology. See also a recent scale (D’Italia and Okulicz-Kozaryn 2025). Another useful review is Kouba (2016) and also relevant here as it points to our WVS measure.

There are theoretical reasons to expect more freedom in cities v ruralities. In rurality there is mechanical solidarity—rural folks function like parts of a simple machine—groups are held together through homogeneity and similarity: similar lifestyles, roles, shared values and morality (Durkheim [1893] 1997). Hence, ruralities force social conformity not conducive for individual freedom. On the other hand, organic solidarity in cities allows for more differentiation and individualism. Urbanites are also more tolerant (Tuch 1987). Simmel observed that as urbanism overloads a person it forces to distance oneself from the environment (Simmel 1903), which also may be conducive for differentiation, individualism, and then perhaps freedom. On the other hand, urban hunkering down, withdrawal, and blase attitude (Simmel 1903) are not agentic. Cities provide freedom from traditional rural social constraints—*Gesellschaft* v *Gemeinschaft*—deviation from community norms is more likely to be reprimanded and ostracized in rural v urban. An urbanite may be more internally driven rather than externally constrained rural folk (Yamagishi et al. 2012). City provides more opportunities for alternative jobs and social relations, so urbanite is more free by being less concerned about social exclusion as she has plenty of alternatives (Yamagishi et al. 2012). Also see discussion in Luca et al. (2023).

One mechanism is higher social pressure for conformity in ruralities—cities provide anonymity—indeed concern for moral reputation is greater in rural than urban areas (Danielson et al 2023 cited in Okulicz-Kozaryn 2026). The higher the population density, the lower the frequency of social comparison measuring social comparison with searches on Google (Okulicz-Kozaryn 2026).

But there is also theory suggesting less freedom in cities. City is stressed, hectic, anxious, materialistic, consumerist (Okulicz-Kozaryn 2015), and overstimulating (Simmel 1903, Lederbogen et al. 2011)—in such conditions freedom/agency is difficult. Steve Pile in his colorful writing argues that old cities can be a weight on its inhabitants with their history and past (Pile 2005a,b, Pile et al. 1999), and old cities carry melancholia (Pile 2005b). Weight of history and melancholia can inhibit how free one feels.

Many studies approach freedom as a predictor of subjective wellbeing, but freedom is also an end-in-itself (Okulicz-Kozaryn 2014), as recognized by Fromm (2013, 2012, 1992, 1964, 1944, [1941] 1994) and Sen (2000). The Universal Declaration of Human Rights states that freedom is a human right: freedom for the realization of one’s human potential (art. 26, 29).

There are studies on freedom’s predictors using WVS variable that we use in our study (e.g., Okulicz-Kozaryn 2014, Pitlik 2016, Nikolaev and Bennett 2016), but they do not examine urbanicity. There are many studies on the effect of urban-rural on subjective wellbeing (SWB) (e.g., Sørensen 2024, 2014, Knies 2026, Okulicz-Kozaryn and Valente 2021), but the effect of urban-rural on freedom has only been studied few times, and in few countries only.

Lolle (2023) finds similar non-differences in agency across different sized communities in Denmark. Atherton et al. (2024) find that psychological well-being—which entails agency as a subcomponent—is lower in the most rural places compared to the most urban places in the USA; however, these effects turn insignificant once socio-demographic factors are controlled for. Kusnali et al. (2025) find that women in Indonesia have more decision making agency in many (household expenditures, food expenditures, time allocation to social activities) but not all (contraceptive use) aspects of daily life in urban compared to rural areas. Indian females have more autonomy in city v rural, both in personal and household decisions (Mishra 2014). Baage et al find no urban-rural difference in peoples perceived reproductive agency, i.e. how much freedom of choice and control they have over family planning, in Ethiopia, Kenya, and Zimbabwe (Båge et al. 2024) and sub-Saharan Africa (Båge et al. 2026).

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<sup>4</sup>The degree of perceived control over choice is called “locus of control” in psychology and Verme (2009) argues that WVS item used here measures locus of control in addition to freedom.

## 2 Perceived political efficacy

We turn to political efficacy. It's a person's belief that she has a say in politics, a perception that government will respond to one's voice.

As the world continues to urbanize, ruralities are disappearing (Pospěch et al. 2024). Rurality in the developed world is increasingly a periphery, losing power and voice v urban centrality, being abandoned and disrespected—there is rural discontent, resentment, and shame. See superb accounts in Hanson (2015), Fuller (2017). This would suggest that ruralities in rich countries may have less political efficacy—but that's rather because ruralities have been abandoned and disrespected, not because they are inherently less free or agentic. In general, while there has been a loss of power and respect in rural v urban, not necessarily the autonomy has been lost, ruralities still have a good deal of agency.

Like in Czech Republic, in the US there is rural resentment, e.g., people feel that policy is done to them, not with or for them; and there is racial resentment where rural Whites resent urban minorities and immigrants getting better treatment than them (Diamond 2023). And while there is a reliance on community and tight connection, there is also self-reliance, and desire for autonomy and freedom as a core value (Diamond 2023).

Rural areas are 'left-behind' (Ulrich-Schad and Duncan 2021), 'places that don't matter' (Rodríguez-Pose 2018)—they are second-tier areas with limited or deteriorating opportunities. When people's opportunities are constrained, their freedom to exercise agency is likewise restricted. Exercising agency requires access to real and substantively different opportunities from which individuals can choose in pursuing lives that are consistent with their own goals and values. Limited opportunities may therefore undermine both, people's perceived ability to shape their own lives in accordance with these goals and values and their perceived ability to shape the conditions in which their lives are embedded, such as institutions, infrastructure, and the built and natural environment. We therefore expect people living in 'left-behind' rural areas to perceive themselves as having less influence over the conditions in which they are embedded and that shape their lives, and consequently to perceive lower levels of political efficacy.

Only few studies have investigated urban-rural differences of political efficacy, suggesting people from rural areas and small towns show lower levels of political efficacy compared to urbanites, (e.g., Rowland et al. 2024) for the US and Luukkonen et al. (e.g., 2022) for Europe. There is more political efficacy in urban areas and least in rural in Europe (García Del Horno et al. 2024, Luukkonen et al. 2022), and USA (Rowland et al. 2024).<sup>5</sup>here are also related studies. Margalit et al. (2025) examine political agency in relation to rurality, conceptualizing "rural resentment" as the perception among rural citizens that the political system gives people like them little or no say in what the government does. Across 10 European countries, they consistently find that rural resentment is associated with a higher likelihood of voting for populist parties, while in the United States it is associated with a higher likelihood of having voted for Trump. Political efficacy has been shown to be positively linked to political engagement (Accetta 2023). Self-efficacy and political efficacy have been shown to have opposing effects on the justification of terrorism and political violence: people who perceive themselves as highly self-efficacious and those who feel they have little political efficacy are less likely to justify such actions (Julkif 2022). But no one has investigated whether this relationship is universal across different contexts (such as high/low GDP).

## 3 Hypotheses

These are in terms of both personal agency and political efficacy:

$H_1$  : "City Air Makes You Free" and "Left Behind Ruralities" Hypothesis: People in cities are freer than people in rural areas.

$H_2$  : This difference is more pronounced in less developed countries

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<sup>5</sup>T

$H_3$  : The "more" of freedom in cities mostly stems from people satisfied with finances (following the Toennies quote)

## 4 Data and model

World Values Survey (WVS) is a fantastic and free dataset available from <https://www.worldvaluessurvey.org>. It is a great source of data as compared to \$30k+ closed source Gallup data. It is helpful to use open source public data amidst corporatization of research and academia (Mills 2012a, Cox 2013, Mills 2012b, Catropa and Andrews 2020, Schmidlin 2015). WVS is representative of each country. Countries in the sample and sample sizes are shown in appendix.

Urban-rural is a gradient, not a dichotomy (Okulicz-Kozaryn 2017), but here for simplicity we use 3 or 4 collapsed categories and focus on the endpoints of the spectrum:  $> 500k$  v small  $< 10k$  or  $< 20k$  (separately for each country).  $> 500k$  represents city—indeed it could even be bigger than 500k—urban unhappiness starts around several hundred thousand (Okulicz-Kozaryn 2017).

In appendix we do robustness checks dropping countries with fewer than 50 observations per category.

### 4.1 Personal Agency

Autonomy/freedom variable reads: "Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out."

All variables including controls are listed in appendix and there are also shown distributions of all the variables. We are controlling for married, employment type dummies (full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction,<sup>6</sup> health, kids, and importance of religion.

Per controls we follow Pitlik (2016), Nikolaev and Bennett (2016) except that we use financial satisfaction in most specifications instead of income as it arguably predicts freedom more than income and indeed in our sample correlates higher, and also we don't use their country level democracy index, and use PCGDP in appendix only. Eventually we decided against age and age<sup>2</sup>, as it is rather urbanicity predicting age composition than the other way round, hence we would risk overcontrol bias. Neither we include gender as it likely differs less than age across urban-rural, and again rather predicted by than predicting urban-rural.

We were tempted to add trust, crime victimization and fear of crime, as these correlate with urbanicity and may predict freedom. But eventually we do not include them because those are rather caused by urbanicity, not the other way round, and hence may lead to overcontrol bias:

What we really need, to identify/estimate  $X \rightarrow Y$ , is to control for other determinants of  $Y$  that are also antecedents of  $X$  (so,  $W \rightarrow X$ ). If we include controls where the relationship goes the other way ( $X \rightarrow W$ ), we will exacerbate bias in our estimate of  $X \rightarrow Y$ . (The relevant term for that situation is 'overcontrol bias') (Bartram et al. 2024, p. 5).

### 4.2 Political Efficacy

The political efficacy item reads "how much would you say the political system in your country allows people like you to have a say in what the government does?" There are quite a few countries where the question was not asked: the United States, Germany, South Korea, Greece, Turkey, Thailand, China, Chile, Ecuador, Serbia, Romania, Russia, and Tunisia.<sup>7</sup>

<sup>6</sup>Financial satisfaction is the key—not everyone needs the same amount of money to be financially satisfied and feel free. It is measured as "How satisfied are you with the financial situation of your household?"

<sup>7</sup>Although Tunisian respondents were asked this question, the data for Tunisia were not curated properly and so could not be analyzed in this study. <https://onlinelibrary.wiley.com/doi/full/10.1111/ssqu.13120>

Controls. Education: more educated people more efficacy (Emmenegger et al. 2015, García Del Horno et al. 2024, Marx and Nguyen 2016, Rowland et al. 2024). Employment: compared to paid employment people in education ??? more political efficacy, permanently sick/disabled less (Luukkonen et al. 2022), unemployment or precarious employment are negatively linked to political efficacy (Emmenegger et al. 2015, Marx and Nguyen 2016). Financial situation: People who do well (making ends meet), have more political efficacy (Luukkonen et al. 2022, Marx and Nguyen 2016). Higher income = more political efficacy (Emmenegger et al. 2015, Rowland et al. 2024). Health: positive for political efficacy (García Del Horno et al. 2024, Gidengil and Wass 2024, Shore et al. 2019). Religiosity: more religious people have less political efficacy (Marx and Nguyen 2016, Rowland et al. 2024). Why these controls: There are indications for urban-rural divides in education (Van Maarseveen 2021, Zahl-Thanem and Rye 2024), employment (Eurostat 2026, Ananian and Dellaferrera 2024), financial situation (Kpatcha et al. 2026, Meloni et al. 2024, Finnemann et al. 2024), health (MacKinnon et al. 2023, Menashe-Oren and Masquelier 2022), and religiosity (Balazka 2020, Meulemann and Schmidt-Catran 2020).

## 5 Results

Means of personal agency and political efficacy across urban-rural and level of development are in table 1. It is worth highlighting that a finding of even no difference in personal agency between urban and rural would be worth reporting as we have hypothesised a positive relationship.

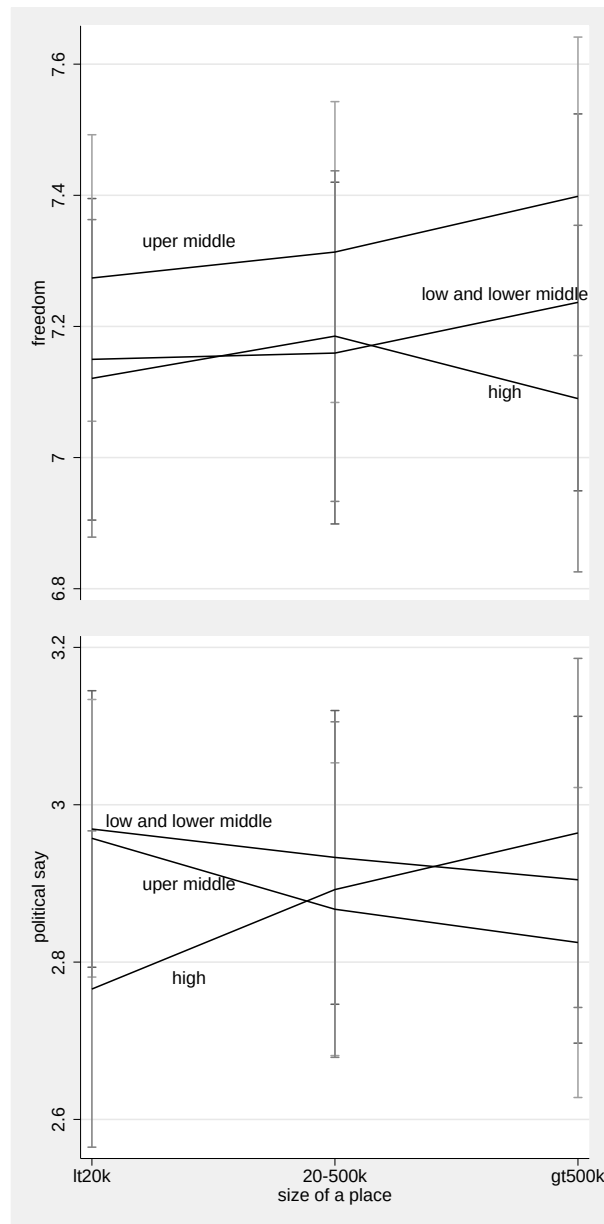
Overall personal agency is the lowest in the largest places, including in less developed countries, and the difference being especially large for high development countries. But in upper middle it is actually highest in cities. But overall, the differences are not that large and go away when controlling for predictors of freedom.

Political efficacy is also overall slightly lower in largest cities. But as opposed to freedom differences are large by level of development. In less developed countries cities have much less political say; in upper middle less, but in high almost the opposite of low/lower, and that last difference holds when controlling for predictors of political say. Margins are plotted in figure 1.

While the confidence intervals are wide, the interaction between >500k and high development countries is significant v the base case of <20k and low/lower development countries. The results for personal agency are insignificant.

**Table 1:** means

| free    |         |           |        |       |        | political say |         |           |        |       |        |      |
|---------|---------|-----------|--------|-------|--------|---------------|---------|-----------|--------|-------|--------|------|
| place   | overall | low/lower | middle | upper | middle | high          | overall | low/lower | middle | upper | middle | high |
| <20k    | 7.20    |           | 7.11   |       | 7.26   | 7.24          | 2.96    |           | 3.03   |       | 3.06   | 2.67 |
| 20-500k | 7.27    |           | 7.21   |       | 7.21   | 7.36          | 2.88    |           | 2.95   |       | 2.83   | 2.87 |
| >500k   | 7.12    |           | 6.96   |       | 7.40   | 6.98          | 2.89    |           | 2.67   |       | 2.86   | 2.99 |



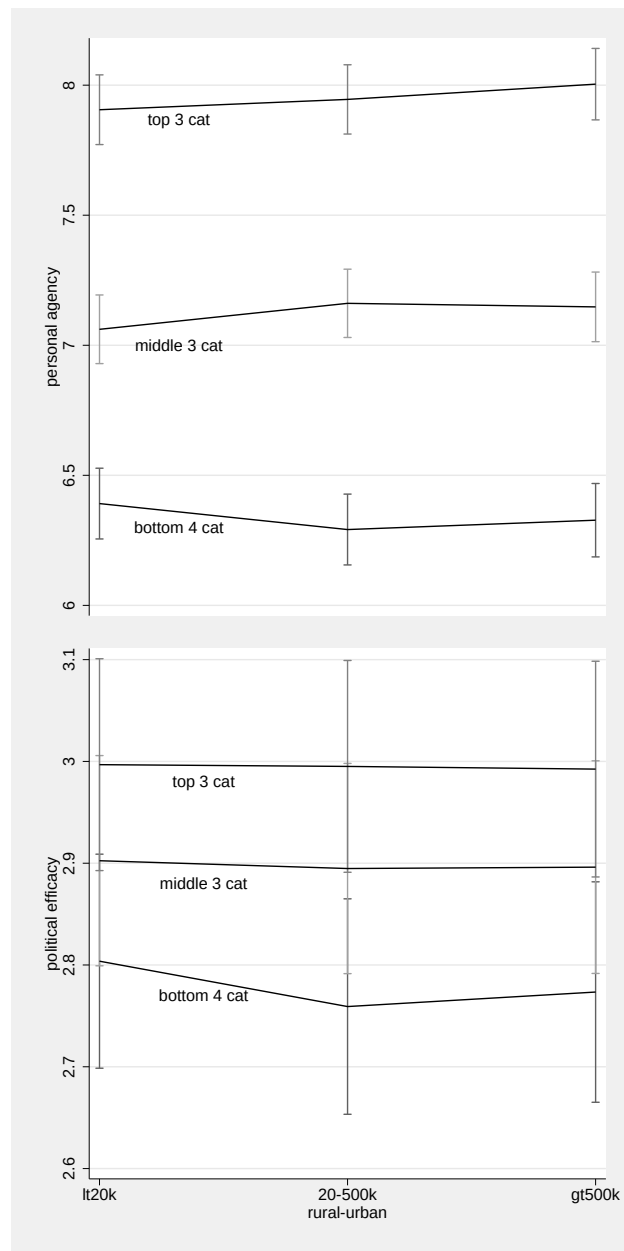
**Figure 1**

## 5.1 Interactions with financial satisfaction

Results are visualized in figure 5.<sup>8</sup>

In panel a by financial satisfaction those in top and also middle 3 categories increase from  $< 20k$  to  $> 500k$ . In other words, freedom differences are magnified in largest cities for financial satisfaction—the financially satisfied enjoy higher freedom premium over financially dissatisfied in cities. The results for political efficacy are insignificant.

<sup>8</sup>We focus here on visual interpretations using marginsplot, but underlying regressions are in appendix. The relevant significant interactions are for smallest places  $< 10k$  (against the base case of  $> 500k$  with top category(ies) for financial satisfaction in first column a for middle financial satisfaction (v bottom base case) at -0.12 and the top category at -0.13. For political efficacy in panel b results are insignificant.



**Figure 2**

We also separate results by region and income group for financial satisfaction and postpone these results to appendix<sup>9</sup>

## 6 Discussion and Conclusion

It is worth highlighting that finding no urban-rural effect for personal agency in most countries is a finding worth reporting—the expectation from theory and literature is that the bigger places have more freedom.

But political efficacy is significantly higher in urban v rural in rich countries as hypothesised in “left behind ruralities” hypothesis.

We find support for Toennies’ idea: “in the city and therefore, where general conditions characteristic of the *Gesellschaft* prevail, only the upper strata, the rich and the cultured, are really active and alive” (Tönnies [1887] 2002, p. 227). There are significant

<sup>9</sup>Here just real quick: Financial satisfaction panel a results are stronger in East Asia/Pacific and Sub-Saharan Africa (due to sharp increase for the top tertile), and show similar pattern in Europe/Central Asia except that the middle tertile for financial satisfaction is stronger than the top. Latin America/Caribbean and North America shows the opposite pattern where the bottom tertile has the steepest slope, and in Middle East/North Africa middle and top tertiles even reverse to negative. in South Asia the slopes are parallel—no differences there. By income group: High income is similar to the overall pattern.

interactions in the pooled sample for financial satisfaction.

Like in Okulicz-Kozaryn (2024) urbanites probably actually have in a way fuller and more free life, but perhaps they realize more that they miss out relatively to other urbanites, and so they feel relatively less free. One mechanism could be that people tend to compare upwards (Frey and Stutzer 2002) and there are more people to compare to in urban areas, inequality ladders are taller and the pond is bigger—small fish in big pond (city) seem to be less free than big fish in a small pond (rural area).

In poorer/less economically developed countries there are more differences in freedom across rural-urban, mostly less freedom in smaller places<sup>10</sup>—this makes sense—smaller places in developing countries can be very traditional and very poor—freedom is difficult in such places. In richer countries, on the other hand, one can already be free just about anywhere, no need for a large city. This is a similar finding as in literature with respect to happiness (Okulicz-Kozaryn 2015)—only in the very poorest places of all such as Sub-Saharan Africa, happiness is greater in larger cities as life is unbearable and lacking necessities such as shelter and clean water in ruralities. Everywhere else urbanites tend to be less happy (Okulicz-Kozaryn and Valente 2021).

# ONLINE APPENDIX (SOM)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead as SOM 'Supplementary Online Material']

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<sup>10</sup>(but sometimes more)

# 1 More recent results

## 1.1 Literature

### 1.1.1 Freedom/personal agency

While Ester and Vinken (1993) is not about urban-rural freedom specifically, it is indirectly related. Ester and Vinken (1993) uses similar dataset to ours, EVS/WVS, and tests yuppie value syndrome (yuppies are young urban professionals), and finds that yuppies tend to hold liberal social and ethical views, but are not: economically conservative, absent from politics, and don't have distinctive lifestyle emphasis.

Yamagishi et al. (2012) claims to test our “city free air” hypothesis, but Yamagishi et al. (2012) test is very different from ours—they only show that urbanites are more likely to chose extravagant pens than rural folks. Still, the study is useful in at least 2 ways. First they show that indeed there is more freedom in city (using extravagant pens test), and they provide theory and literature summarized in the next paragraph.

Perception of freedom can be counterintuitive and deceptive—a place may seem free, or indeed actively advertise itself as an oasis of freedom, but actually not be that free after all. The US advertises itself as “the land of the free” and yet people in the US feel less free at 7.7 than Mexicans and Colombians at above 8.1 (WVS Wave 7). Likewise, city may seem more free than ruralites, but as this paper finds, usually that's not the case. In a city v rural (or in the US v Latin America) there is often more social pressure (Okulicz-Kozaryn and Martinez 2025, Okulicz-Kozaryn et al. 2026)<sup>11</sup> and so an urbanite is more of an “automaton” (to use Fromm's term) (Okulicz-Kozaryn 2014).

Money is important for personal freedom/agency. Indeed as many financial gurus on YouTube would tell you freedom is the only or main reason to accumulate wealth. For instance, the FIRE movement (Financial Independence, Retire Early) is all about freedom/autonomy. Hence in our analyses we will focus on income, class and financial satisfaction.

## 1.2 Original (and before the final) results from the paper postponed here

A natural first approach would be to find overall pattern on pooled complete dataset across all countries to see if there are universal patterns or at least by income group or region. Indeed this is how we have started, but we postpone these results to the appendix as we have learned through our sensitivity/robustness checks that inclusion of country dummies yields all urban-rural results insignificant. Hence we simply start with the results by country.

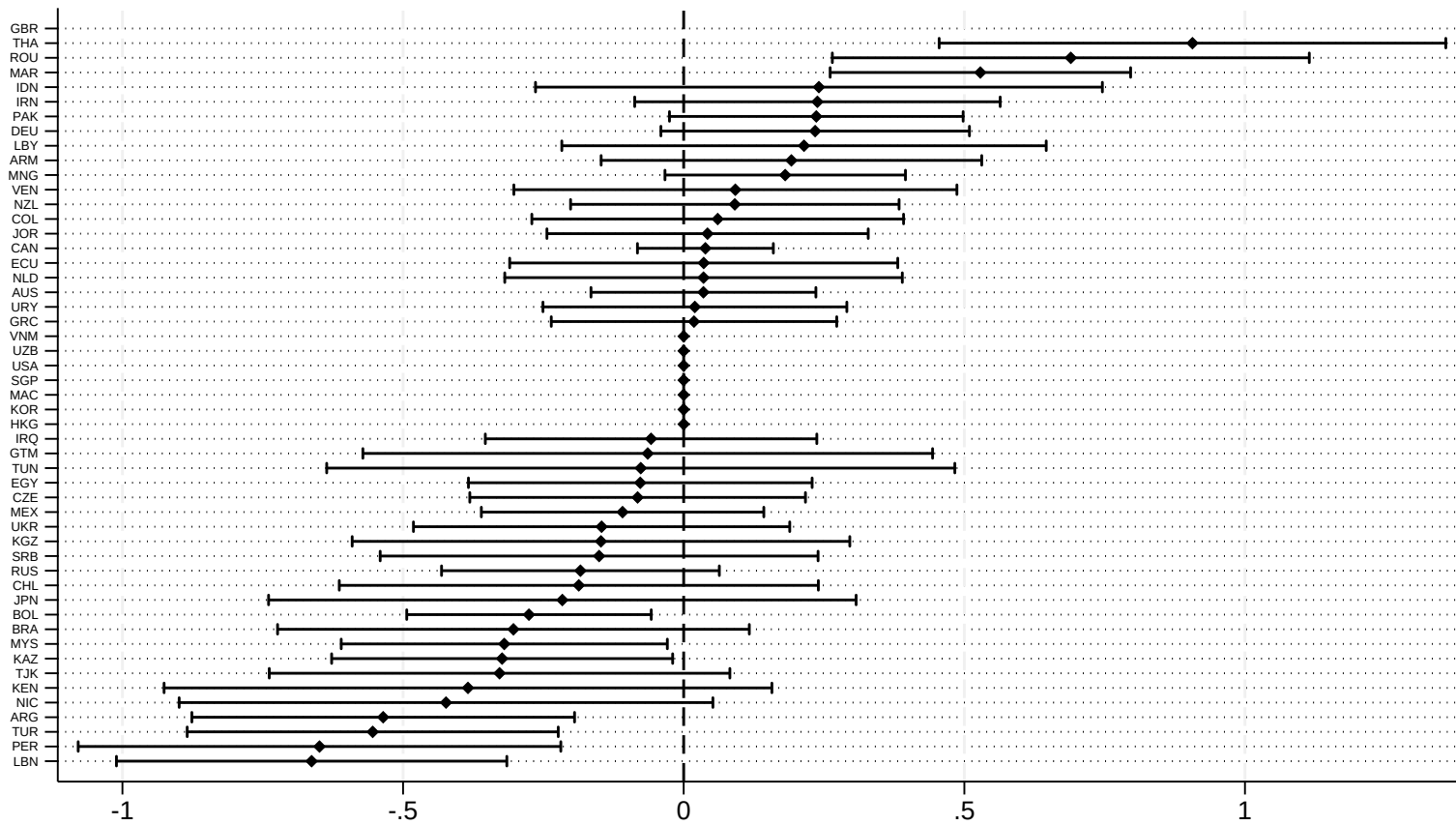
Financial satisfaction is the strongest correlate of freedom (0.33), stronger than health (0.21), class (0.14), and income (0.09). Financial satisfaction is also related to class (0.26) and income (0.34).

### **By country**

Results are in figure 4. Is there less freedom in smaller areas as often argued? Are cities more free? No. Most countries show insignificant urban-rural differences at  $< 0.05$  level of significance. Again, it is already a finding worth reporting that people don't have more freedom/autonomy in cities, because there are theories, reasons, and arguments that they do.

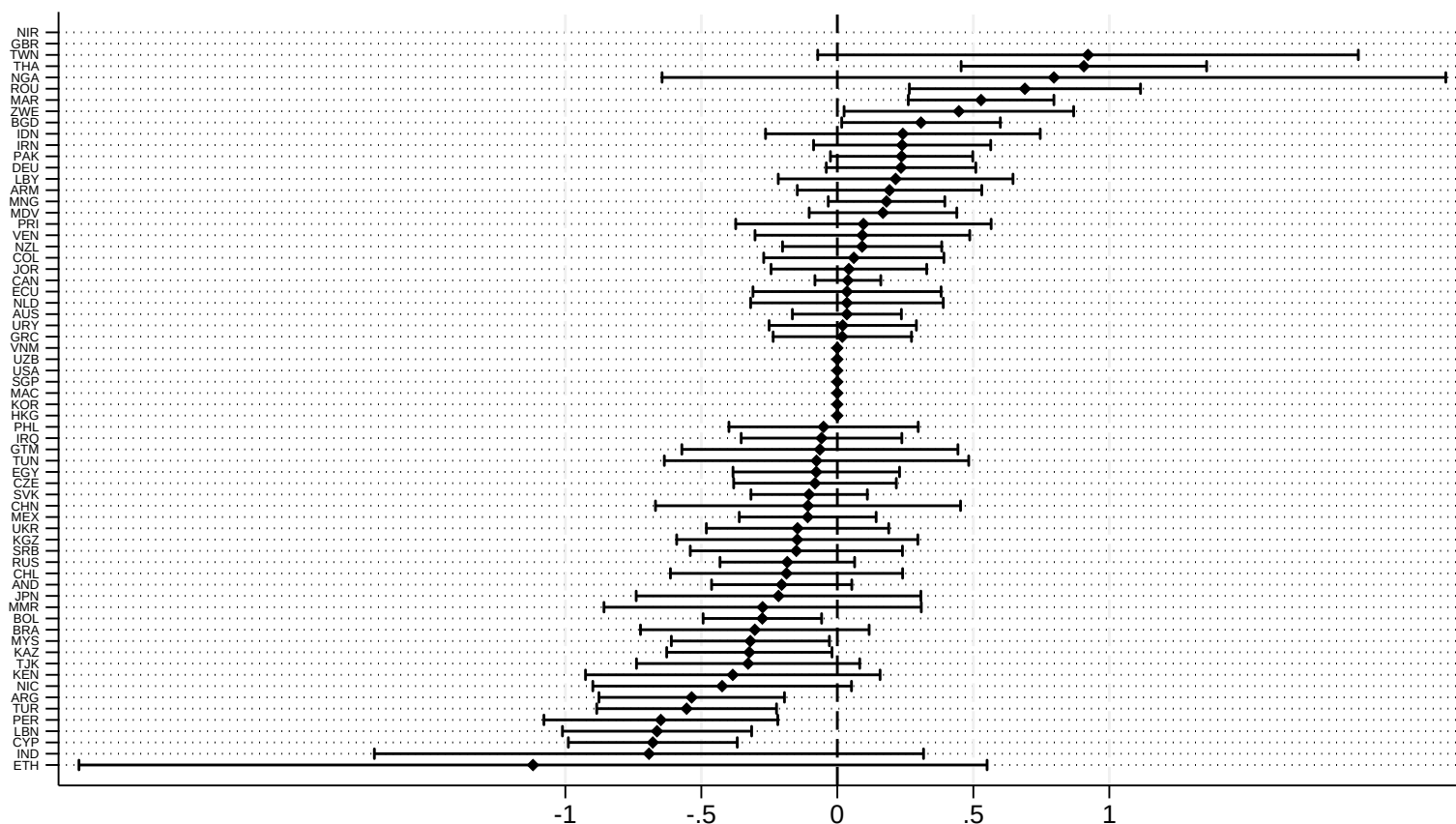
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<sup>11</sup>Also see a fantastic book by Frank (2012) on keeping up with the Joneses in the US.



b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

**Figure 3:** A more final version, where country count goes down from 66 to 51 when dropped if gt500k missing; same pattern as earlier: most insignificant, some positive but most negative.



b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

**Figure 4:** Original version. Using model controlling for age, age<sup>2</sup>, male, married, employment type dummies(full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and religion important. See appendix for results in table also including omitted here 20 – 500k category. See appendix model for pooled sample in table 4 except that original 8 step urbanicity variable is now categorized into 3 bins as there are smaller cell sizes separately by country. Still note that few countries lack the estimates as even this simplified 3 step urbanicity spectrum is not available. Overall the results are mostly insignificant, but more negative (11) than positive (6). So in general within countries for few significant relationships there is less freedom in smaller areas as theorized in the literature.

Only 16 out of 64 countries have more freedom in cities. They have at least one coefficient on either  $< 20k$  or  $20 - 500k$  ( $v > 500k$ ) negative and significant at  $< 0.05$ . 6 countries, on the other hand, have more freedom in ruralities—they have positive effect coefficients on smaller areas dummies. Note that figure 4 only shows  $< 20k$  coefficients; for  $20 - 500k$  coefficients see appendix.

It is difficult to see any consistent country patterns in the table. Perhaps what stands out is that among the richer or more economically developed countries there are smaller differences. In poorer/less economically developed countries there are more differences: mostly less freedom in smaller places.<sup>12</sup> This makes sense—smaller places in developing countries can be very traditional and very poor—freedom is difficult in such places. In richer countries, on the other hand, one can already be free just about anywhere, no need for a large city. This is similar to what happiness literature have found with respect to happiness (Okulicz-Kozaryn 2015)—only in the very poorest places of all such as Sub-Saharan Africa happiness is greater in larger cities as life is unbearable and lacking necessities such as shelter and clean water in ruralities.

Countries could be classified by region (see appendix). About half of Latin countries in this sample have lower freedom in smaller areas: ARG, BOL, CHL, NIC, and PER; and about the other half are insignificant: BRA, COL, ECU, GTM, PRI, URY, and VEN.

South Asia does not have a clear pattern: BGD is positive and PAK marginally positive, but IND highly negative, and MDV is insignificant.

<sup>12</sup>But sometimes more.

Europe/Central Asia: mostly insignificant: ARM, CZE, GRC, KGZ, NLD, RUS, SRB, SVK, and UKR. But otherwise mostly negative: AND marginally, CYP, KAZ, TJK marginally, and TUR; and positive: ROU, DEU marginally. Note that GBR has missing obs and does not appear in the table at all, and UZB has missing urbanicity data.

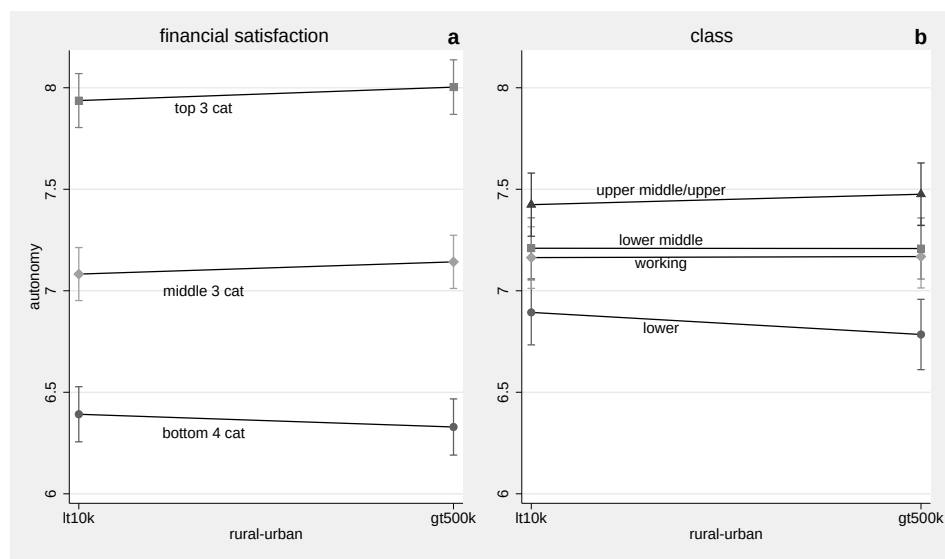
East Asia/Pacific is mostly insignificant: AUS, CHN, IDN, MMR, NZL, PHL. And several countries in East Asia/Pacific are without respondents in smaller areas: HKG, KOR, MAC, SGP, and VNM. Positive: MNG, and highly positive THA; and negative: JPN, MYS.

Middle East/North Africa mostly insignificant: EGY, IRN, JOR, LBY, and TUN. Two negative: IRQ, LBN; and one positive: MAR.

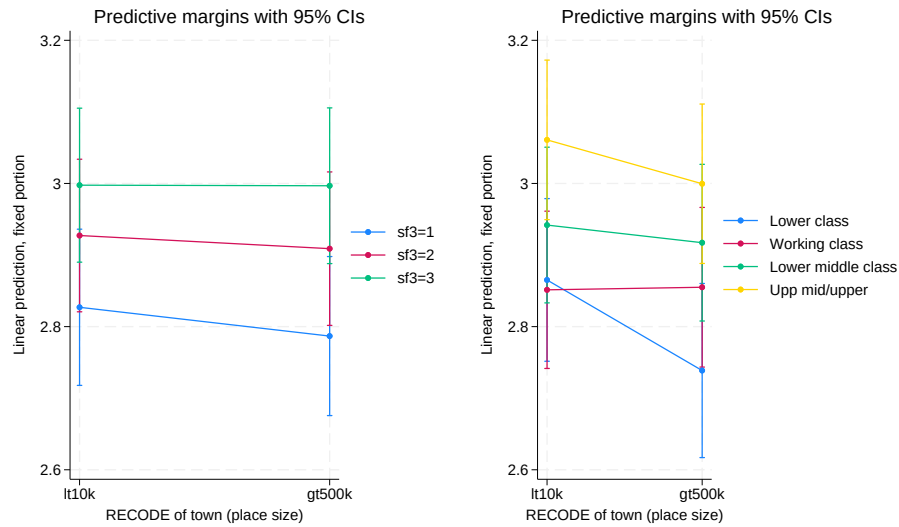
Sub-Saharan Africa: two insignificant: KEN, NGA; ETH is negative, but ZWE positive.

Two countries missing the region classification by the World Bank: NIR and TWN. NIR has a small sample size of less than 500 and missing observations on the variables used in regression so it does not appear in the table. TWN is insignificant.

### 1.3 Exploring political say



**Figure 5:** Multilevel country random intercept regressions of freedom on rural-urban dummies interacted with: a) financial satisfaction, and b) class. For underlying regressions see appendix. Controls are the same as in the previous section.



**Figure 6:** mixed polSay i.t4##i.sf3 age age2 male mar i.emp health kids rel\_imp ||| cc: and the second one same just interacted with soc class

GDP took avg for 2015-2025. And same results as for the World Bank income groups.

Political say variable—the sample size drops quite a lot—overall there are about 94k observations. But this variable has 69k only; with freedom have more like 85k

```
mixed polSay i.t3##c.ny_gdp_pcap_kd mar i.emp satFin health kids rel_imp || cc: i.t3 ,mle
ta cc t3 if e(sample)==1
```

| ISO 3166-1<br>alpha-3<br>country<br>code | RECODE of town (place size) |         |        | Total |
|------------------------------------------|-----------------------------|---------|--------|-------|
|                                          | lt20k                       | 20-500k | gt500k |       |
| AND                                      | 718                         | 270     | 0      | 988   |
| ARG                                      | 124                         | 135     | 467    | 726   |
| ARM                                      | 497                         | 211     | 418    | 1,126 |
| AUS                                      | 412                         | 514     | 656    | 1,582 |
| BGD                                      | 942                         | 201     | 0      | 1,143 |
| BRA                                      | 164                         | 778     | 640    | 1,582 |
| CAN                                      | 1,098                       | 1,694   | 1,226  | 4,018 |
| COL                                      | 304                         | 722     | 494    | 1,520 |
| CYP                                      | 528                         | 358     | 0      | 886   |
| CZE                                      | 668                         | 360     | 148    | 1,176 |
| EGY                                      | 336                         | 388     | 261    | 985   |
| ETH                                      | 812                         | 371     | 9      | 1,192 |
| CTM                                      | 67                          | 450     | 590    | 1,107 |
| HKG                                      | 0                           | 0       | 2,040  | 2,040 |
| IDN                                      | 2,631                       | 331     | 89     | 3,051 |
| IRN                                      | 388                         | 571     | 472    | 1,431 |
| IRQ                                      | 305                         | 219     | 625    | 1,149 |
| JOR                                      | 363                         | 287     | 490    | 1,140 |
| JPN                                      | 43                          | 613     | 403    | 1,059 |
| KAZ                                      | 367                         | 373     | 316    | 1,056 |
| KEN                                      | 479                         | 413     | 116    | 1,008 |
| KGZ                                      | 666                         | 244     | 193    | 1,103 |
| LBN                                      | 407                         | 657     | 130    | 1,194 |
| LBY                                      | 397                         | 409     | 112    | 918   |
| MAC                                      | 0                           | 0       | 986    | 986   |
| MAR                                      | 380                         | 480     | 340    | 1,200 |
| MDV                                      | 705                         | 294     | 0      | 999   |
| MEX                                      | 801                         | 435     | 384    | 1,620 |
| MMR                                      | 1,116                       | 84      | 0      | 1,200 |
| MNG                                      | 378                         | 257     | 1,003  | 1,638 |
| MYS                                      | 744                         | 428     | 141    | 1,313 |
| NGA                                      | 798                         | 352     | 14     | 1,164 |
| NIC                                      | 381                         | 709     | 110    | 1,200 |
| NLD                                      | 81                          | 1,350   | 162    | 1,593 |
| NZL                                      | 363                         | 333     | 164    | 860   |
| PAK                                      | 1,185                       | 230     | 439    | 1,854 |
| PER                                      | 336                         | 928     | 100    | 1,364 |
| PHL                                      | 908                         | 290     | 0      | 1,198 |
| SGP                                      | 0                           | 0       | 1,940  | 1,940 |
| SVK                                      | 723                         | 435     | 0      | 1,158 |
| TJK                                      | 880                         | 210     | 110    | 1,200 |
| TUN                                      | 433                         | 529     | 77     | 1,039 |
| UKR                                      | 535                         | 301     | 250    | 1,086 |
| URY                                      | 255                         | 254     | 426    | 935   |
| VEN                                      | 179                         | 659     | 349    | 1,187 |
| VNM                                      | 0                           | 1,049   | 151    | 1,200 |

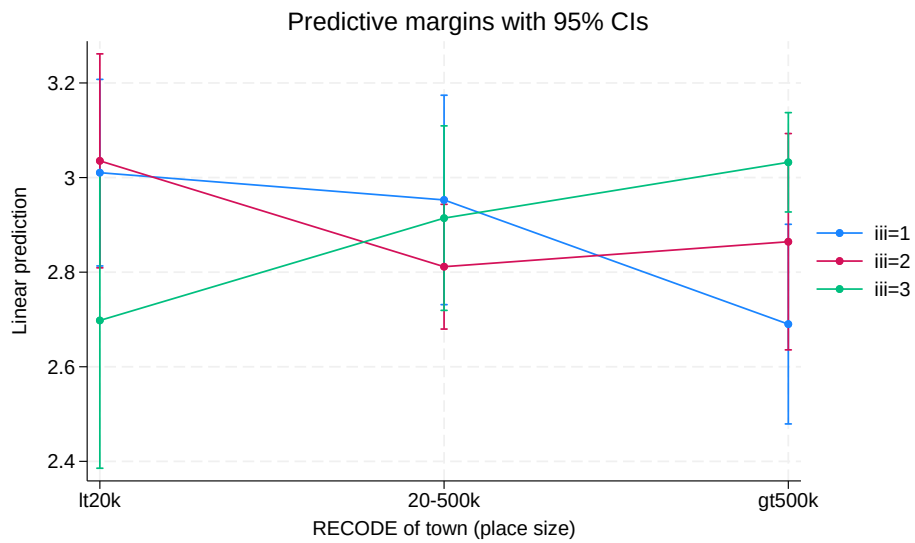
|       |        |        |        |        |
|-------|--------|--------|--------|--------|
| ZWE   | 915    | 265    | 0      | 1,180  |
| Total | 24,812 | 20,441 | 17,041 | 62,294 |

.19

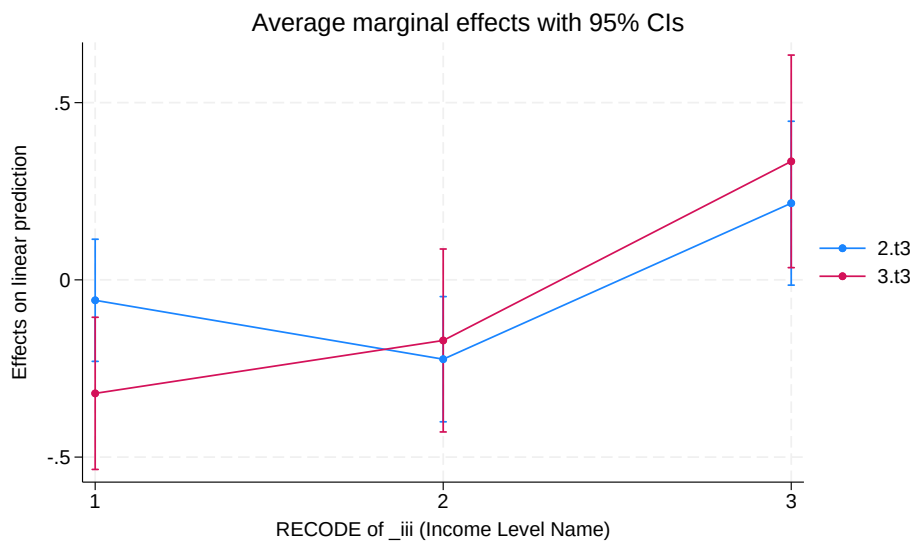
```
mixed free i.t3##c.ny_gdp_pcap_kd mar i.emp satFin health kids rel_imp || cc: i.t3 ,mle
ta cc t3 if e(sample)==1
```

| country<br>code | RECODE of town (place size) |         |        | Total  |
|-----------------|-----------------------------|---------|--------|--------|
|                 | lt20k                       | 20-500k | gt500k |        |
| AND             | 725                         | 272     | 0      | 997    |
| ARG             | 145                         | 158     | 546    | 849    |
| ARM             | 532                         | 226     | 434    | 1,192  |
| AUS             | 411                         | 515     | 658    | 1,584  |
| BGD             | 988                         | 209     | 0      | 1,197  |
| BOL             | 590                         | 711     | 689    | 1,990  |
| BRA             | 169                         | 826     | 693    | 1,688  |
| CAN             | 1,098                       | 1,694   | 1,226  | 4,018  |
| CHL             | 141                         | 436     | 326    | 903    |
| CHN             | 48                          | 2,727   | 0      | 2,775  |
| COL             | 304                         | 722     | 494    | 1,520  |
| CYP             | 571                         | 372     | 0      | 943    |
| CZE             | 675                         | 361     | 150    | 1,186  |
| DEU             | 693                         | 604     | 194    | 1,491  |
| ECU             | 375                         | 462     | 347    | 1,184  |
| EGY             | 416                         | 449     | 310    | 1,175  |
| ETH             | 830                         | 376     | 9      | 1,215  |
| GRC             | 406                         | 324     | 431    | 1,161  |
| GTM             | 67                          | 454     | 592    | 1,113  |
| HKG             | 0                           | 0       | 2,060  | 2,060  |
| IDN             | 2,722                       | 341     | 91     | 3,154  |
| IRN             | 396                         | 598     | 483    | 1,477  |
| IRQ             | 311                         | 228     | 638    | 1,177  |
| JOR             | 375                         | 305     | 522    | 1,202  |
| JPN             | 45                          | 654     | 431    | 1,130  |
| KAZ             | 385                         | 399     | 332    | 1,116  |
| KEN             | 479                         | 413     | 119    | 1,011  |
| KGZ             | 678                         | 258     | 195    | 1,131  |
| KOR             | 0                           | 364     | 881    | 1,245  |
| LBN             | 410                         | 660     | 130    | 1,200  |
| LBY             | 403                         | 412     | 115    | 930    |
| MAC             | 0                           | 0       | 984    | 984    |
| MAR             | 380                         | 480     | 340    | 1,200  |
| MDV             | 710                         | 300     | 0      | 1,010  |
| MEX             | 802                         | 435     | 384    | 1,621  |
| MMR             | 1,116                       | 84      | 0      | 1,200  |
| MNG             | 378                         | 257     | 1,003  | 1,638  |
| MYS             | 744                         | 428     | 141    | 1,313  |
| NGA             | 829                         | 369     | 14     | 1,212  |
| NIC             | 381                         | 709     | 110    | 1,200  |
| NLD             | 86                          | 1,416   | 173    | 1,675  |
| NZL             | 361                         | 329     | 166    | 856    |
| PAK             | 1,261                       | 243     | 471    | 1,975  |
| PER             | 347                         | 940     | 100    | 1,387  |
| PHL             | 909                         | 290     | 0      | 1,199  |
| PRI             | 72                          | 993     | 0      | 1,065  |
| ROU             | 630                         | 368     | 101    | 1,099  |
| RUS             | 539                         | 577     | 560    | 1,676  |
| SGP             | 0                           | 0       | 1,966  | 1,966  |
| SRB             | 372                         | 427     | 184    | 983    |
| SVK             | 735                         | 440     | 0      | 1,175  |
| THA             | 1,064                       | 96      | 107    | 1,267  |
| TJK             | 880                         | 210     | 110    | 1,200  |
| TUN             | 448                         | 548     | 77     | 1,073  |
| TUR             | 232                         | 1,724   | 402    | 2,358  |
| UKR             | 553                         | 309     | 254    | 1,116  |
| URY             | 258                         | 272     | 436    | 966    |
| USA             | 0                           | 1,395   | 363    | 1,758  |
| VEN             | 179                         | 659     | 349    | 1,187  |
| VNM             | 0                           | 1,049   | 151    | 1,200  |
| ZWE             | 933                         | 269     | 0      | 1,202  |
| Total           | 30,587                      | 32,146  | 22,042 | 84,775 |

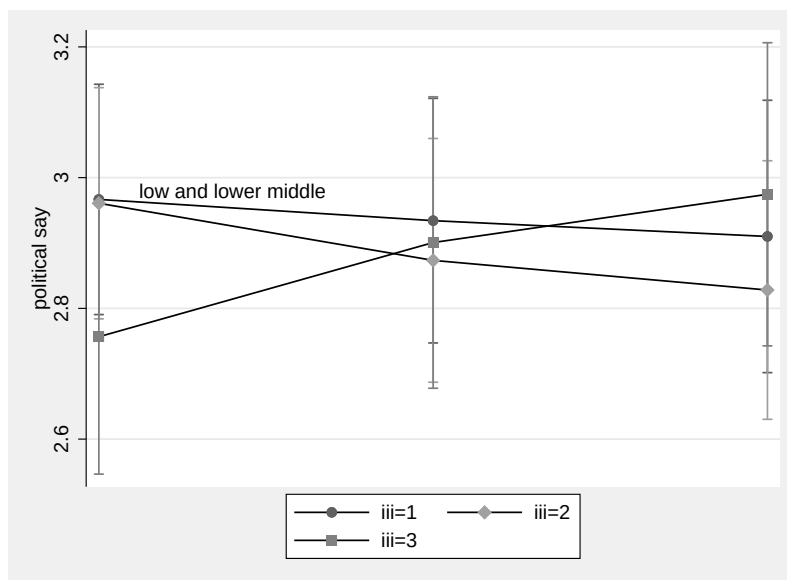
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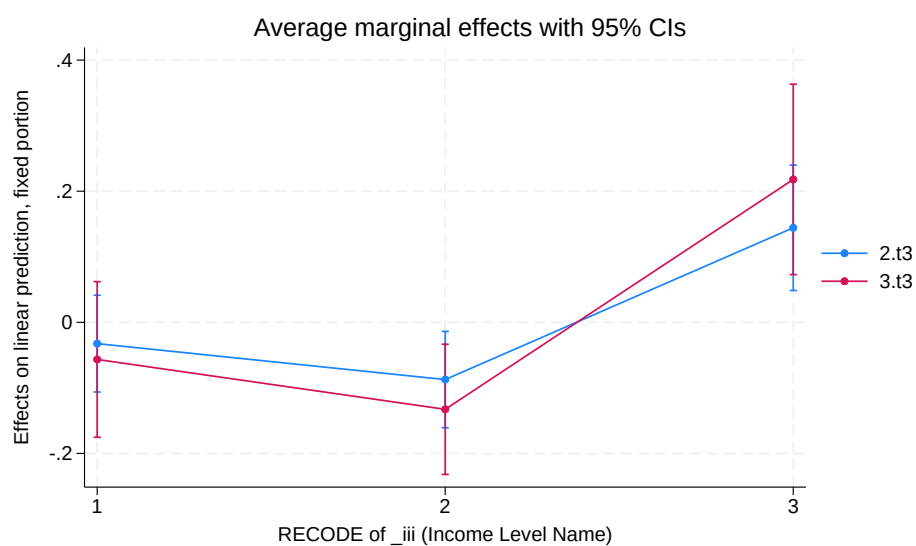
**Figure 7:** `reg polSay i.t3##i.iii mar i.emp satFin health kids rel_imp , robustcluster(ccc)`



**Figure 8:** `reg polSay i.t3##i.iii mar i.emp satFin health kids rel_imp , robustcluster(ccc)`



**Figure 9:** mixed polSay *i.t3##i.iii* mar i.emp satFin health kids rel\_imp ||| cc: i.t3 ,mle



**Figure 10:** mixed polSay *i.t3##i.iii* mar i.emp satFin health kids rel\_imp ||| cc: i.t3 ,mle

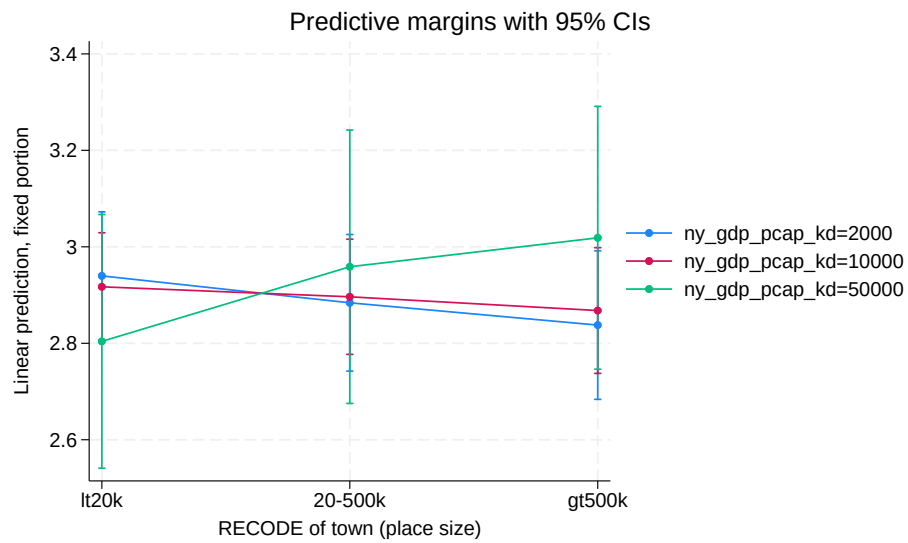


Figure 11

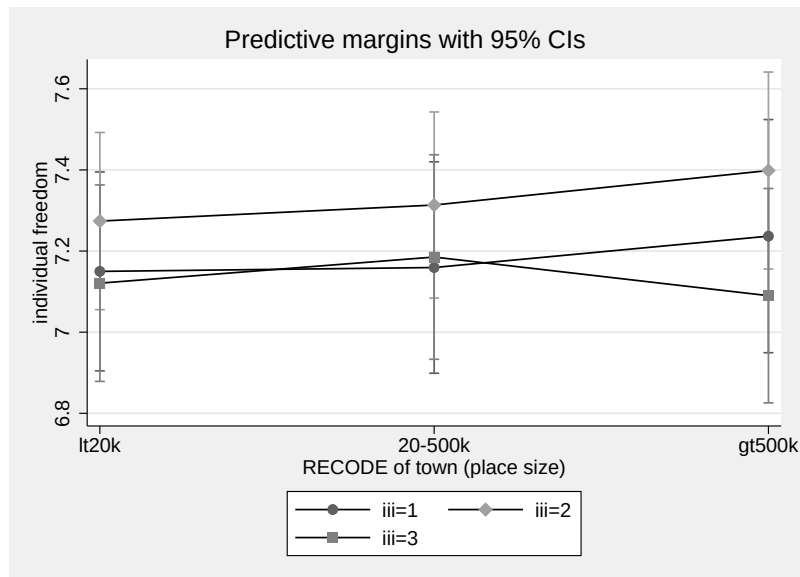
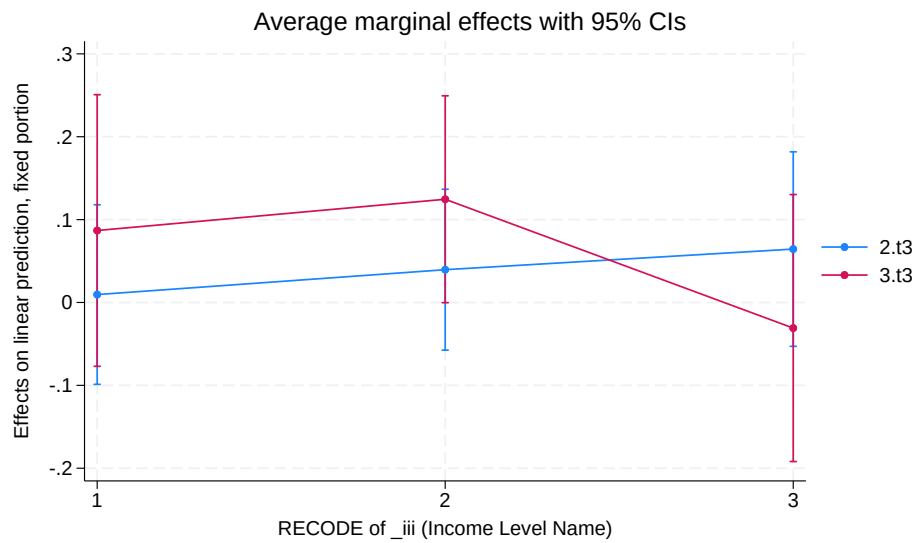
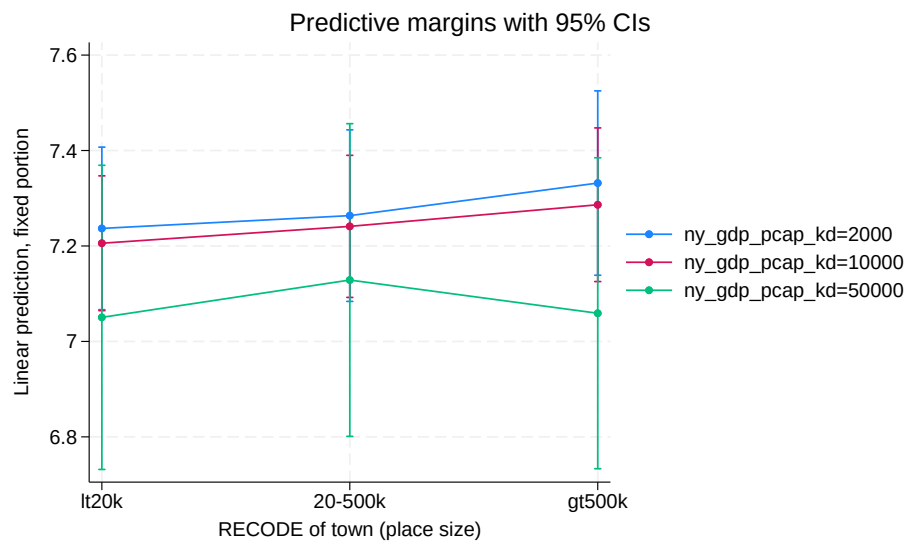


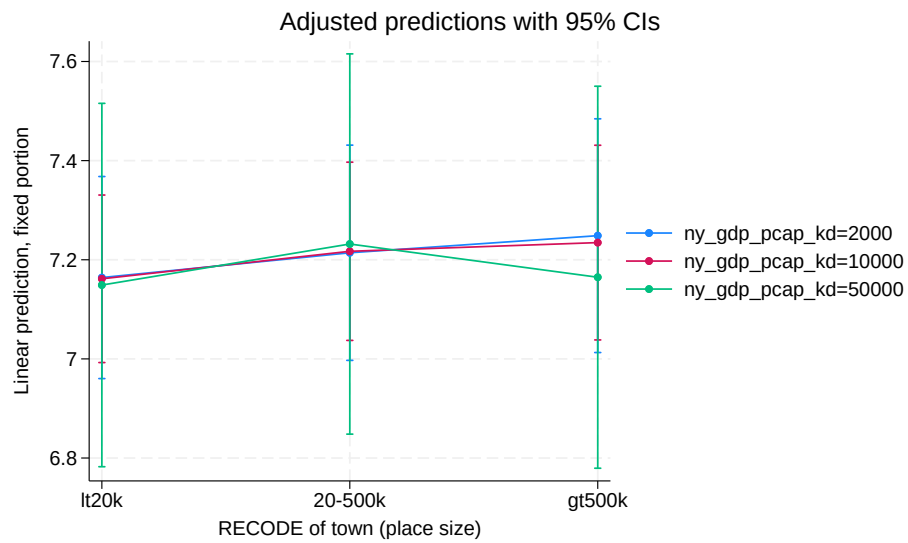
Figure 12: mixed free  $i.t3 \# \# i.iii$  mar i.emp satFin health kids rel\_imp ||| cc: i.t3 ,mle



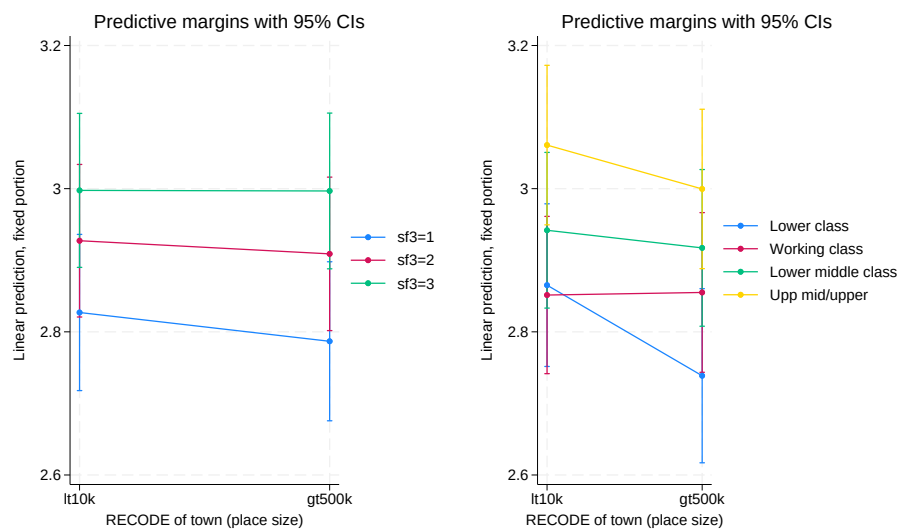
**Figure 13:** mixed free *i.t3##i.iii* mar i.emp satFin health kids relImp ||| cc: i.t3 ,mle



**Figure 14**



**Figure 15:** without controls freedom is not lowest for richest places



**Figure 16:** mixed polSay *i.t4##i.sf3* age age2 male mar i.emp health kids *rel.imp* ||| cc: and the second one same just interacted with soc class

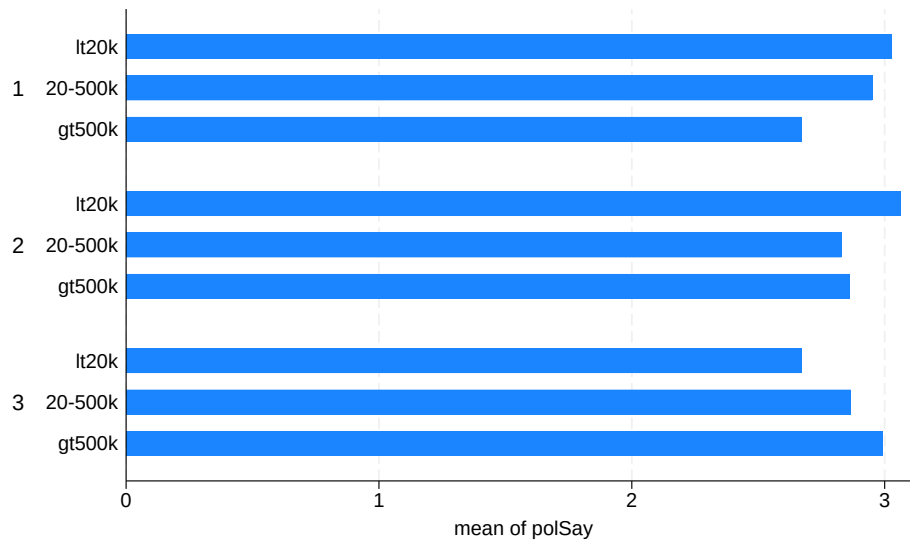


Figure 17

```
. reg polSay TTT2 TTT3 mar i.emp satFin health kids rel_imp i.ccc if iii==1, robust
Linear regression
Number of obs   = 20,590
F(30, 20559)    = 114.51
Prob > F        = 0.0000
R-squared       = 0.1176
Root MSE       = 1.2078
```

| polSay        | Coefficient | Robust<br>std. err. | t      | P> t | [95% conf. interval] |       |
|---------------|-------------|---------------------|--------|------|----------------------|-------|
| TTT2          | -0.04       | 0.02                | -1.92  | 0.05 | -0.08                | 0.00  |
| TTT3          | -0.06       | 0.03                | -2.24  | 0.02 | -0.12                | -0.01 |
| mar           | 0.07        | 0.02                | 3.25   | 0.00 | 0.03                 | 0.12  |
| emp           |             |                     |        |      |                      |       |
| Part time     | -0.06       | 0.03                | -1.70  | 0.09 | -0.12                | 0.01  |
| Self employed | -0.04       | 0.03                | -1.74  | 0.08 | -0.09                | 0.01  |
| Retired       | -0.02       | 0.04                | -0.58  | 0.56 | -0.10                | 0.05  |
| Housewife     | -0.13       | 0.03                | -4.80  | 0.00 | -0.19                | -0.08 |
| Students      | -0.01       | 0.04                | -0.32  | 0.75 | -0.09                | 0.06  |
| Unemployed    | -0.07       | 0.03                | -2.25  | 0.02 | -0.14                | -0.01 |
| Other         | 0.06        | 0.13                | 0.43   | 0.67 | -0.21                | 0.32  |
| satFin        | 0.03        | 0.00                | 7.74   | 0.00 | 0.02                 | 0.04  |
| health        | 0.08        | 0.01                | 7.95   | 0.00 | 0.06                 | 0.10  |
| kids          | -0.00       | 0.01                | -0.27  | 0.79 | -0.01                | 0.01  |
| rel_imp       | 0.03        | 0.01                | 2.10   | 0.04 | 0.00                 | 0.06  |
| ccc           |             |                     |        |      |                      |       |
| 104           | -0.72       | 0.05                | -14.37 | 0.00 | -0.82                | -0.62 |
| 231           | -0.11       | 0.05                | -2.29  | 0.02 | -0.21                | -0.02 |
| 404           | -0.54       | 0.05                | -10.13 | 0.00 | -0.64                | -0.43 |
| 417           | -0.58       | 0.05                | -10.91 | 0.00 | -0.68                | -0.48 |
| 496           | -0.88       | 0.05                | -17.23 | 0.00 | -0.98                | -0.78 |
| 504           | -1.11       | 0.04                | -24.79 | 0.00 | -1.20                | -1.02 |
| 558           | -0.51       | 0.06                | -9.21  | 0.00 | -0.62                | -0.40 |
| 566           | -0.82       | 0.06                | -14.69 | 0.00 | -0.93                | -0.71 |
| 586           | -0.42       | 0.05                | -9.29  | 0.00 | -0.51                | -0.33 |
| 608           | 0.28        | 0.04                | 6.47   | 0.00 | 0.20                 | 0.36  |
| 704           | -0.09       | 0.05                | -1.67  | 0.09 | -0.20                | 0.02  |
| 716           | -0.64       | 0.05                | -11.99 | 0.00 | -0.75                | -0.54 |
| 762           | -0.08       | 0.05                | -1.76  | 0.08 | -0.18                | 0.01  |
| 788           | -0.48       | 0.05                | -9.71  | 0.00 | -0.58                | -0.38 |
| 804           | -1.17       | 0.05                | -21.77 | 0.00 | -1.28                | -1.07 |
| 818           | -0.97       | 0.05                | -20.06 | 0.00 | -1.06                | -0.87 |
| _cons         | 2.91        | 0.08                | 36.31  | 0.00 | 2.76                 | 3.07  |

```
end of do-file
. do "/tmp/SD18984.000000"
. reg polSay TTT2 TTT3 mar i.emp satFin health kids rel_imp i.ccc if iii==3, robust
Linear regression
Number of obs   = 19,221
F(26, 19194)    = 82.06
Prob > F        = 0.0000
R-squared       = 0.1055
Root MSE       = 1.0325
```

| polSay        | Coefficient | Robust<br>std. err. | t     | P> t | [95% conf. interval] |       |
|---------------|-------------|---------------------|-------|------|----------------------|-------|
| TTT2          | 0.13        | 0.02                | 5.93  | 0.00 | 0.09                 | 0.18  |
| TTT3          | 0.21        | 0.03                | 7.97  | 0.00 | 0.16                 | 0.26  |
| mar           | 0.08        | 0.02                | 4.89  | 0.00 | 0.05                 | 0.12  |
| emp           |             |                     |       |      |                      |       |
| Part time     | -0.03       | 0.03                | -1.26 | 0.21 | -0.09                | 0.02  |
| Self employed | -0.07       | 0.03                | -2.11 | 0.03 | -0.14                | -0.01 |
| Retired       | -0.04       | 0.02                | -1.96 | 0.05 | -0.08                | -0.00 |

|            |       |      |        |      |       |       |
|------------|-------|------|--------|------|-------|-------|
| Housewife  | -0.13 | 0.03 | -3.65  | 0.00 | -0.19 | -0.06 |
| Students   | 0.11  | 0.03 | 3.25   | 0.00 | 0.05  | 0.18  |
| Unemployed | -0.11 | 0.04 | -2.63  | 0.01 | -0.19 | -0.03 |
| Other      | -0.13 | 0.06 | -2.16  | 0.03 | -0.26 | -0.01 |
| satFin     | 0.06  | 0.00 | 14.19  | 0.00 | 0.05  | 0.07  |
| health     | 0.06  | 0.01 | 5.45   | 0.00 | 0.04  | 0.08  |
| kids       | -0.01 | 0.01 | -1.72  | 0.09 | -0.03 | 0.00  |
| rel_imp    | 0.08  | 0.01 | 10.77  | 0.00 | 0.07  | 0.10  |
| ccc        |       |      |        |      |       |       |
| 36         | -0.62 | 0.05 | -12.64 | 0.00 | -0.72 | -0.52 |
| 124        | -0.35 | 0.04 | -7.73  | 0.00 | -0.43 | -0.26 |
| 196        | -0.86 | 0.06 | -15.29 | 0.00 | -0.97 | -0.75 |
| 203        | -1.00 | 0.05 | -19.61 | 0.00 | -1.10 | -0.90 |
| 344        | -0.55 | 0.05 | -10.76 | 0.00 | -0.65 | -0.45 |
| 392        | 0.09  | 0.06 | 1.50   | 0.13 | -0.03 | 0.20  |
| 446        | -0.35 | 0.05 | -6.33  | 0.00 | -0.46 | -0.24 |
| 528        | -0.61 | 0.05 | -12.20 | 0.00 | -0.71 | -0.52 |
| 554        | -0.49 | 0.05 | -9.64  | 0.00 | -0.59 | -0.39 |
| 702        | -0.52 | 0.05 | -9.86  | 0.00 | -0.63 | -0.42 |
| 703        | -1.06 | 0.05 | -20.09 | 0.00 | -1.16 | -0.95 |
| 858        | -0.50 | 0.06 | -8.72  | 0.00 | -0.61 | -0.39 |
| _cons      | 2.46  | 0.06 | 40.03  | 0.00 | 2.34  | 2.58  |

end of do-file

. reg polSay i.t3##i.iii mar i.emp satFin health kids rel\_imp i.ccc, robust

note: 818.ccc omitted because of collinearity.

note: 862.ccc omitted because of collinearity.

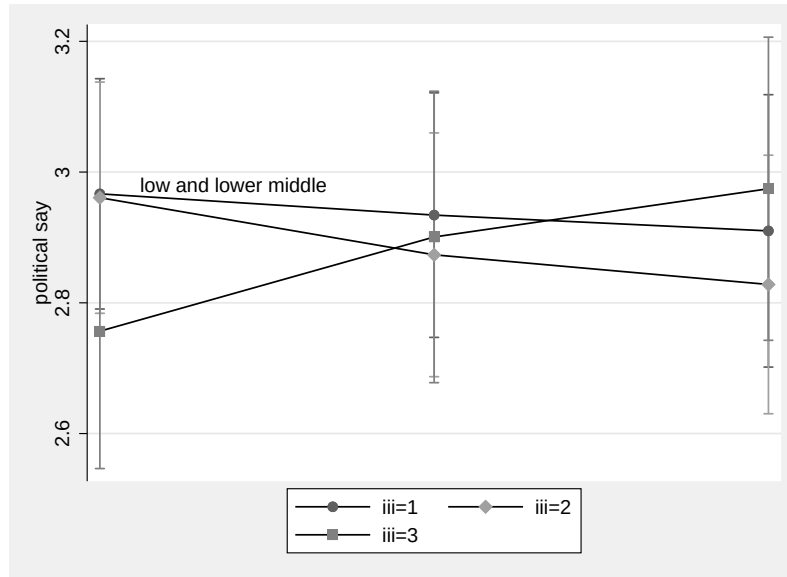
Linear regression

Number of obs = 62,294  
F(64, 62229) = 125.62  
Prob > F = 0.0000  
R-squared = 0.1008  
Root MSE = 1.1565

| polSay        | Coefficient | Robust<br>std. err. | t      | P> t | [95% conf. interval] |       |
|---------------|-------------|---------------------|--------|------|----------------------|-------|
| t3            |             |                     |        |      |                      |       |
| 20-500k       | -0.04       | 0.02                | -1.83  | 0.07 | -0.08                | 0.00  |
| gt500k        | -0.06       | 0.03                | -2.14  | 0.03 | -0.11                | -0.01 |
| iii           |             |                     |        |      |                      |       |
| 2             | 0.24        | 0.05                | 4.60   | 0.00 | 0.14                 | 0.35  |
| 3             | 0.79        | 0.06                | 14.20  | 0.00 | 0.68                 | 0.90  |
| t3#iii        |             |                     |        |      |                      |       |
| 20-500k#2     | -0.04       | 0.03                | -1.32  | 0.19 | -0.10                | 0.02  |
| 20-500k#3     | 0.18        | 0.03                | 5.67   | 0.00 | 0.11                 | 0.24  |
| gt500k#2      | -0.04       | 0.04                | -1.22  | 0.22 | -0.12                | 0.03  |
| gt500k#3      | 0.27        | 0.04                | 7.11   | 0.00 | 0.20                 | 0.35  |
| mar           | 0.06        | 0.01                | 5.61   | 0.00 | 0.04                 | 0.09  |
| emp           |             |                     |        |      |                      |       |
| Part time     | -0.00       | 0.02                | -0.12  | 0.91 | -0.04                | 0.03  |
| Self employed | -0.04       | 0.02                | -2.33  | 0.02 | -0.07                | -0.01 |
| Retired       | -0.02       | 0.02                | -0.93  | 0.35 | -0.05                | 0.02  |
| Housewife     | -0.09       | 0.02                | -5.51  | 0.00 | -0.12                | -0.06 |
| Students      | 0.04        | 0.02                | 1.91   | 0.06 | -0.00                | 0.08  |
| Unemployed    | -0.05       | 0.02                | -2.61  | 0.01 | -0.09                | -0.01 |
| Other         | -0.04       | 0.05                | -0.69  | 0.49 | -0.14                | 0.07  |
| satFin        | 0.03        | 0.00                | 15.08  | 0.00 | 0.03                 | 0.04  |
| health        | 0.07        | 0.01                | 11.71  | 0.00 | 0.06                 | 0.08  |
| kids          | 0.00        | 0.00                | 0.13   | 0.90 | -0.01                | 0.01  |
| rel_imp       | 0.06        | 0.01                | 11.16  | 0.00 | 0.05                 | 0.08  |
| ccc           |             |                     |        |      |                      |       |
| 32            | 1.21        | 0.05                | 26.33  | 0.00 | 1.12                 | 1.30  |
| 36            | -0.63       | 0.05                | -12.97 | 0.00 | -0.73                | -0.54 |
| 50            | 0.96        | 0.05                | 20.24  | 0.00 | 0.87                 | 1.05  |
| 51            | 0.18        | 0.05                | 3.47   | 0.00 | 0.08                 | 0.29  |
| 76            | -0.30       | 0.05                | -6.38  | 0.00 | -0.39                | -0.21 |
| 104           | 0.26        | 0.05                | 4.93   | 0.00 | 0.15                 | 0.36  |
| 124           | -0.35       | 0.04                | -7.74  | 0.00 | -0.43                | -0.26 |
| 170           | 0.01        | 0.05                | 0.20   | 0.84 | -0.09                | 0.11  |
| 196           | -0.86       | 0.06                | -15.39 | 0.00 | -0.97                | -0.75 |
| 203           | -1.00       | 0.05                | -19.83 | 0.00 | -1.10                | -0.91 |
| 231           | 0.87        | 0.05                | 17.58  | 0.00 | 0.77                 | 0.97  |
| 320           | 0.13        | 0.05                | 2.51   | 0.01 | 0.03                 | 0.23  |
| 344           | -0.55       | 0.05                | -10.75 | 0.00 | -0.65                | -0.45 |
| 360           | 0.60        | 0.04                | 13.81  | 0.00 | 0.52                 | 0.69  |
| 364           | 0.22        | 0.05                | 4.65   | 0.00 | 0.13                 | 0.32  |
| 368           | 0.09        | 0.05                | 1.69   | 0.09 | -0.01                | 0.19  |
| 392           | 0.06        | 0.06                | 0.99   | 0.32 | -0.05                | 0.17  |
| 398           | 0.65        | 0.05                | 13.65  | 0.00 | 0.56                 | 0.75  |
| 400           | 0.48        | 0.05                | 9.43   | 0.00 | 0.38                 | 0.58  |
| 404           | 0.45        | 0.05                | 8.56   | 0.00 | 0.34                 | 0.55  |
| 417           | 0.41        | 0.05                | 7.87   | 0.00 | 0.30                 | 0.51  |
| 422           | -0.03       | 0.05                | -0.61  | 0.54 | -0.13                | 0.07  |
| 434           | 0.07        | 0.05                | 1.21   | 0.23 | -0.04                | 0.17  |
| 446           | -0.35       | 0.05                | -6.34  | 0.00 | -0.45                | -0.24 |
| 458           | 0.46        | 0.05                | 9.66   | 0.00 | 0.37                 | 0.56  |
| 462           | -0.28       | 0.05                | -5.29  | 0.00 | -0.38                | -0.18 |
| 484           | 0.02        | 0.04                | 0.53   | 0.59 | -0.06                | 0.11  |
| 496           | 0.16        | 0.04                | 3.59   | 0.00 | 0.07                 | 0.24  |
| 504           | -0.14       | 0.04                | -3.20  | 0.00 | -0.23                | -0.05 |
| 528           | -0.61       | 0.05                | -12.19 | 0.00 | -0.71                | -0.51 |
| 554           | -0.50       | 0.05                | -9.84  | 0.00 | -0.60                | -0.40 |
| 558           | 0.47        | 0.05                | 8.65   | 0.00 | 0.37                 | 0.58  |
| 566           | 0.16        | 0.06                | 2.86   | 0.00 | 0.05                 | 0.27  |
| 586           | 0.55        | 0.05                | 11.79  | 0.00 | 0.46                 | 0.64  |
| 604           | 0.27        | 0.05                | 5.73   | 0.00 | 0.18                 | 0.36  |
| 608           | 1.25        | 0.04                | 27.97  | 0.00 | 1.16                 | 1.34  |

|       |       |           |        |      |       |       |
|-------|-------|-----------|--------|------|-------|-------|
| 702   | -0.52 | 0.05      | -9.79  | 0.00 | -0.62 | -0.41 |
| 703   | -1.07 | 0.05      | -20.44 | 0.00 | -1.17 | -0.97 |
| 704   | 0.94  | 0.05      | 18.73  | 0.00 | 0.84  | 1.03  |
| 716   | 0.34  | 0.05      | 6.32   | 0.00 | 0.23  | 0.44  |
| 762   | 0.90  | 0.05      | 18.66  | 0.00 | 0.80  | 0.99  |
| 788   | 0.49  | 0.05      | 10.10  | 0.00 | 0.40  | 0.59  |
| 804   | -0.15 | 0.05      | -3.02  | 0.00 | -0.25 | -0.05 |
| 818   | 0.00  | (omitted) |        |      |       |       |
| 858   | -0.51 | 0.06      | -8.96  | 0.00 | -0.62 | -0.40 |
| 862   | 0.00  | (omitted) |        |      |       |       |
| _cons | 1.80  | 0.05      | 36.71  | 0.00 | 1.71  | 1.90  |

.19



**Figure 18:** por some reason margins dont work for me with cc dummies this is without cc dummies, but above in stata printout interactions significant with cc dummies

## 2 Additional info

### 2.1 AI literature review

Gemini query “scholarly literature on individual autonomy or freedom across urban-rural”<sup>13</sup>

| Autonomy Dimension | Rural                                                                    | Urban                                                                           |
|--------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Social Control     | Strict; governed by kinship, mutual surveillance, and traditional norms. | Loose; regulated by formal systems (laws, courts) and shielded by anonymity.    |
| Core Values        | Interdependence, community solidarity, and generational stability.       | Individualism, open-mindedness, and socioeconomic mobility.                     |
| Family Dynamics    | High value placed on obedience and material/financial cooperation.       | Focus on psychological independence and diverse career/lifestyle paths.         |
| Physical Liberty   | High geographic isolation; dependency on personal transit for survival.  | High physical access via transit; restricted by crowd density and safety rules. |

### 2.2 Country sample sizes; Regional country classification; Variables' definitions (survey questions); Histograms

We use World Bank regions classification into several regions: East Asia/Pacific, Europe/Central Asia, Latin America/Caribbean, Middle East/North Africa, North America, South Asia, and Sub-Saharan Africa. Countries in each region and sample sizes are in appendix.

```
. bys region: ta cc
```

```
-----> regionname =
```

```
ISO 3166-1 |  
alpha-3 |
```

<sup>13</sup>[https://www.google.com/search?q=scholarly+literature+on+individual+autonomy+or+freedom+across+urban-rural&rlz=1CAHWSA\\_enUS1125US1125&oq=scholarly+literature+on+individual+autonomy+or+freedom+across+urban-rural&gs\\_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCTIyNDYyYjBqN6gCCLACAfEFAI5zaHyURfbxBQC0c2h81EX2&sourceid=chrome&source=chrome.ob&ie=UTF-8](https://www.google.com/search?q=scholarly+literature+on+individual+autonomy+or+freedom+across+urban-rural&rlz=1CAHWSA_enUS1125US1125&oq=scholarly+literature+on+individual+autonomy+or+freedom+across+urban-rural&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCTIyNDYyYjBqN6gCCLACAfEFAI5zaHyURfbxBQC0c2h81EX2&sourceid=chrome&source=chrome.ob&ie=UTF-8)

| country<br>code | Freq. | Percent | Cum.   |
|-----------------|-------|---------|--------|
| NIR             | 447   | 26.77   | 26.77  |
| TWN             | 1,223 | 73.23   | 100.00 |
| Total           | 1,670 | 100.00  |        |

-> regionname = East Asia/Pacific

| ISO 3166-1<br>alpha-3<br>country<br>code | Freq.  | Percent | Cum.   |
|------------------------------------------|--------|---------|--------|
| AUS                                      | 1,813  | 7.29    | 7.29   |
| CHN                                      | 3,036  | 12.21   | 19.50  |
| HKG                                      | 2,075  | 8.35    | 27.85  |
| IDN                                      | 3,200  | 12.87   | 40.72  |
| JPN                                      | 1,353  | 5.44    | 46.16  |
| KOR                                      | 1,245  | 5.01    | 51.16  |
| MAC                                      | 1,023  | 4.11    | 55.28  |
| MMR                                      | 1,200  | 4.83    | 60.10  |
| MNG                                      | 1,638  | 6.59    | 66.69  |
| MYS                                      | 1,313  | 5.28    | 71.97  |
| NZL                                      | 1,057  | 4.25    | 76.22  |
| PHL                                      | 1,200  | 4.83    | 81.05  |
| SGP                                      | 2,012  | 8.09    | 89.14  |
| THA                                      | 1,500  | 6.03    | 95.17  |
| VNM                                      | 1,200  | 4.83    | 100.00 |
| Total                                    | 24,865 | 100.00  |        |

-> regionname = Europe/Central Asia

| ISO 3166-1<br>alpha-3<br>country<br>code | Freq.  | Percent | Cum.   |
|------------------------------------------|--------|---------|--------|
| AND                                      | 1,004  | 3.88    | 3.88   |
| ARM                                      | 1,223  | 4.73    | 8.61   |
| CYP                                      | 1,000  | 3.87    | 12.48  |
| CZE                                      | 1,200  | 4.64    | 17.12  |
| DEU                                      | 1,528  | 5.91    | 23.03  |
| GBR                                      | 2,609  | 10.09   | 33.13  |
| GRC                                      | 1,200  | 4.64    | 37.77  |
| KAZ                                      | 1,276  | 4.94    | 42.70  |
| KGZ                                      | 1,200  | 4.64    | 47.35  |
| NLD                                      | 2,145  | 8.30    | 55.64  |
| ROU                                      | 1,257  | 4.86    | 60.51  |
| RUS                                      | 1,810  | 7.00    | 67.51  |
| SRB                                      | 1,046  | 4.05    | 71.55  |
| SVK                                      | 1,200  | 4.64    | 76.20  |
| TJK                                      | 1,200  | 4.64    | 80.84  |
| TUR                                      | 2,415  | 9.34    | 90.18  |
| UKR                                      | 1,289  | 4.99    | 95.16  |
| UZB                                      | 1,250  | 4.84    | 100.00 |
| Total                                    | 25,852 | 100.00  |        |

-> regionname = Latin America/Caribbean

| ISO 3166-1<br>alpha-3<br>country<br>code | Freq.  | Percent | Cum.   |
|------------------------------------------|--------|---------|--------|
| ARG                                      | 1,003  | 5.75    | 5.75   |
| BOL                                      | 2,067  | 11.85   | 17.60  |
| BRA                                      | 1,762  | 10.10   | 27.71  |
| CHL                                      | 1,000  | 5.73    | 33.44  |
| COL                                      | 1,520  | 8.72    | 42.16  |
| ECU                                      | 1,200  | 6.88    | 49.04  |
| GTM                                      | 1,229  | 7.05    | 56.09  |
| MEX                                      | 1,741  | 9.98    | 66.07  |
| NIC                                      | 1,200  | 6.88    | 72.95  |
| PER                                      | 1,400  | 8.03    | 80.98  |
| PRI                                      | 1,127  | 6.46    | 87.44  |
| URY                                      | 1,000  | 5.73    | 93.18  |
| VEN                                      | 1,190  | 6.82    | 100.00 |
| Total                                    | 17,439 | 100.00  |        |

-> regionname = Middle East/North Africa

| ISO 3166-1<br>alpha-3<br>country<br>code | Freq. | Percent | Cum.   |
|------------------------------------------|-------|---------|--------|
| EGY                                      | 1,200 | 12.11   | 12.11  |
| IRN                                      | 1,499 | 15.13   | 27.25  |
| IRQ                                      | 1,200 | 12.11   | 39.36  |
| JOR                                      | 1,203 | 12.14   | 51.50  |
| LBN                                      | 1,200 | 12.11   | 63.62  |
| LBY                                      | 1,196 | 12.07   | 75.69  |
| MAR                                      | 1,200 | 12.11   | 87.81  |
| TUN                                      | 1,208 | 12.19   | 100.00 |
| Total                                    | 9,906 | 100.00  |        |

-> regionname = North America

| ISO 3166-1<br>alpha-3<br>country<br>code | Freq. | Percent | Cum. |
|------------------------------------------|-------|---------|------|
|------------------------------------------|-------|---------|------|

|                                    |       |         |        |
|------------------------------------|-------|---------|--------|
| CAN                                | 4,018 | 60.75   | 60.75  |
| USA                                | 2,596 | 39.25   | 100.00 |
| Total                              | 6,614 | 100.00  |        |
| -----                              |       |         |        |
| -> regionname = South Asia         |       |         |        |
| ISO 3166-1                         |       |         |        |
| alpha-3                            |       |         |        |
| country                            |       |         |        |
| code                               | Freq. | Percent | Cum.   |
| BGD                                | 1,200 | 20.25   | 20.25  |
| IND                                | 1,692 | 28.55   | 48.80  |
| MDV                                | 1,039 | 17.53   | 66.33  |
| PAK                                | 1,995 | 33.67   | 100.00 |
| Total                              | 5,926 | 100.00  |        |
| -----                              |       |         |        |
| -> regionname = Sub-Saharan Africa |       |         |        |
| ISO 3166-1                         |       |         |        |
| alpha-3                            |       |         |        |
| country                            |       |         |        |
| code                               | Freq. | Percent | Cum.   |
| ETH                                | 1,230 | 24.86   | 24.86  |
| KEN                                | 1,266 | 25.59   | 50.44  |
| NGA                                | 1,237 | 25.00   | 75.44  |
| ZWE                                | 1,215 | 24.56   | 100.00 |
| Total                              | 4,948 | 100.00  |        |

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**Table 2:** Variable definitions.

| name                                  | description                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| freedom                               | "Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out." |
| place size                            | "OBSERVATIONS BY THE INTERVIEWER; Code size of town where interview was conducted"                                                                                                                                                                                                                                                                           |
| age                                   | age                                                                                                                                                                                                                                                                                                                                                          |
| male                                  | male                                                                                                                                                                                                                                                                                                                                                         |
| married or living together as married | "Are you currently ....(READ OUT AND CODE ONE ONLY) 1 'Married' 2 'Living together as married' 3 'Divorced' 4 'Separated' 5 'Widowed' 6 'Single/Never married' 7 'Divorced, Separated or Widow' 8 'Living apart but steady relation (married,cohabitation)'"                                                                                                 |
| Employment status                     | "Are you employed now or not? IF YES: About how many hours a week? If more than one job: only for the main job 1 'Full time' 2 'Part time' 3 'Self employed' 4 'Retired' 5 'Housewife' 6 'Students' 7 'Unemployed' 8 'Other'"                                                                                                                                |
| financial satisfaction                | "How satisfied are you with the financial situation of your household?"                                                                                                                                                                                                                                                                                      |
| class                                 | "Social class (subjective)"                                                                                                                                                                                                                                                                                                                                  |
| income                                | "Scale of incomes"                                                                                                                                                                                                                                                                                                                                           |
| health                                | "State of health (subjective)"                                                                                                                                                                                                                                                                                                                               |
| religion important                    | "WVS2000: For each of the following aspects, indicate how important it is in your life. Would you say it is: EVS1999: Please say, for each of the following, how important it is in your life. Religion"                                                                                                                                                     |

Per distributions it is worth noting left-skewed distribution on freedom, unexpectedly most people feel maximally free at 10 out of 1-10, likewise but as expected with urbanism—most people are in the top category  $> 500k$ , and again left-skewed on religion, probably not unexpected—religiosity is declining, but still strong.

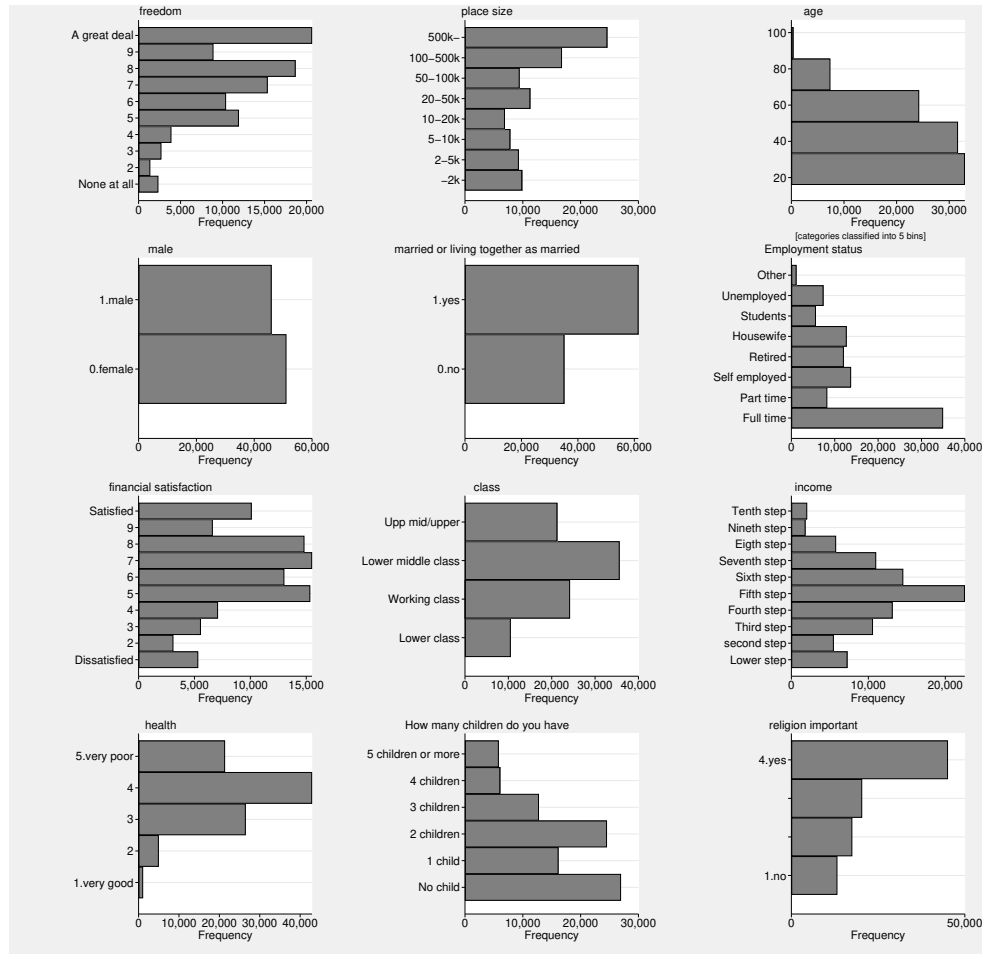


Figure 19: Variables' distribution.

## 2.3 First/initial results

### 2.3.1 by region

These were our first/initial results but they do not hold when including country dummies and so we postpone them to appendix.

First for all regions in general there is more freedom in smaller areas v top bin, the base case ( $> 500k$ ) in table 3 but controlling for other predictors in table 4 it is just the 2nd (2-5k) and 3rd (5-10k) smallest categories, and smaller values in 6th and 7th categories. But the very smallest areas ( $< 2k$ ) are not more free. Next we separate by region.

East Asia/Pacific stands out with high effect sizes at about .4 or even .5, also controlling for other predictors of freedom. Europe is mostly less free in smaller areas. Latin America/Caribbean stands out with negative effect in the very smallest areas. Middle East/North Africa has mostly negative effects. North America is positive in 2nd largest category (100-500k). South Asia and Sub-Saharan Africa mostly do not show significant differences.<sup>14</sup>

<sup>14</sup>Except .3 for South Asia for the smallest places ( $< 2k$ ).

|                | All region dummies | (with dummies) | East Asia/Pacific | Europe/Central Asia | Latin America/Caribbean | Middle East/North Africa | North America | South Asia | Sub-Saharan Africa |
|----------------|--------------------|----------------|-------------------|---------------------|-------------------------|--------------------------|---------------|------------|--------------------|
| town==2k       | -0.05+             |                | 0.05              | -0.13*              | -0.28***                | 0.22                     | -0.07         | 0.01       | -0.29              |
| town==2-5k     | 0.14***            |                | 0.36***           | 0.09                | -0.13                   | -0.09                    | -0.07         | 0.02       | 0.06               |
| town==5-10k    | 0.20***            |                | 0.51***           | 0.03                | -0.01                   | -0.12                    | 0.13          | 0.03       | 0.35               |
| town==10-20k   | 0.09**             |                | 0.52***           | -0.15**             | 0.07                    | -0.34***                 | -0.03         | -0.02      | 0.19               |
| town==20-50k   | 0.12***            |                | 0.28***           | 0.08+               | 0.08                    | -0.17*                   | 0.09          | 0.03       | -0.22              |
| town==50-100k  | 0.12***            |                | 0.41***           | -0.10+              | 0.04                    | -0.02                    | 0.12          | -0.41+     | -0.06              |
| town==100-500k | 0.08***            |                | 0.55***           | -0.12**             | -0.08+                  | -0.27***                 | 0.18**        | 0.04       | -0.45+             |
| constant       | 6.94***            |                | 6.76***           | 7.06***             | 7.85***                 | 6.88***                  | 7.44***       | 7.52***    | 7.14***            |
| N              | 93414              |                | 24560             | 25289               | 17285                   | 9846                     | 5833          | 5907       | 4694               |

+ 0.10 \* 0.05 \*\*  
0.01 \*\*\* 0.001; robust std err

Table 3: OLS regressions of freedom/agency

|                                       | All region dummies | (with dummies) | East Asia/Pacific | Europe/Central Asia | Latin America/Caribbean | Middle East/North Africa | North America | South Asia | Sub-Saharan Africa |
|---------------------------------------|--------------------|----------------|-------------------|---------------------|-------------------------|--------------------------|---------------|------------|--------------------|
| town==2k                              | -0.03              |                | -0.00             | -0.03               | -0.18**                 | 0.02                     | -0.06         | 0.28*      | -0.17              |
| town==2-5k                            | 0.09**             |                | 0.28***           | 0.00                | -0.14                   | -0.13                    | -0.10         | 0.24+      | 0.18               |
| town==5-10k                           | 0.11***            |                | 0.36***           | 0.02                | -0.03                   | -0.15+                   | -0.03         | 0.13       | 0.33               |
| town==10-20k                          | 0.03               |                | 0.41***           | -0.20***            | -0.00                   | -0.33***                 | -0.00         | 0.27       | 0.19               |
| town==20-50k                          | 0.04               |                | 0.24***           | -0.01               | -0.01                   | -0.22**                  | 0.05          | 0.02       | -0.17              |
| town==50-100k                         | 0.06*              |                | 0.34***           | -0.21***            | -0.01                   | -0.07                    | 0.13+         | 0.10       | 0.02               |
| town==100-500k                        | 0.06**             |                | 0.48***           | -0.18***            | -0.03                   | -0.30***                 | 0.13*         | 0.07       | -0.43+             |
| age                                   | 0.01***            |                | 0.01*             | 0.00                | 0.02**                  | 0.00                     | 0.03***       | 0.02       | 0.02               |
| age2                                  | -0.00***           |                | -0.00**           | -0.00               | -0.00+                  | 0.00                     | -0.00**       | -0.00      | -0.00              |
| male                                  | 0.00               |                | -0.09***          | 0.05                | -0.08*                  | 0.23***                  | -0.13**       | 0.16+      | 0.14               |
| married or living together as married | -0.01              |                | -0.00             | -0.06+              | -0.05                   | 0.01                     | -0.00         | 0.06       | 0.27*              |
| Full time                             | 0.00               |                | 0.00              | 0.00                | 0.00                    | 0.00                     | 0.00          | 0.00       | 0.00               |
| Part time                             | -0.10***           |                | -0.04             | -0.09+              | -0.24***                | -0.05                    | 0.08          | -0.32**    | -0.06              |
| Self employed                         | -0.07**            |                | -0.12**           | 0.08                | -0.04                   | -0.10                    | 0.32***       | -0.47***   | 0.15               |
| Retired                               | -0.01              |                | 0.15**            | -0.08               | 0.05                    | 0.22*                    | -0.07         | -0.31      | -0.15              |
| Housewife                             | -0.22***           |                | -0.19***          | -0.19***            | -0.05                   | -0.17*                   | -0.05         | -0.54***   | -0.15              |
| Students                              | -0.21***           |                | -0.22***          | -0.08               | -0.27***                | -0.21*                   | 0.09          | -0.31      | -0.12              |
| Unemployed                            | -0.10**            |                | -0.11+            | -0.16*              | -0.02                   | -0.14                    | -0.14         | -0.28+     | 0.12               |
| Other                                 | -0.23**            |                | -0.29+            | -0.34*              | 0.01                    | -0.20                    | -0.03         | -1.87***   | 1.53**             |
| financial satisfaction                | 0.27***            |                | 0.31***           | 0.31***             | 0.19***                 | 0.31***                  | 0.26***       | 0.31***    | 0.10***            |
| health                                | 0.34***            |                | 0.32***           | 0.25***             | 0.37***                 | 0.47***                  | 0.51***       | 0.21***    | 0.24***            |
| How many children do you have         | 0.04***            |                | -0.01             | 0.05***             | 0.02                    | 0.10***                  | 0.05**        | 0.03       | -0.10*             |
| religion important                    | 0.06***            |                | 0.09***           | 0.01                | 0.03                    | 0.37***                  | 0.08***       | 0.17**     | 0.24*              |
| constant                              | 3.55***            |                | 3.23***           | 4.26***             | 4.79***                 | 1.76***                  | 2.89***       | 3.28***    | 4.26***            |
| N                                     | 86653              |                | 23321             | 21600               | 16670                   | 9427                     | 5776          | 5231       | 4628               |

+ 0.10 \* 0.05 \*\*  
0.01 \*\*\* 0.001; robust std err

Table 4: OLS regressions of freedom/agency

### 2.3.2 Insignificant results with country dummies

Country dummies render cross-country patterns insignificant.

Here we conduct important sensitivity/robustness checks. First in a1 just urbanicity dummies. Then a1a adds region dummies like earlier in the first table little change. A1b has income level dummies TODO spell out; again little change; but addition of country dummies in a1c kills the significance and turns many categories negative notably the first smallest one, similar with controls in models a2\* and country dummies make categories even more negative in a2c than in a1c. And then a1c2 and a2c2 add clustered standard errors on country (stata: robust cluster(country))—all significance disappears except in a2c2 on the smallest areas—hence only the very smallest areas < 2k are less free than the largest ones > 500k.

in table 5 we conduct additional robustness checks, a1 and a2 are repeated from the body; a1a and a2a include region dummies: in a1a 100 – 500k results are attenuated, but others are stronger; and in a2a the top three categories spanning 20 – 500k are weaker but other categories are stronger and the smaller category < 2k positive effect goes away. But then in a1c and a2c addition of country dummies kills the effects unfortunately. And having clustered standard errors yields all results insignificant

|                                       | a1      | ala     | alb     | alc      | alc2    | a2       | a2a      | a2b      | a2c      | a2c2     |
|---------------------------------------|---------|---------|---------|----------|---------|----------|----------|----------|----------|----------|
| town==2k                              | -0.03   | -0.05+  | -0.03   | -0.07*   | -0.07   | -0.06*   | -0.03    | -0.06*   | -0.06+   | -0.06    |
| town==2-5k                            | 0.08**  | 0.14*** | 0.10**  | -0.05    | -0.05   | 0.03     | 0.09**   | 0.04     | -0.03    | -0.03    |
| town==5-10k                           | 0.16*** | 0.20*** | 0.16*** | 0.00     | 0.00    | 0.07*    | 0.11***  | 0.07*    | -0.02    | -0.02    |
| town==10-20k                          | 0.03    | 0.09**  | 0.02    | -0.12*** | -0.12   | 0.00     | 0.03     | -0.01    | -0.09**  | -0.09    |
| town==20-50k                          | 0.14*** | 0.12*** | 0.13*** | -0.02    | -0.02   | 0.07**   | 0.04     | 0.07**   | -0.04    | -0.04    |
| town==50-100k                         | 0.13*** | 0.12*** | 0.11*** | -0.01    | -0.01   | 0.10***  | 0.06*    | 0.08**   | -0.01    | -0.01    |
| town==100-500k                        | 0.13*** | 0.08*** | 0.11*** | -0.02    | -0.02   | 0.11***  | 0.06**   | 0.09***  | -0.03    | -0.03    |
| age                                   |         |         |         |          |         | 0.01+    | 0.01***  | 0.00     | 0.02***  | 0.02***  |
| age2                                  |         |         |         |          |         | -0.00*   | -0.00*** | -0.00+   | -0.00*** | -0.00*** |
| male                                  |         |         |         |          |         | 0.02     | 0.00     | 0.02     | 0.01     | 0.01     |
| married or living together as married |         |         |         |          |         | -0.08*** | -0.01    | -0.07*** | 0.00     | 0.00     |
| Full time                             |         |         |         |          |         | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     |
| Part time                             |         |         |         |          |         | -0.09*** | -0.10*** | -0.09**  | -0.09*** | -0.09*   |
| Self employed                         |         |         |         |          |         | 0.02     | -0.07**  | 0.02     | -0.02    | -0.02    |
| Retired                               |         |         |         |          |         | 0.02     | -0.01    | 0.01     | -0.05+   | -0.05    |
| Housewife                             |         |         |         |          |         | -0.16*** | -0.22*** | -0.16*** | -0.20*** | -0.20*** |
| Students                              |         |         |         |          |         | -0.19*** | -0.21*** | -0.19*** | -0.14*** | -0.14**  |
| Unemployed                            |         |         |         |          |         | -0.01    | -0.10**  | -0.00    | -0.11*** | -0.11*   |
| Other                                 |         |         |         |          |         | -0.07    | -0.23**  | -0.04    | -0.25*** | -0.25*   |
| financial satisfaction                |         |         |         |          |         | 0.27***  | 0.27***  | 0.27***  | 0.26***  | 0.26***  |
| health                                |         |         |         |          |         | 0.35***  | 0.34***  | 0.34***  | 0.34***  | 0.34***  |
| How many children do you have         |         |         |         |          |         | 0.07***  | 0.04***  | 0.07***  | 0.02*    | 0.02     |
| religion important                    |         |         |         |          |         | 0.07***  | 0.06***  | 0.07***  | 0.03***  | 0.03*    |
| region dummies                        | no      | yes     | no      | no       | no      | no       | yes      | no       | no       | no       |
| country dummies                       | no      | no      | no      | yes      | yes     | no       | no       | no       | yes      | yes      |
| income level dummies                  | no      | no      | yes     | no       | no      | no       | no       | yes      | no       | no       |
| constant                              | 7.14*** | 6.94*** | 7.14*** | 7.57***  | 7.57*** | 3.84***  | 3.55***  | 3.87***  | 3.99***  | 3.99***  |
| N                                     | 95083   | 93414   | 93414   | 95083    | 95083   | 87874    | 86653    | 86653    | 87874    | 87874    |

+ 0.10 \* 0.05 \*\*  
0.01 \*\*\* 0.001; robust std err

Table 5: OLS regressions of freedom/agency

## 2.4 Freedom main results from paper

### 2.4.1 By country

First let's start with sample sizes by country by 3 urbanicity categories—in general they are sizable > 100, and in cases where they are not CI is wider on the estimates such as with ETH at -1.12 on lt20k that is insignificant

| ISO 3166-1<br>alpha-3<br>country<br>code | RECODE of town (place size) |         |        | Total |
|------------------------------------------|-----------------------------|---------|--------|-------|
|                                          | lt20k                       | 20-500k | gt500k |       |
| AND                                      | 730                         | 274     | 0      | 1,004 |
| ARG                                      | 200                         | 180     | 623    | 1,003 |
| ARM                                      | 545                         | 232     | 446    | 1,223 |
| AUS                                      | 453                         | 567     | 705    | 1,725 |
| BGD                                      | 991                         | 209     | 0      | 1,200 |
| BOL                                      | 625                         | 741     | 701    | 2,067 |
| BRA                                      | 185                         | 864     | 713    | 1,762 |
| CAN                                      | 1,098                       | 1,694   | 1,226  | 4,018 |
| CHL                                      | 152                         | 487     | 361    | 1,000 |
| CHN                                      | 50                          | 2,986   | 0      | 3,036 |
| COL                                      | 304                         | 722     | 494    | 1,520 |
| CYP                                      | 615                         | 385     | 0      | 1,000 |
| CZE                                      | 680                         | 370     | 150    | 1,200 |
| DEU                                      | 712                         | 616     | 200    | 1,528 |
| ECU                                      | 384                         | 466     | 350    | 1,200 |
| EGY                                      | 424                         | 456     | 320    | 1,200 |
| ETH                                      | 841                         | 380     | 9      | 1,230 |
| GBR                                      | 495                         | 1,713   | 397    | 2,605 |
| GRC                                      | 420                         | 340     | 440    | 1,200 |
| GTM                                      | 71                          | 505     | 653    | 1,229 |
| HKG                                      | 0                           | 0       | 2,075  | 2,075 |
| IDN                                      | 2,761                       | 346     | 93     | 3,200 |
| IND                                      | 1,415                       | 235     | 42     | 1,692 |
| IRN                                      | 400                         | 609     | 490    | 1,499 |
| IRQ                                      | 320                         | 232     | 648    | 1,200 |
| JOR                                      | 375                         | 306     | 522    | 1,203 |
| JPN                                      | 51                          | 786     | 516    | 1,353 |
| KAZ                                      | 457                         | 456     | 363    | 1,276 |
| KEN                                      | 499                         | 421     | 122    | 1,042 |
| KGZ                                      | 710                         | 290     | 200    | 1,200 |
| KOR                                      | 0                           | 364     | 881    | 1,245 |
| LBN                                      | 410                         | 660     | 130    | 1,200 |
| LBY                                      | 480                         | 544     | 172    | 1,196 |
| MAC                                      | 0                           | 0       | 1,023  | 1,023 |
| MAR                                      | 380                         | 480     | 340    | 1,200 |
| MDV                                      | 738                         | 301     | 0      | 1,039 |

|       |        |        |        |        |
|-------|--------|--------|--------|--------|
| MEX   | 858    | 463    | 403    | 1,724  |
| MMR   | 1,116  | 84     | 0      | 1,200  |
| MNG   | 378    | 257    | 1,003  | 1,638  |
| MYS   | 744    | 428    | 141    | 1,313  |
| NGA   | 848    | 375    | 14     | 1,237  |
| NIC   | 381    | 709    | 110    | 1,200  |
| NIR   | 39     | 407    | 0      | 446    |
| NLD   | 104    | 1,710  | 217    | 2,031  |
| NZL   | 420    | 387    | 187    | 994    |
| PAK   | 1,272  | 249    | 474    | 1,995  |
| PER   | 355    | 945    | 100    | 1,400  |
| PHL   | 910    | 290    | 0      | 1,200  |
| PRI   | 73     | 1,054  | 0      | 1,127  |
| ROU   | 703    | 435    | 119    | 1,257  |
| RUS   | 594    | 626    | 590    | 1,810  |
| SGP   | 0      | 0      | 2,012  | 2,012  |
| SRB   | 403    | 447    | 196    | 1,046  |
| SVK   | 750    | 450    | 0      | 1,200  |
| THA   | 1,244  | 116    | 140    | 1,500  |
| TJK   | 880    | 210    | 110    | 1,200  |
| TUN   | 514    | 614    | 80     | 1,208  |
| TUR   | 240    | 1,767  | 408    | 2,415  |
| TWN   | 40     | 1,165  | 18     | 1,223  |
| UKR   | 638    | 355    | 296    | 1,289  |
| URY   | 270    | 290    | 440    | 1,000  |
| USA   | 0      | 1,443  | 385    | 1,828  |
| UZB   | 0      | 0      | 1,250  | 1,250  |
| VEN   | 180    | 661    | 349    | 1,190  |
| VNM   | 0      | 1,049  | 151    | 1,200  |
| ZWE   | 945    | 270    | 0      | 1,215  |
| ----- |        |        |        |        |
| Total | 33,870 | 37,473 | 24,598 | 95,941 |

.19

|                                             | lt20k  | 20-500k | N    |
|---------------------------------------------|--------|---------|------|
| AND                                         | -0.20+ | 0.00    | 997  |
| ARG                                         | -0.54* | -0.34+  | 849  |
| ARM                                         | 0.19   | -0.03   | 1192 |
| AUS                                         | 0.04   | -0.02   | 1575 |
| BGD                                         | 0.31*  | 0.00    | 1197 |
| BOL                                         | -0.28* | -0.25*  | 1990 |
| BRA                                         | -0.30  | -0.03   | 1687 |
| CAN                                         | 0.04   | 0.08    | 4018 |
| CHL                                         | -0.19  | -0.38*  | 903  |
| CHN                                         | -0.11  | 0.00    | 2775 |
| COL                                         | 0.06   | 0.05    | 1520 |
| CYP                                         | -0.68* | 0.00    | 943  |
| CZE                                         | -0.08  | -0.05   | 1186 |
| DEU                                         | 0.23   | 0.28+   | 1491 |
| ECU                                         | 0.04   | 0.02    | 1184 |
| EGY                                         | -0.08  | 0.03    | 1175 |
| ETH                                         | -1.12  | -1.43*  | 1215 |
| GRC                                         | 0.02   | 0.15    | 1161 |
| GTM                                         | -0.06  | -0.01   | 1113 |
| HKG                                         |        |         | 2044 |
| IDN                                         | 0.24   | 0.39    | 3154 |
| IND                                         | -0.69* | -1.12*  | 1063 |
| IRN                                         | 0.24   | 0.18    | 1477 |
| IRQ                                         | -0.06  | -0.36*  | 1177 |
| JOR                                         | 0.04   | 0.10    | 1202 |
| JPN                                         | -0.22  | -0.31*  | 1130 |
| KAZ                                         | -0.32* | -0.15   | 1116 |
| KEN                                         | -0.38  | -0.02   | 1003 |
| KGZ                                         | -0.15  | -0.05   | 1131 |
| KOR                                         |        | -0.03   | 1245 |
| LBN                                         | -0.66* | -0.39*  | 1200 |
| LBY                                         | 0.21   | -0.29   | 923  |
| MAC                                         |        |         | 792  |
| MAR                                         | 0.53*  | 0.28*   | 1200 |
| MDV                                         | 0.17   | 0.00    | 1005 |
| MEX                                         | -0.11  | -0.07   | 1619 |
| MMR                                         | -0.27  | 0.00    | 1200 |
| MNG                                         | 0.18   | 0.50*   | 1638 |
| MYS                                         | -0.32* | -0.17   | 1313 |
| NGA                                         | 0.80   | 0.16    | 1212 |
| NIC                                         | -0.42+ | -0.20   | 1200 |
| NLD                                         | 0.04   | -0.12   | 1675 |
| NZL                                         | 0.09   | 0.02    | 832  |
| PAK                                         | 0.24+  | -0.00   | 1966 |
| PER                                         | -0.65* | -0.33+  | 1387 |
| PHL                                         | -0.05  | 0.00    | 1199 |
| PRI                                         | 0.10   | 0.00    | 1065 |
| ROU                                         | 0.69*  | 0.56*   | 1088 |
| RUS                                         | -0.18  | -0.06   | 1676 |
| SGP                                         |        |         | 1966 |
| SRB                                         | -0.15  | -0.25   | 980  |
| SVK                                         | -0.10  | 0.00    | 1175 |
| THA                                         | 0.91*  | 1.29*   | 1258 |
| TJK                                         | -0.33+ | -0.28   | 1200 |
| TUN                                         | -0.08  | 0.14    | 1073 |
| TUR                                         | -0.55* | -0.39*  | 2357 |
| TWN                                         | 0.92   | 0.89    | 1221 |
| UKR                                         | -0.15  | 0.09    | 1116 |
| URY                                         | 0.02   | 0.23    | 966  |
| USA                                         |        | -0.03   | 1758 |
| UZB                                         |        |         | 1116 |
| VEN                                         | 0.09   | -0.18   | 1187 |
| VNM                                         |        | 0.22    | 1200 |
| ZWE                                         | 0.45*  | 0.00    | 1198 |
| +0.10 *0.05 **0.01 ***0.001; robust std err |        |         |      |

**Table 6:** OLS regressions of freedom/autonomy on urbanicity dummies (base: > 500k) using model controlling for age, age<sup>2</sup>, male, married, employment type dummies(full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and religion important. See model for pooled sample in table 4 except that original 8 step urbanicity variable is now categorized into 3 bins as there are smaller cell sizes separately by country. Still note that few countries lack the estimates as even this simplified 3 step urbanicity spectrum is not available. Overall the results are mostly insignificant, but more negative (11) than positive (6). So in general within countries for few significant relationships there is less freedom in smaller areas as theorized in the literature.

Then we split by tertile within a country—the idea is that we will have retained more observations as oppose to forcing > 500k only to be a city, and we do within country urbanicity where local conditions may be different and already a place of 100 – 500k may be large. For instance earlier AND had 0 observations in > 500k category, but here we have 274 obs in 3rd tertile.

We were thinking about median split, but we went with tertiles as there are theoretical reasons to contrast rural places with large cities (Okulicz-Kozaryn 2017, 2015).

We start below with counts by tertile and country.

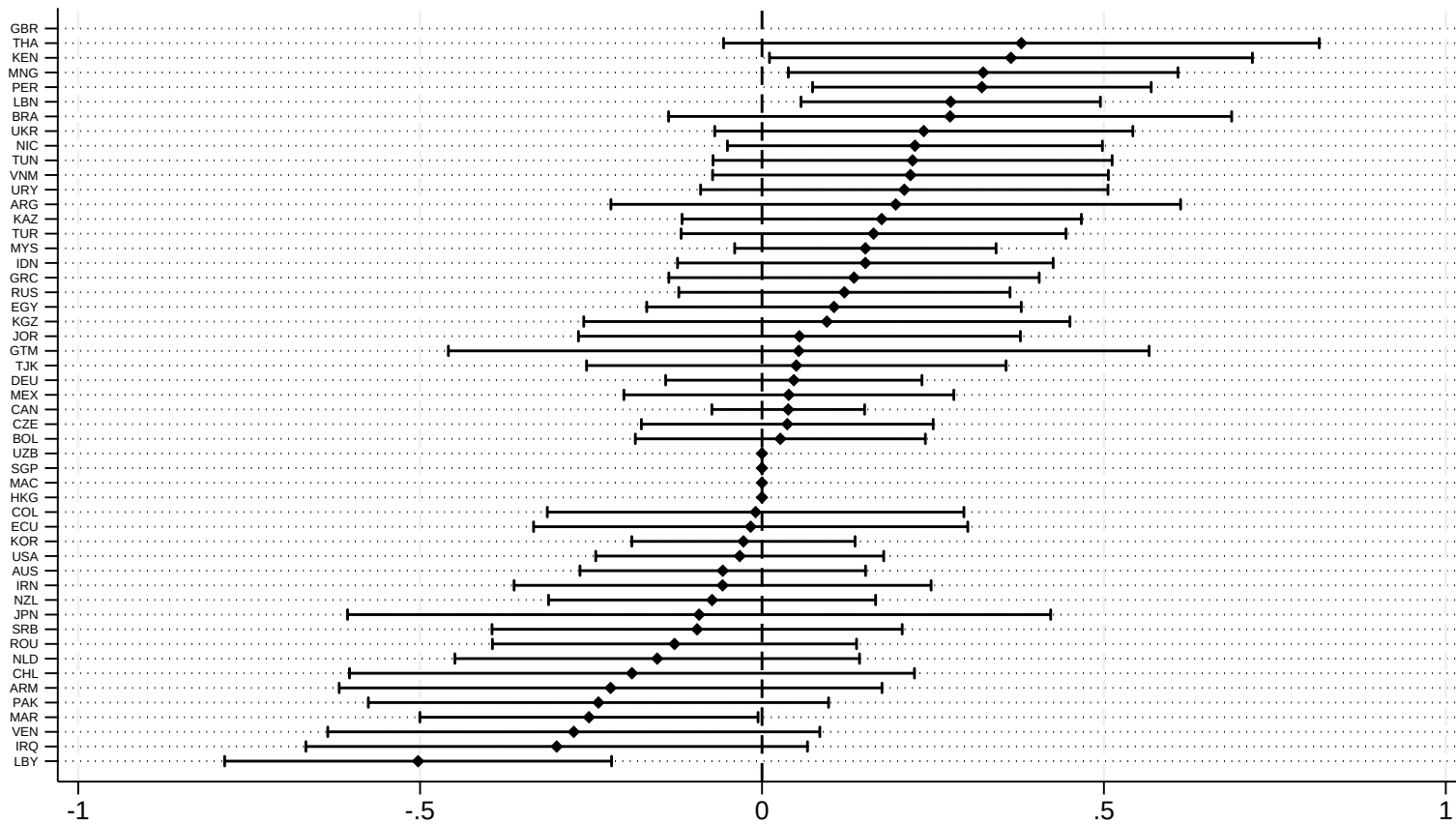
| ta cc tertile |         |       |       |       |
|---------------|---------|-------|-------|-------|
| ISO 3166-1    |         |       |       |       |
| alpha-3       |         |       |       |       |
| country       | tertile |       |       |       |
| code          | 1       | 2     | 3     | Total |
| -----         | -----   | ----- | ----- | ----- |

|       |        |        |        |        |
|-------|--------|--------|--------|--------|
| AND   | 501    | 229    | 274    | 1,004  |
| ARG   | 380    | 623    | 0      | 1,003  |
| ARM   | 448    | 775    | 0      | 1,223  |
| AUS   | 625    | 1,100  | 0      | 1,725  |
| BGD   | 501    | 335    | 364    | 1,200  |
| BOL   | 879    | 1,188  | 0      | 2,067  |
| BRA   | 1,049  | 713    | 0      | 1,762  |
| CAN   | 1,471  | 1,321  | 1,226  | 4,018  |
| CHL   | 639    | 361    | 0      | 1,000  |
| CHN   | 1,980  | 1,056  | 0      | 3,036  |
| COL   | 690    | 336    | 494    | 1,520  |
| CYP   | 367    | 333    | 300    | 1,000  |
| CZE   | 460    | 400    | 340    | 1,200  |
| DEU   | 712    | 359    | 457    | 1,528  |
| ECU   | 509    | 341    | 350    | 1,200  |
| EGY   | 424    | 456    | 320    | 1,200  |
| ETH   | 554    | 287    | 389    | 1,230  |
| GBR   | 989    | 1,219  | 397    | 2,605  |
| GRC   | 420    | 780    | 0      | 1,200  |
| GTM   | 576    | 653    | 0      | 1,229  |
| HKG   | 2,075  | 0      | 0      | 2,075  |
| IDN   | 1,442  | 791    | 967    | 3,200  |
| IND   | 799    | 389    | 504    | 1,692  |
| IRN   | 599    | 410    | 490    | 1,499  |
| IRQ   | 552    | 648    | 0      | 1,200  |
| JOR   | 487    | 716    | 0      | 1,203  |
| JPN   | 837    | 516    | 0      | 1,353  |
| KAZ   | 457    | 456    | 363    | 1,276  |
| KEN   | 395    | 329    | 318    | 1,042  |
| KGZ   | 450    | 380    | 370    | 1,200  |
| KOR   | 1,245  | 0      | 0      | 1,245  |
| LBN   | 410    | 559    | 231    | 1,200  |
| LBY   | 480    | 349    | 367    | 1,196  |
| MAC   | 1,023  | 0      | 0      | 1,023  |
| MAR   | 520    | 340    | 340    | 1,200  |
| MDV   | 494    | 200    | 345    | 1,039  |
| MEX   | 707    | 614    | 403    | 1,724  |
| MMR   | 648    | 240    | 312    | 1,200  |
| MNG   | 575    | 1,063  | 0      | 1,638  |
| MYS   | 580    | 321    | 412    | 1,313  |
| NGA   | 455    | 393    | 389    | 1,237  |
| NIC   | 720    | 220    | 260    | 1,200  |
| NIR   | 161    | 143    | 142    | 446    |
| NLD   | 751    | 1,063  | 217    | 2,031  |
| NZL   | 420    | 387    | 187    | 994    |
| PAK   | 896    | 438    | 661    | 1,995  |
| PER   | 475    | 825    | 100    | 1,400  |
| PHL   | 610    | 210    | 380    | 1,200  |
| PRI   | 726    | 262    | 139    | 1,127  |
| ROU   | 567    | 343    | 347    | 1,257  |
| RUS   | 756    | 464    | 590    | 1,810  |
| SGP   | 2,012  | 0      | 0      | 2,012  |
| SRB   | 365    | 485    | 196    | 1,046  |
| SVK   | 540    | 390    | 270    | 1,200  |
| THA   | 923    | 138    | 439    | 1,500  |
| TJK   | 730    | 80     | 390    | 1,200  |
| TUN   | 442    | 495    | 271    | 1,208  |
| TUR   | 2,007  | 0      | 408    | 2,415  |
| TWN   | 604    | 601    | 18     | 1,223  |
| UKR   | 517    | 476    | 296    | 1,289  |
| URY   | 410    | 590    | 0      | 1,000  |
| USA   | 635    | 808    | 385    | 1,828  |
| UZB   | 1,250  | 0      | 0      | 1,250  |
| VEN   | 401    | 440    | 349    | 1,190  |
| VNM   | 1,049  | 0      | 151    | 1,200  |
| ZWE   | 768    | 152    | 295    | 1,215  |
| Total | 48,139 | 30,589 | 17,213 | 95,941 |

.19

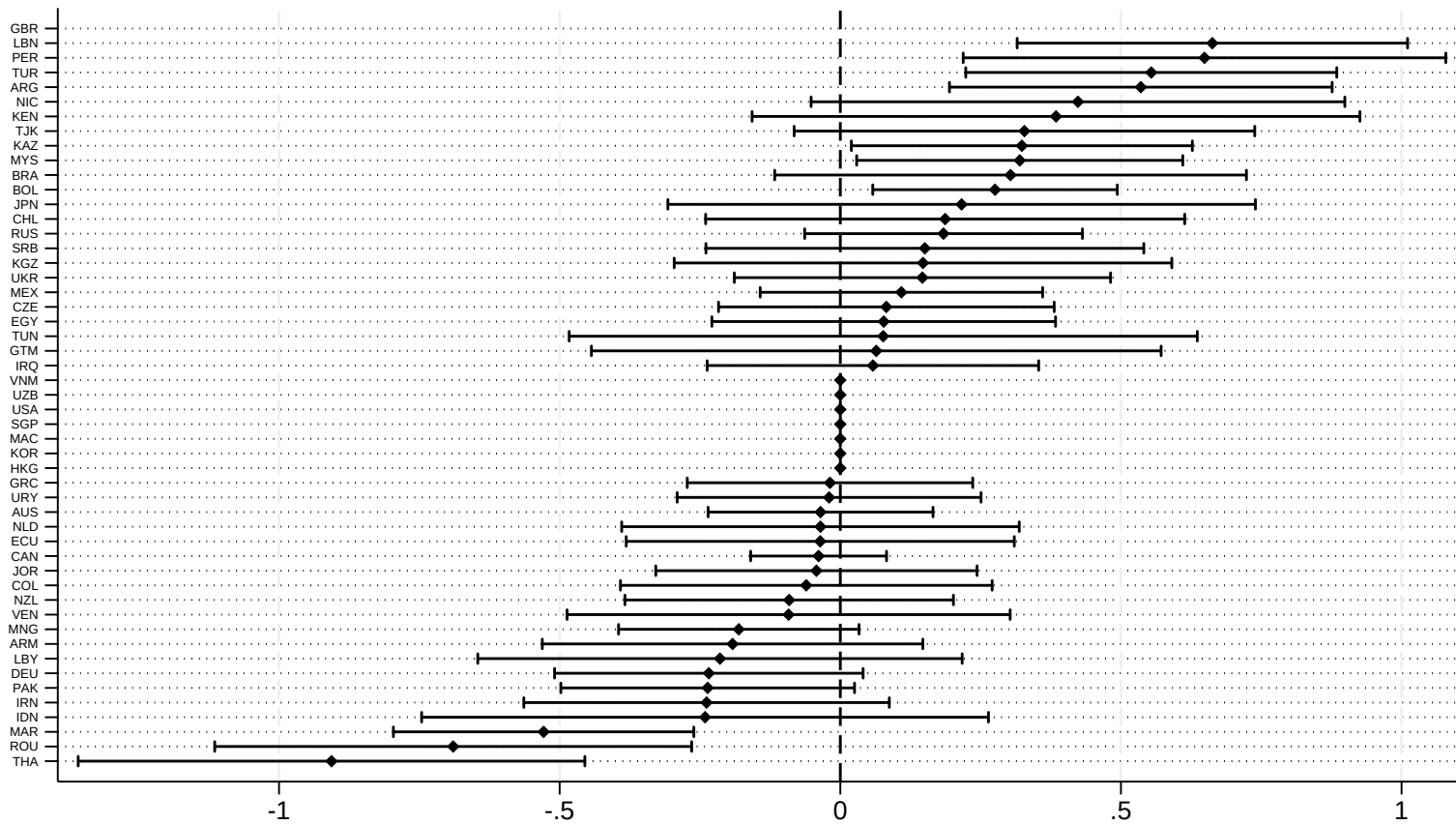
Then we repeat twostep graph from the body just with the 1st category as a base in figure 20 and 21 so we can easier compare it to our tertile estimates.

Overall without controls more freedom in cities as hypothesized, but still most effects are insignificant, and with controls results are even less significant.



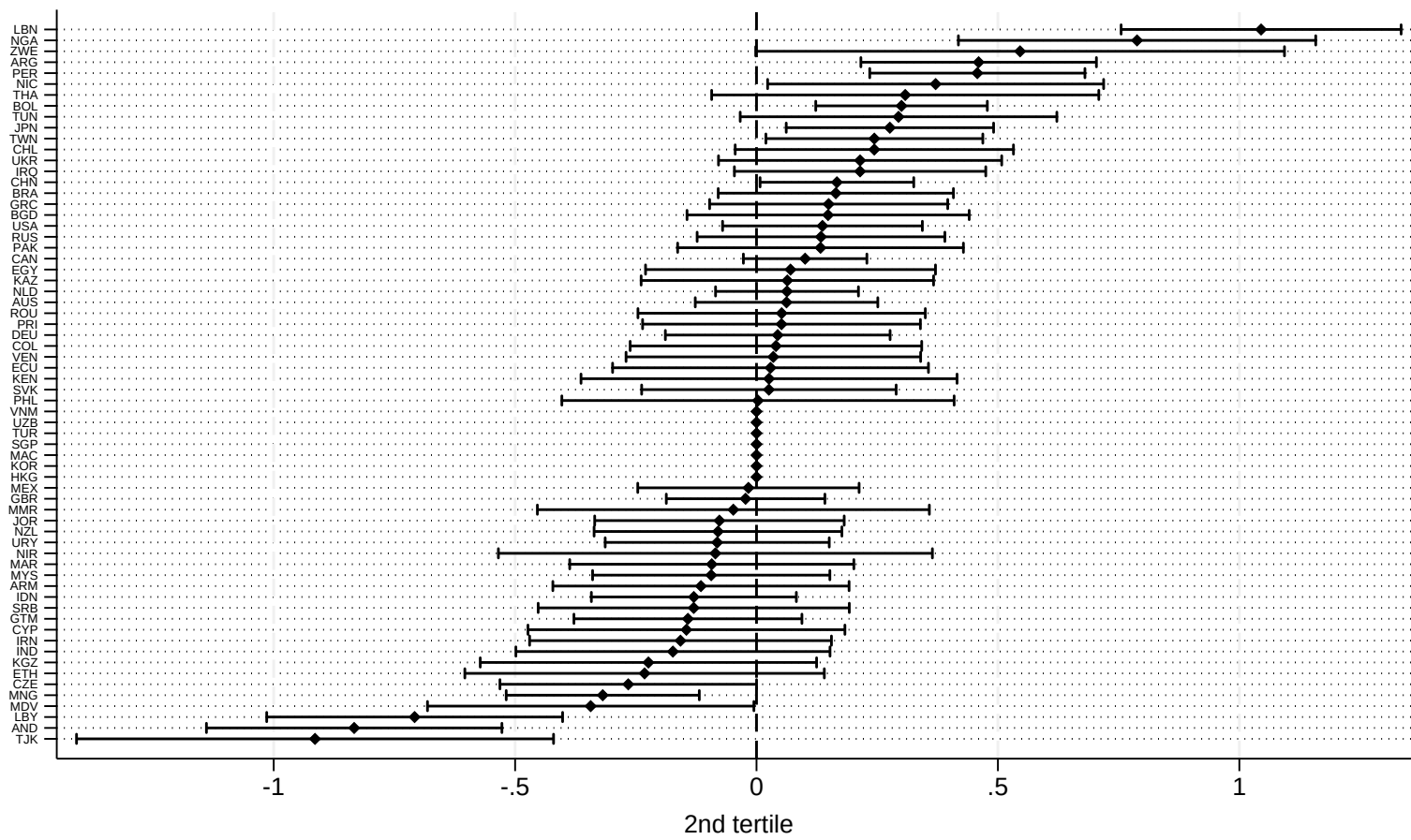
b-coefficients and 95%-CI of life autonomy per country on dummy for 100-500k v lt20k (base case); dropped c if cat gt500k mis

Figure 20: fff2

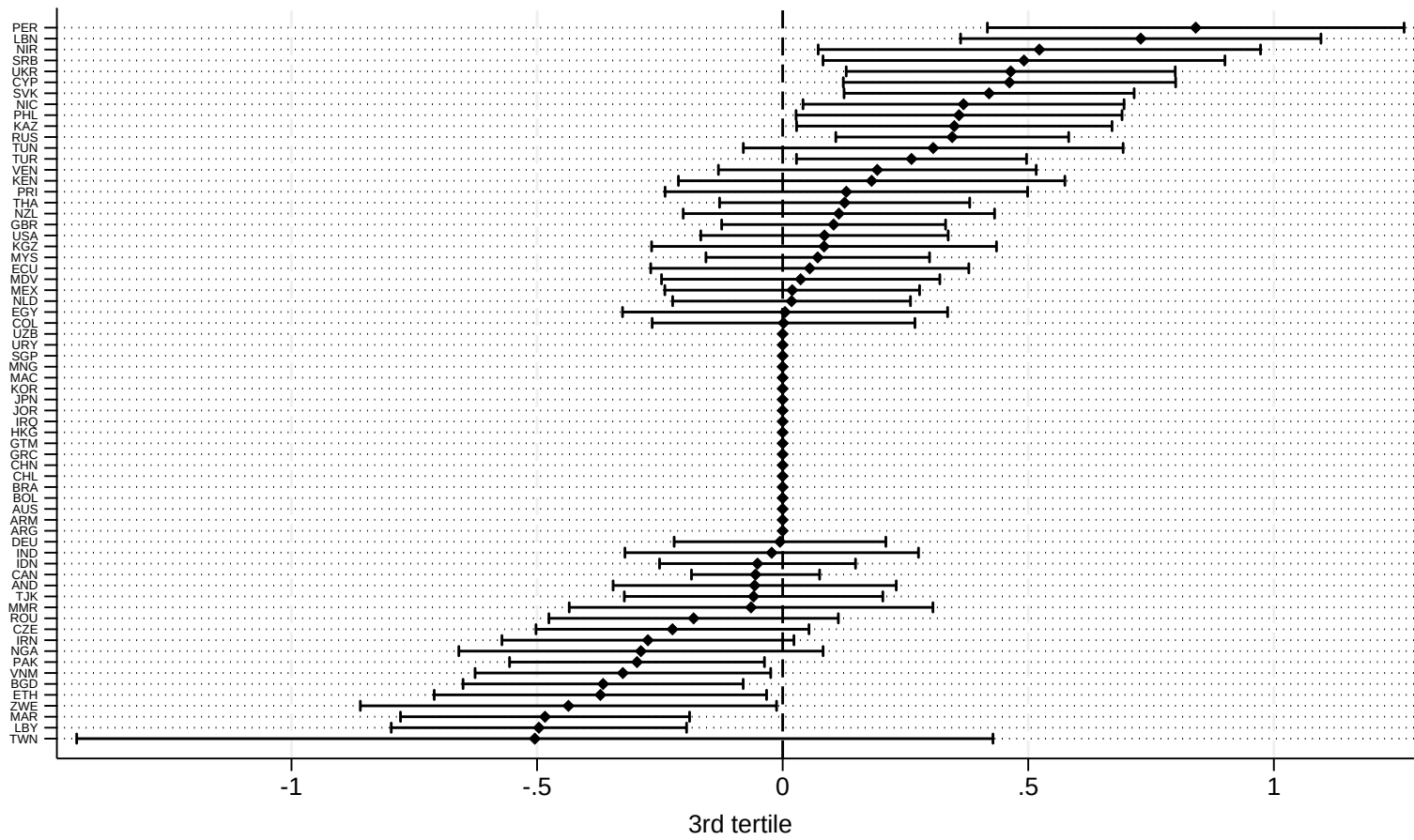


b-coefficients and 95%-CI of life autonomy per country on dummy for 500k v lt20k (base case); dropped c if cat gt500k missing

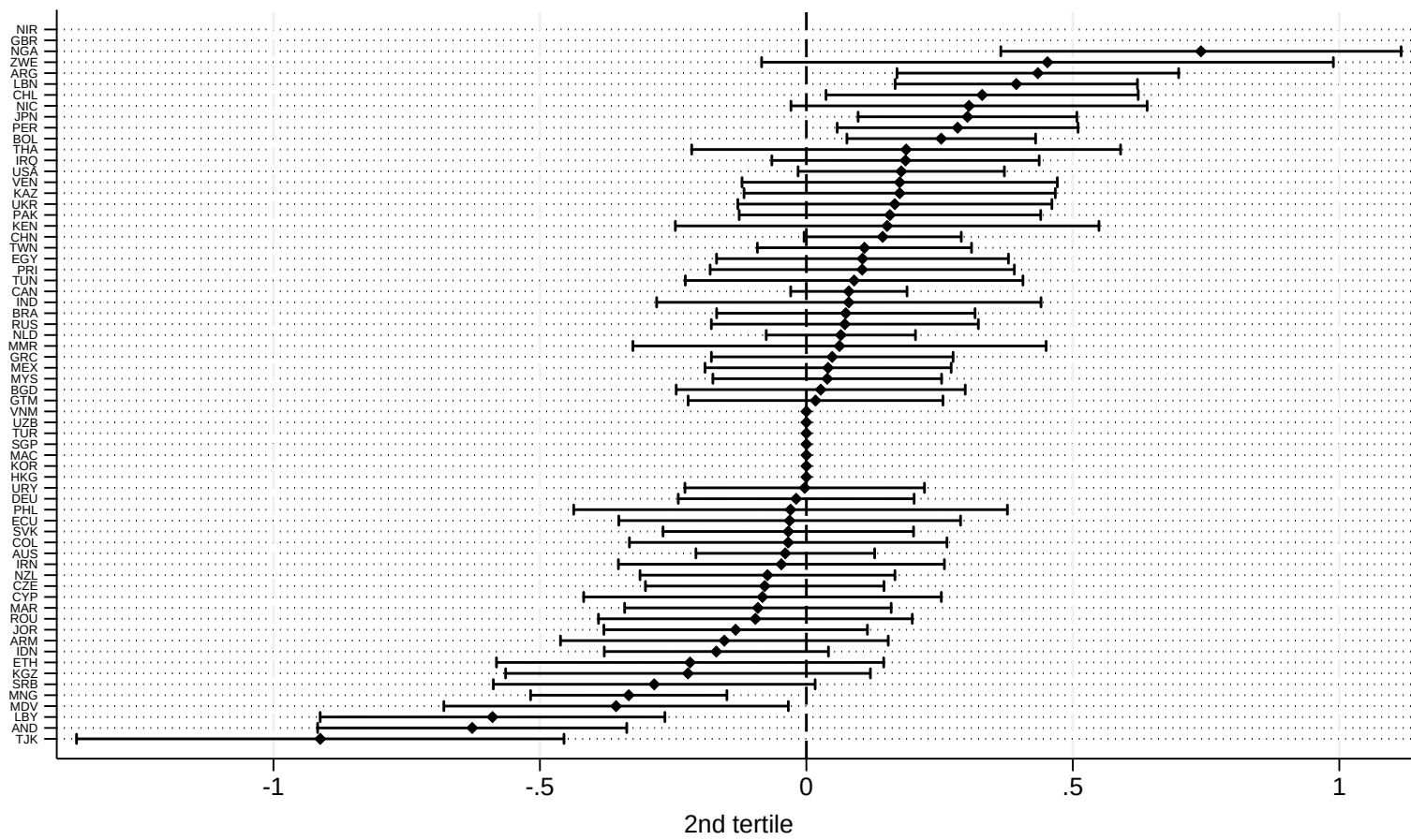
Figure 21: fff3



**Figure 22:** 2nd tertile: just 2 tertile dummies on rhs, 1st tertile base



**Figure 23:** 3rd tertile: just 2 tertile dummies on rhs, 1st tertile base



**Figure 24:** 2nd tertile: all controls

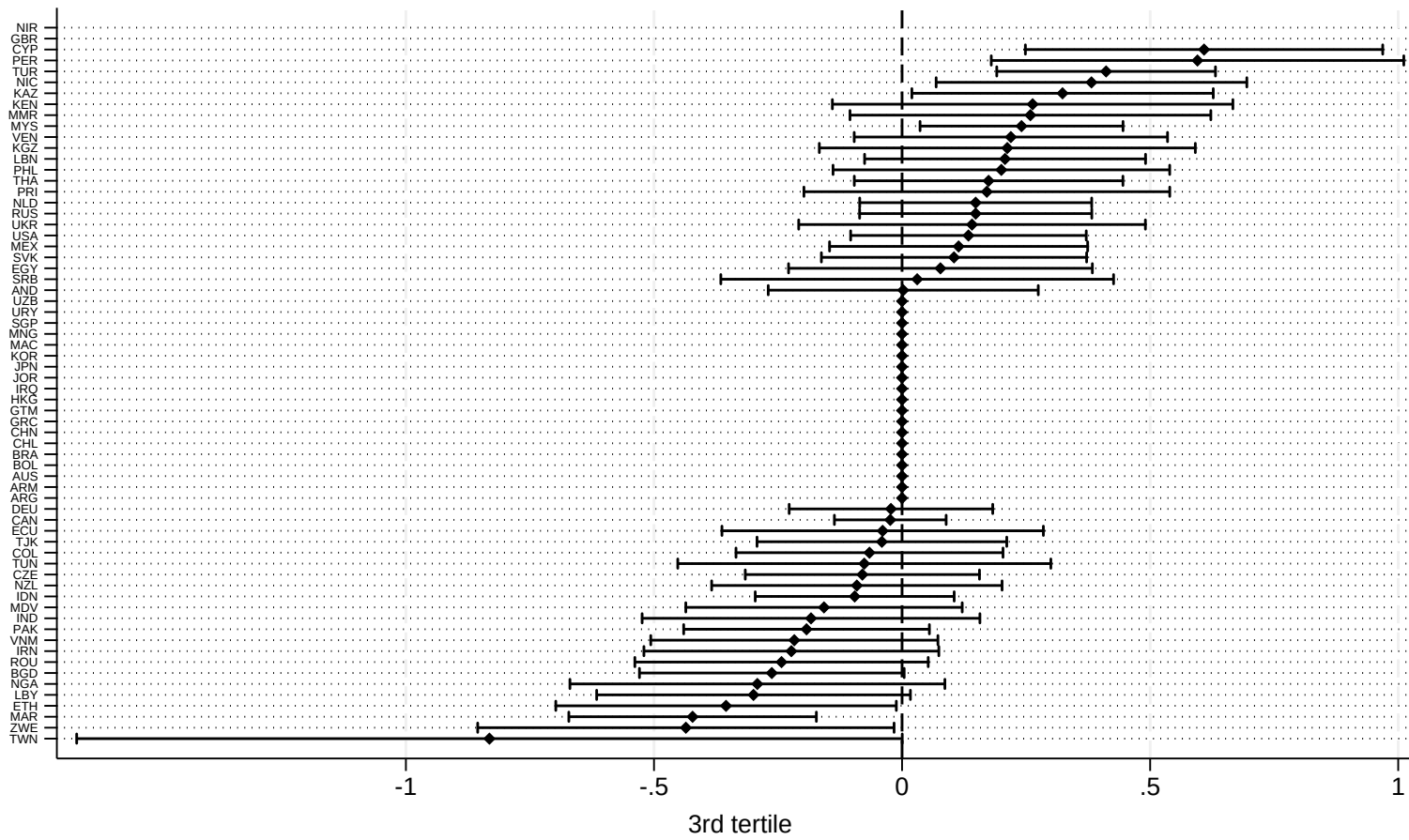


Figure 25: 3rd tertile: all controls

We explore further by interacting region dummies with urban-rural in figure 26. We find significant interactions for Latin America and South Asia (v East Asia/Pacific)—it is difficult to see patterns in figure with all regions—we plot only the significant interactions in figure 27.

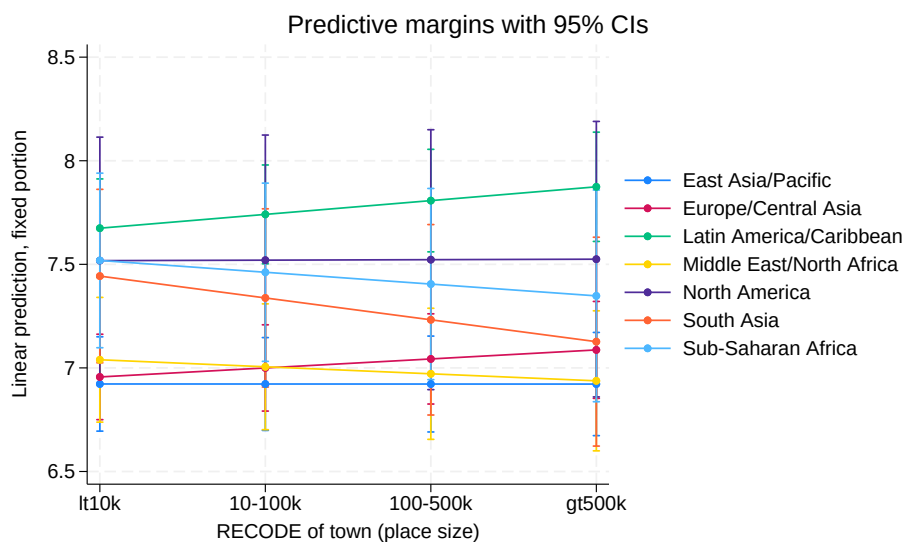
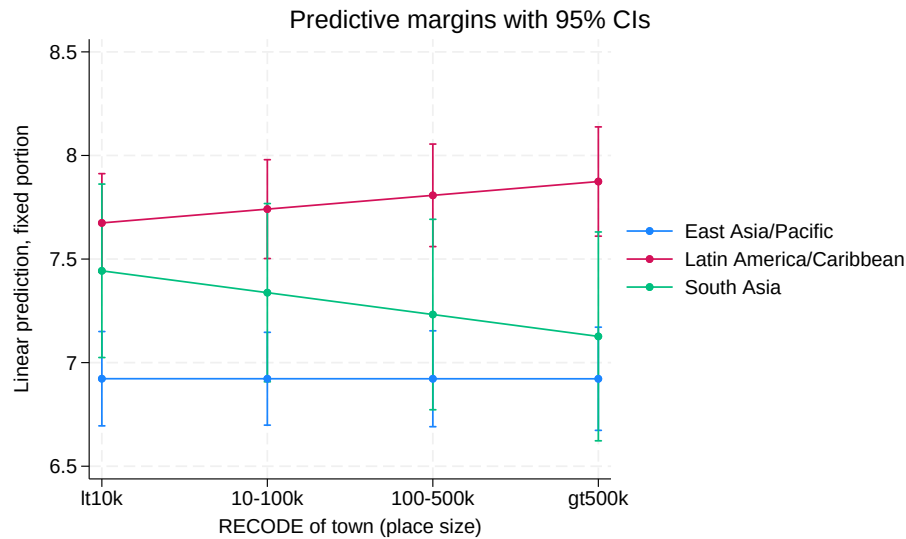
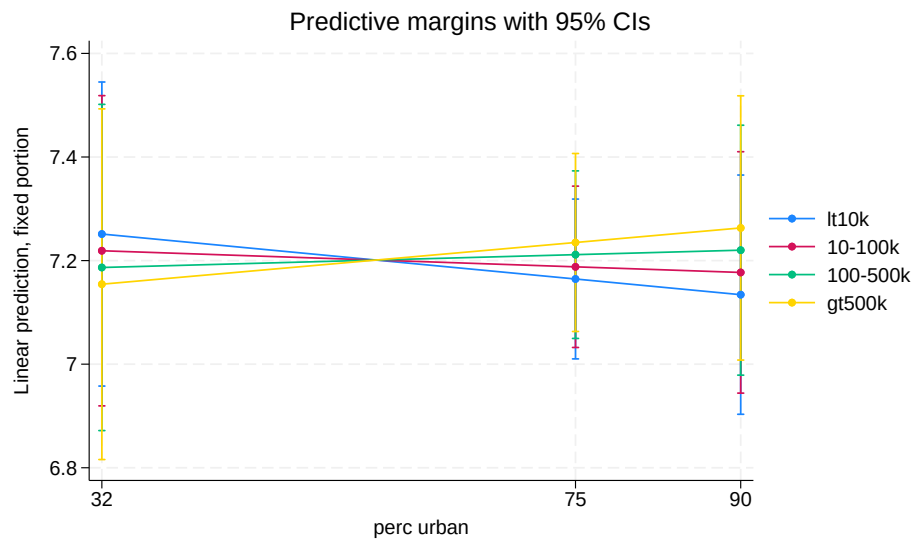


Figure 26: Interactions of region dummies with urban-rural.



**Figure 27:** Interactions of region dummies with urban-rural.

Last, we interacted urban-rural at person level with % urban at country level and the interaction is marginally significant at 0.1 level of significance and we plot results in figure 28. Marginsplot shows that in countries that are less urban, there is more freedom in smaller places, but in more urban countries there is more freedom in large cities. It is an intriguing finding and we do not have a good explanation for it. Perhaps in heavily urbanized countries rural barely exists and it does not help much with freedom, while in more rural countries there is more meaningful distinction between urban and rural and there is more freedom in rural areas.



**Figure 28:** Interaction of urban-rural at person level with % urban at country level. X-axis values are chosen based on the variable's distribution: 31.2 is 5th percentile, median is 75, and 90 is 92nd percentile.

We have used above just the usual World Bank definition, which uses each country's statistical office definition—for discussion see especially Mikhaeil et al. (2024) and also Deb and Okulicz-Kozaryn (2023).

We have also tried similar cross-level interactions with PCGDP and income inequality, but didn't find significant interactions.

Last we check alternative specification if we are not overcontrolling (Bartram et al. 2024)—in the body of the paper we used a set of control variables from the literature, however, we think that we may drop gender, age and age2—as they may not be predictive of

urbanicity. First, we repeat graph from the body for ease of comparison in figure 29. And below it figure without the above mentioned controls 30. Results are substantively the same. And in 31 adding education, eg more educated persons may move to a city—again, very similar results. And in 32 fewer countries but still most insignificant.

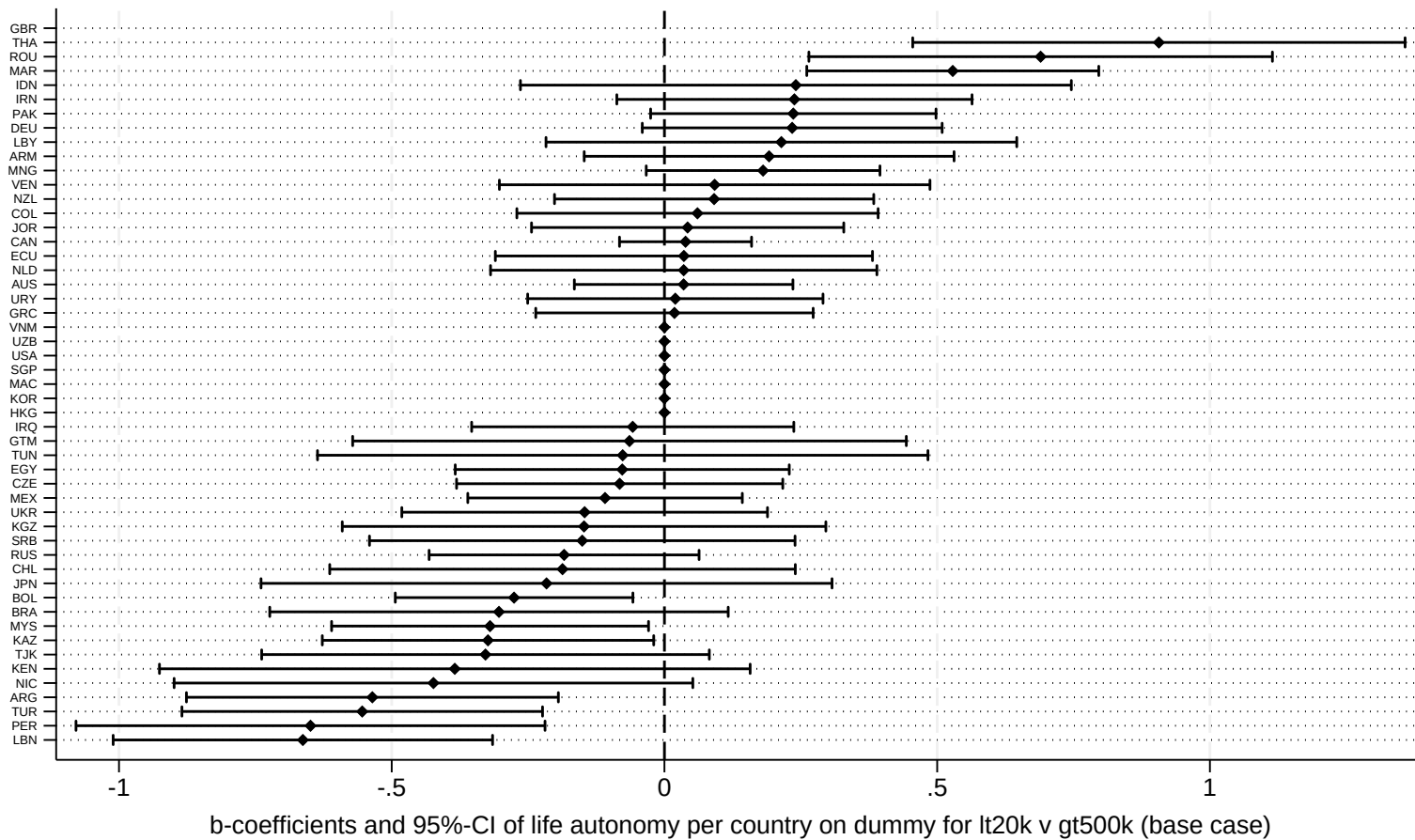
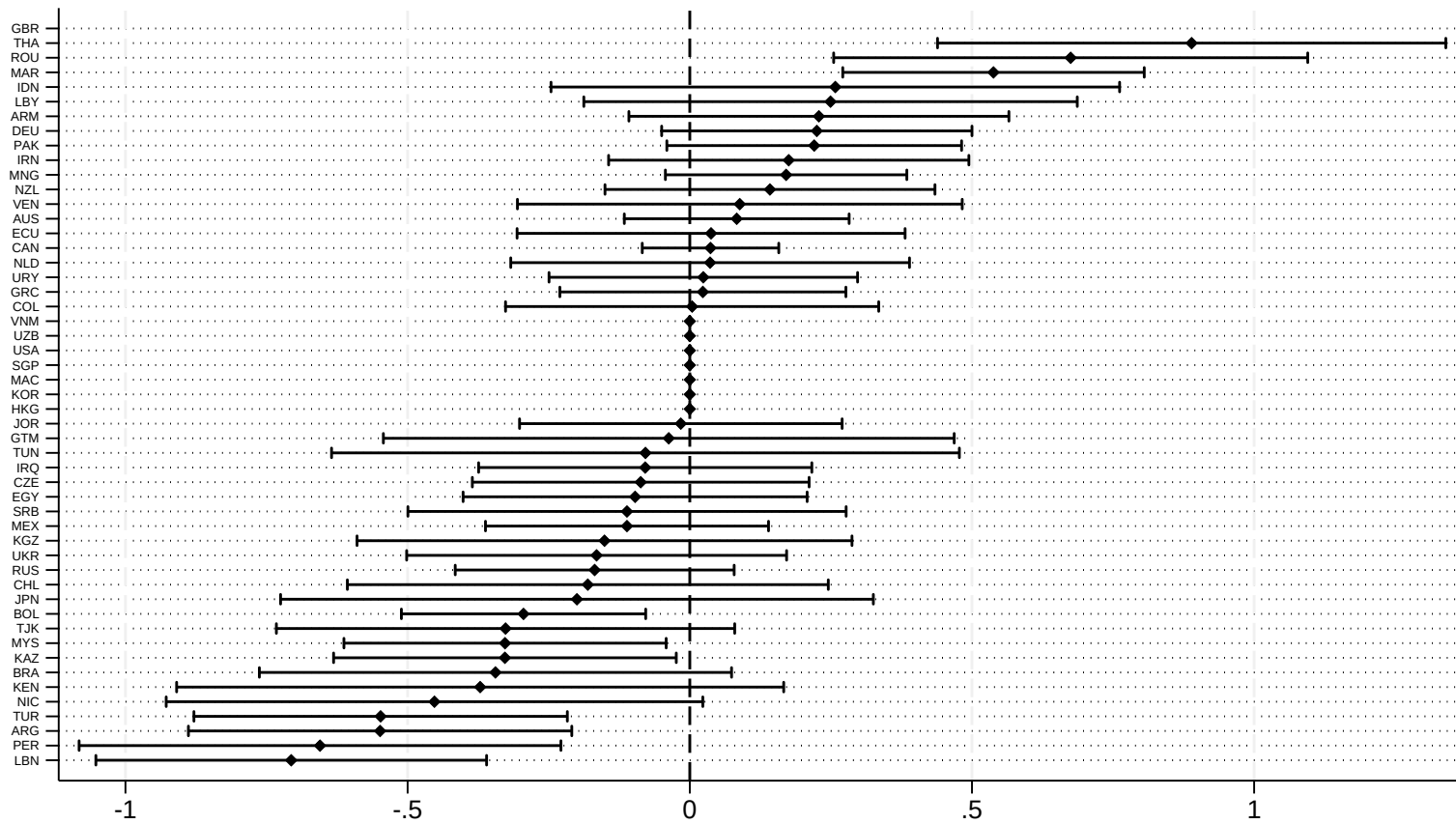
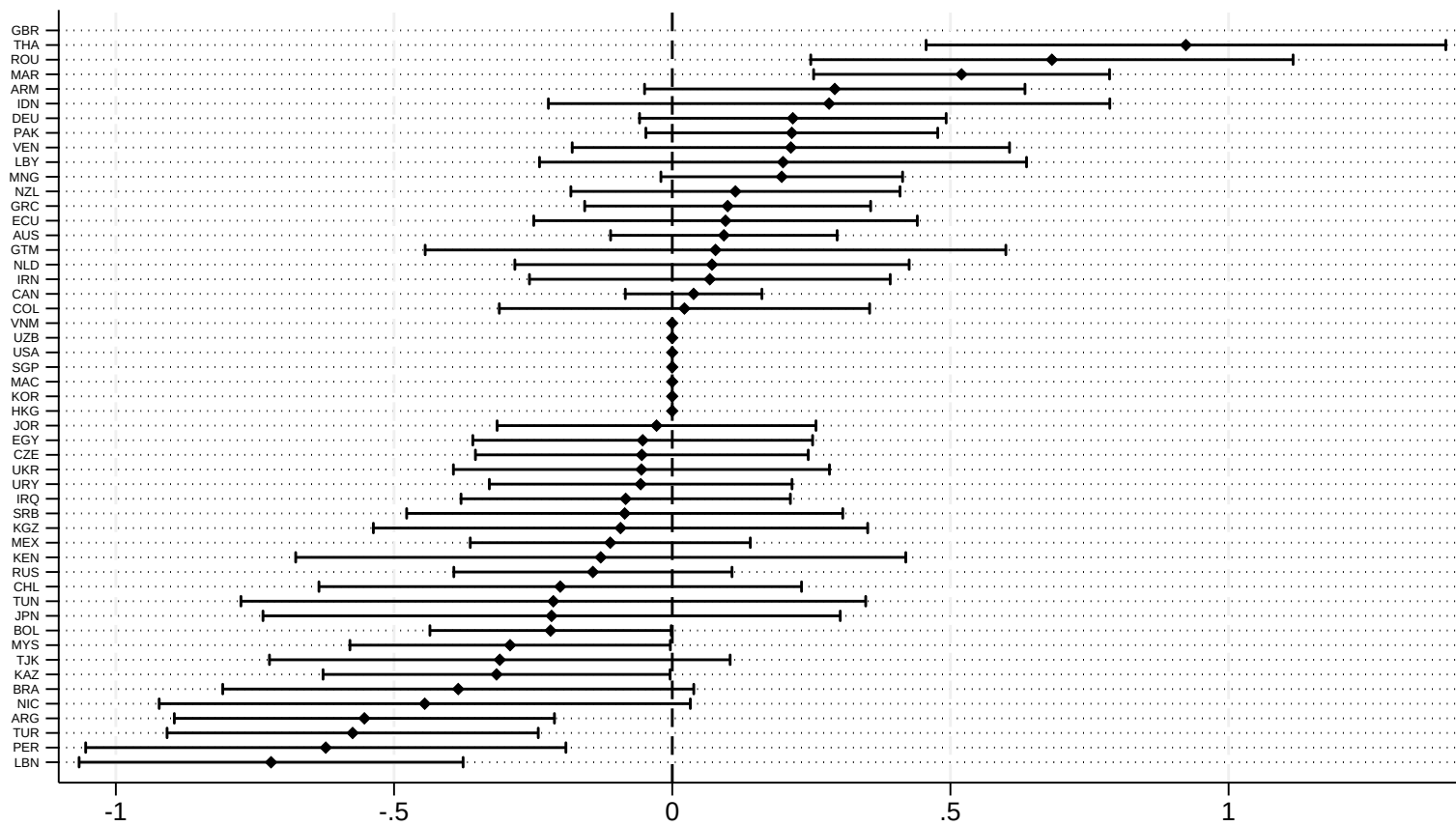


Figure 29



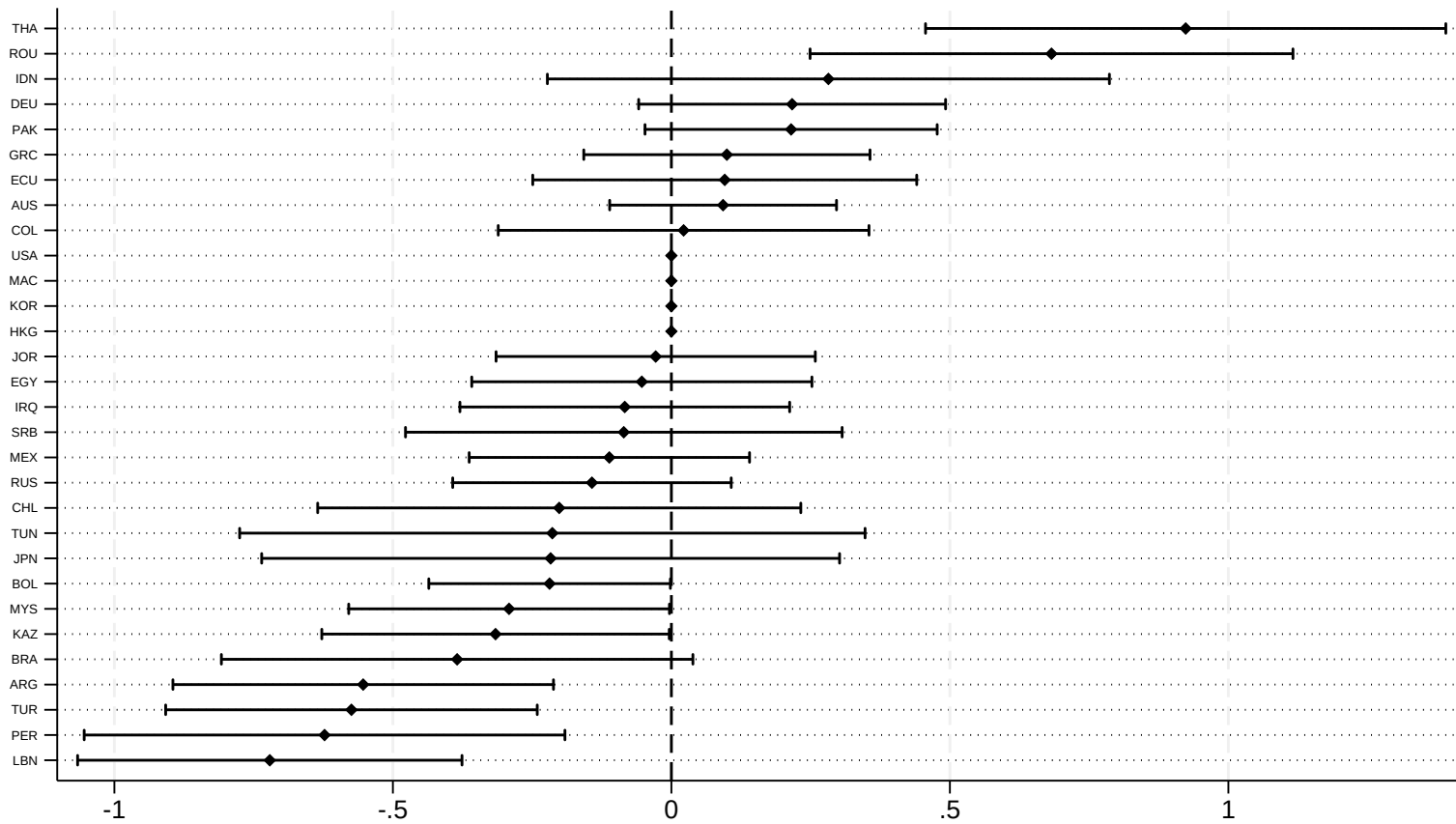
b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

Figure 30: Dropping age age2 and gender



b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

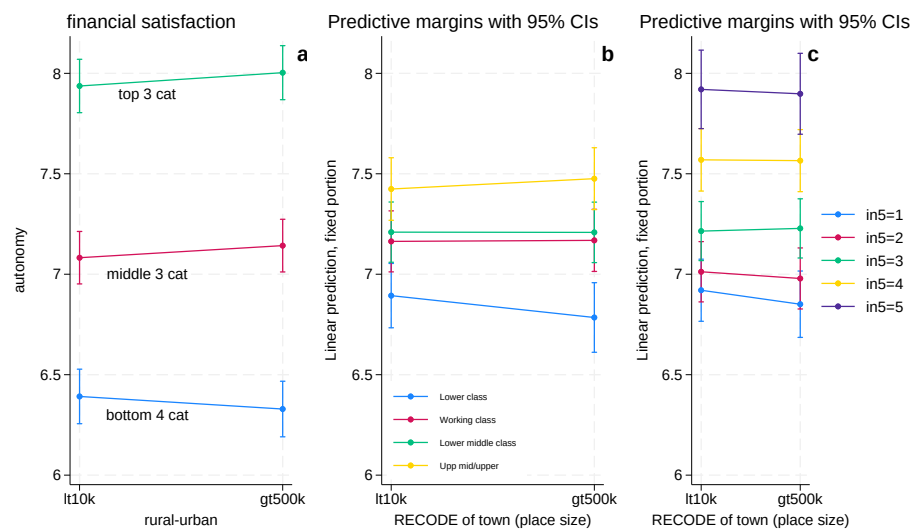
**Figure 31:** Adding education—again, very similar results



b-coefficients and 95%-CI of life autonomy per country on dummy for lt20k v gt500k (base case)

**Figure 32:** only <2020—fewer countries but still most insignificant.

#### 2.4.2 Interactions with financial satisfaction, class, and income



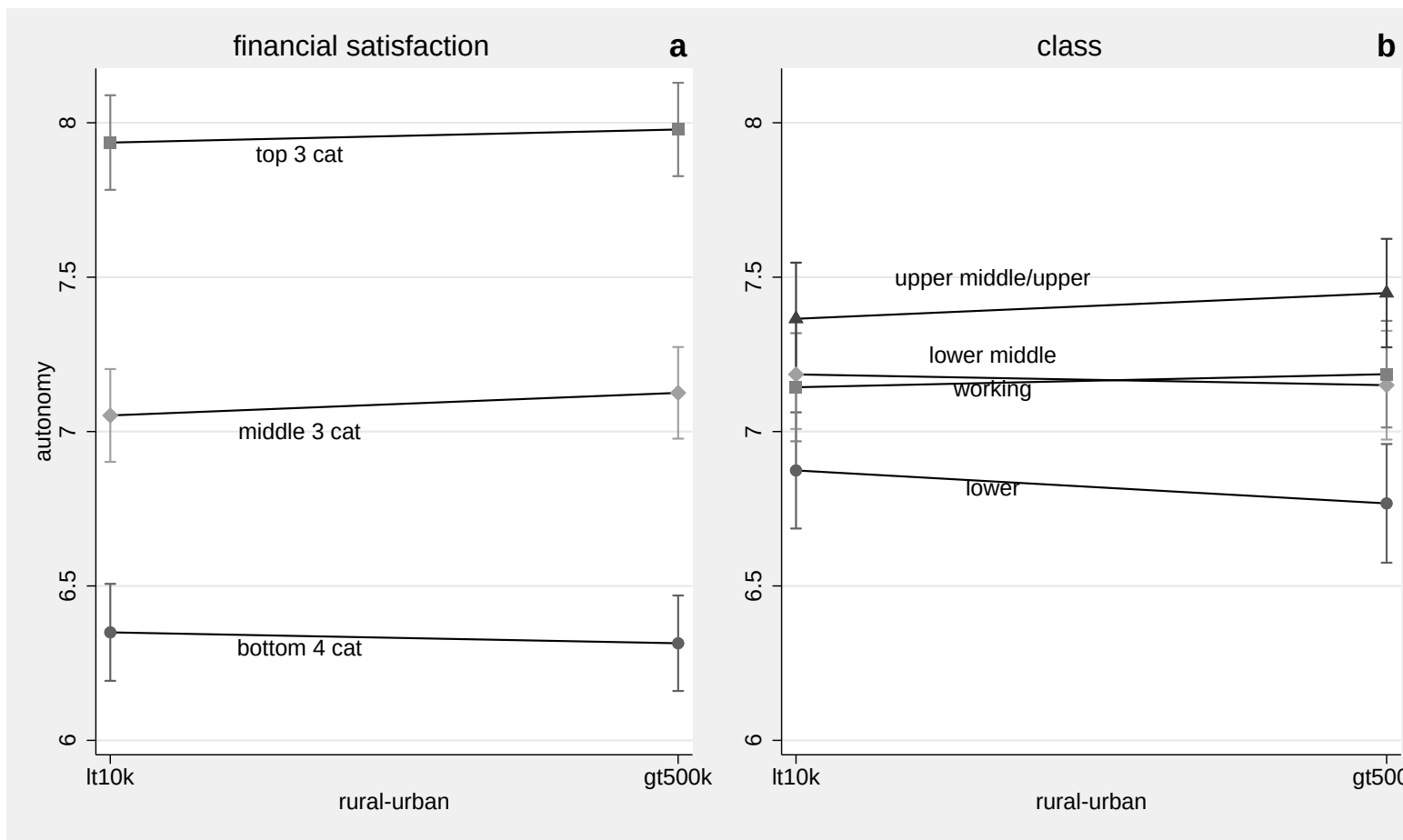
**Figure 33:** Multilevel country random intercept regressions of freedom on rural-urban dummies interacted with: a) financial satisfaction, b) class, c) income. For underlying regressions see appendix. Controls are the same as in the previous section.

|                                                        | a        | b        | c        |
|--------------------------------------------------------|----------|----------|----------|
| freedom                                                |          |          |          |
| lt10k                                                  | 0.06     | 0.11+    | 0.07     |
| 10-100k                                                | -0.07+   | 0.03     | -0.01    |
| 100-500k                                               | 0.01     | -0.05    | 0.07     |
| gt500k                                                 | 0.00     | 0.00     | 0.00     |
| RECODE of satFin (financial satisfaction)=1            | 0.00     |          |          |
| RECODE of satFin (financial satisfaction)=2            | 0.81***  |          |          |
| RECODE of satFin (financial satisfaction)=3            | 1.67***  |          |          |
| lt10k × RECODE of satFin (financial satisfaction)=1    | 0.00     |          |          |
| lt10k × RECODE of satFin (financial satisfaction)=2    | -0.12*   |          |          |
| lt10k × RECODE of satFin (financial satisfaction)=3    | -0.13*   |          |          |
| 10-100k × RECODE of satFin (financial satisfaction)=1  | 0.00     |          |          |
| 10-100k × RECODE of satFin (financial satisfaction)=2  | 0.08+    |          |          |
| 10-100k × RECODE of satFin (financial satisfaction)=3  | -0.01    |          |          |
| 100-500k × RECODE of satFin (financial satisfaction)=1 | 0.00     |          |          |
| 100-500k × RECODE of satFin (financial satisfaction)=2 | -0.04    |          |          |
| 100-500k × RECODE of satFin (financial satisfaction)=3 | -0.06    |          |          |
| gt500k × RECODE of satFin (financial satisfaction)=1   | 0.00     |          |          |
| gt500k × RECODE of satFin (financial satisfaction)=2   | 0.00     |          |          |
| gt500k × RECODE of satFin (financial satisfaction)=3   | 0.00     |          |          |
| age                                                    | 0.01***  | 0.01*    | 0.01*    |
| age2                                                   | -0.00*** | -0.00    | -0.00    |
| male                                                   | 0.01     | 0.01     | -0.00    |
| married or living together as married                  | 0.01     | 0.06**   | 0.04*    |
| Full time                                              | 0.00     | 0.00     | 0.00     |
| Part time                                              | -0.10*** | -0.11*** | -0.10*** |
| Self employed                                          | -0.02    | -0.04+   | -0.03    |
| Retired                                                | -0.06*   | -0.03    | -0.01    |
| Housewife                                              | -0.21*** | -0.20*** | -0.18*** |
| Students                                               | -0.14*** | -0.12*** | -0.11**  |
| Unemployed                                             | -0.14*** | -0.22*** | -0.18*** |
| Other                                                  | -0.27*** | -0.32*** | -0.30*** |
| health                                                 | 0.37***  | 0.50***  | 0.49***  |
| How many children do you have                          | 0.01*    | 0.01+    | 0.01+    |
| religion important                                     | 0.04***  | 0.05***  | 0.05***  |
| Lower class                                            |          | 0.00     |          |
| Working class                                          |          | 0.38***  |          |
| Lower middle class                                     |          | 0.42***  |          |
| Upp mid/upper                                          |          | 0.69***  |          |
| lt10k × Lower class                                    |          | 0.00     |          |
| lt10k × Working class                                  |          | -0.11    |          |
| lt10k × Lower middle class                             |          | -0.11    |          |
| lt10k × Upp mid/upper                                  |          | -0.16*   |          |
| 10-100k × Lower class                                  |          | 0.00     |          |
| 10-100k × Working class                                |          | -0.09    |          |
| 10-100k × Lower middle class                           |          | -0.03    |          |
| 10-100k × Upp mid/upper                                |          | -0.04    |          |
| 100-500k × Lower class                                 |          | 0.00     |          |
| 100-500k × Working class                               |          | 0.01     |          |
| 100-500k × Lower middle class                          |          | 0.03     |          |
| 100-500k × Upp mid/upper                               |          | 0.07     |          |
| gt500k × Lower class                                   |          | 0.00     |          |
| gt500k × Working class                                 |          | 0.00     |          |
| gt500k × Lower middle class                            |          | 0.00     |          |
| gt500k × Upp mid/upper                                 |          | 0.00     |          |
| RECODE of inc (income)=1                               |          |          | 0.00     |
| RECODE of inc (income)=2                               |          |          | 0.13*    |
| RECODE of inc (income)=3                               |          |          | 0.38***  |
| RECODE of inc (income)=4                               |          |          | 0.71***  |
| RECODE of inc (income)=5                               |          |          | 1.05***  |
| lt10k × RECODE of inc (income)=1                       |          |          | 0.00     |
| lt10k × RECODE of inc (income)=2                       |          |          | -0.04    |
| lt10k × RECODE of inc (income)=3                       |          |          | -0.08    |
| lt10k × RECODE of inc (income)=4                       |          |          | -0.07    |
| lt10k × RECODE of inc (income)=5                       |          |          | -0.05    |
| 10-100k × RECODE of inc (income)=1                     |          |          | 0.00     |
| 10-100k × RECODE of inc (income)=2                     |          |          | -0.01    |
| 10-100k × RECODE of inc (income)=3                     |          |          | 0.00     |
| 10-100k × RECODE of inc (income)=4                     |          |          | 0.02     |
| 10-100k × RECODE of inc (income)=5                     |          |          | -0.04    |
| 100-500k × RECODE of inc (income)=1                    |          |          | 0.00     |
| 100-500k × RECODE of inc (income)=2                    |          |          | -0.11    |
| 100-500k × RECODE of inc (income)=3                    |          |          | -0.05    |
| 100-500k × RECODE of inc (income)=4                    |          |          | -0.14    |
| 100-500k × RECODE of inc (income)=5                    |          |          | -0.27*   |
| gt500k × RECODE of inc (income)=1                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=2                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=3                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=4                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=5                      |          |          | 0.00     |
| constant                                               | 4.51***  | 4.52***  | 4.61***  |
| lns1.1.1                                               |          |          |          |
| constant                                               | -0.68*** | -0.54*** | -0.56*** |
| lnsig.e                                                |          |          |          |
| constant                                               | 0.73***  | 0.76***  | 0.76***  |
| N                                                      | 87874    | 86391    | 86471    |

+ 0.10 \* 0.05 \*\* 0.01 \*\*\* 0.001; robust std err

**Table 7:** OLS regressions of freedom/agency

We could have explored more fully here using usual sequential elaboration of the models, but for now just sticking with the final full model as there are already plenty of robustness/sensitivity checks and one could easily get lost in this sea of extra results.



**Figure 34:** Robustness/sensitivity check: on the sample after drooping countries where top gt500k cat missing. Results are mostly similar to those reported in the body of the paper.

Last we check alternative specification if we are not overcontrolling (Bartram et al. 2024)—in the body of the paper we used a set of control variables from the literature, however, we think that we may drop gender, age and age2—as they may not be predictive of urbanicity. Results are in figure 35 and are substantively the same, and so when adding education in figure 36. The very final check is dropping pandemic years 2020 and later—pandemic might have made city life less free, but our results remain similar, if anything stronger in figure 37

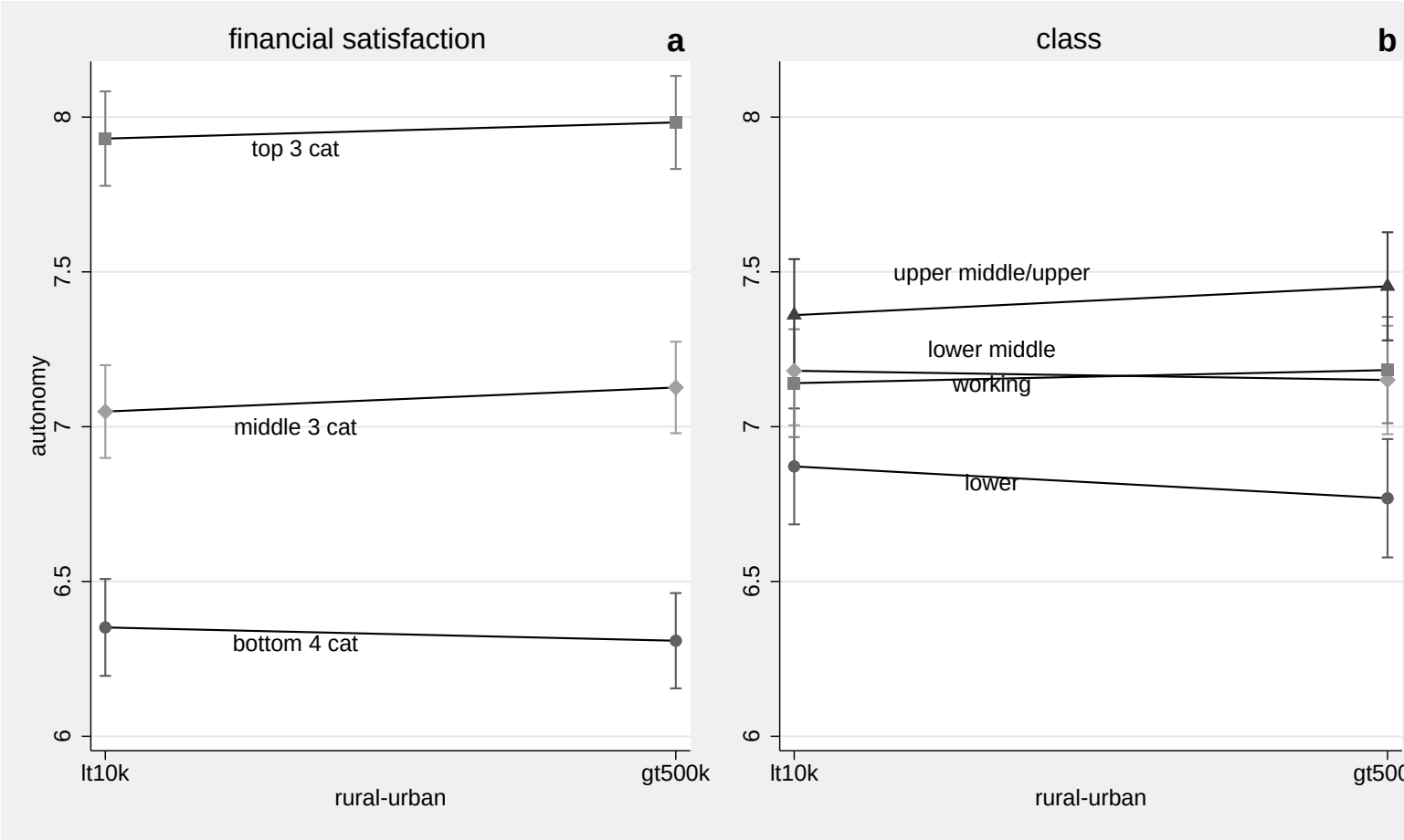


Figure 35: Dropping age age2 and gender.

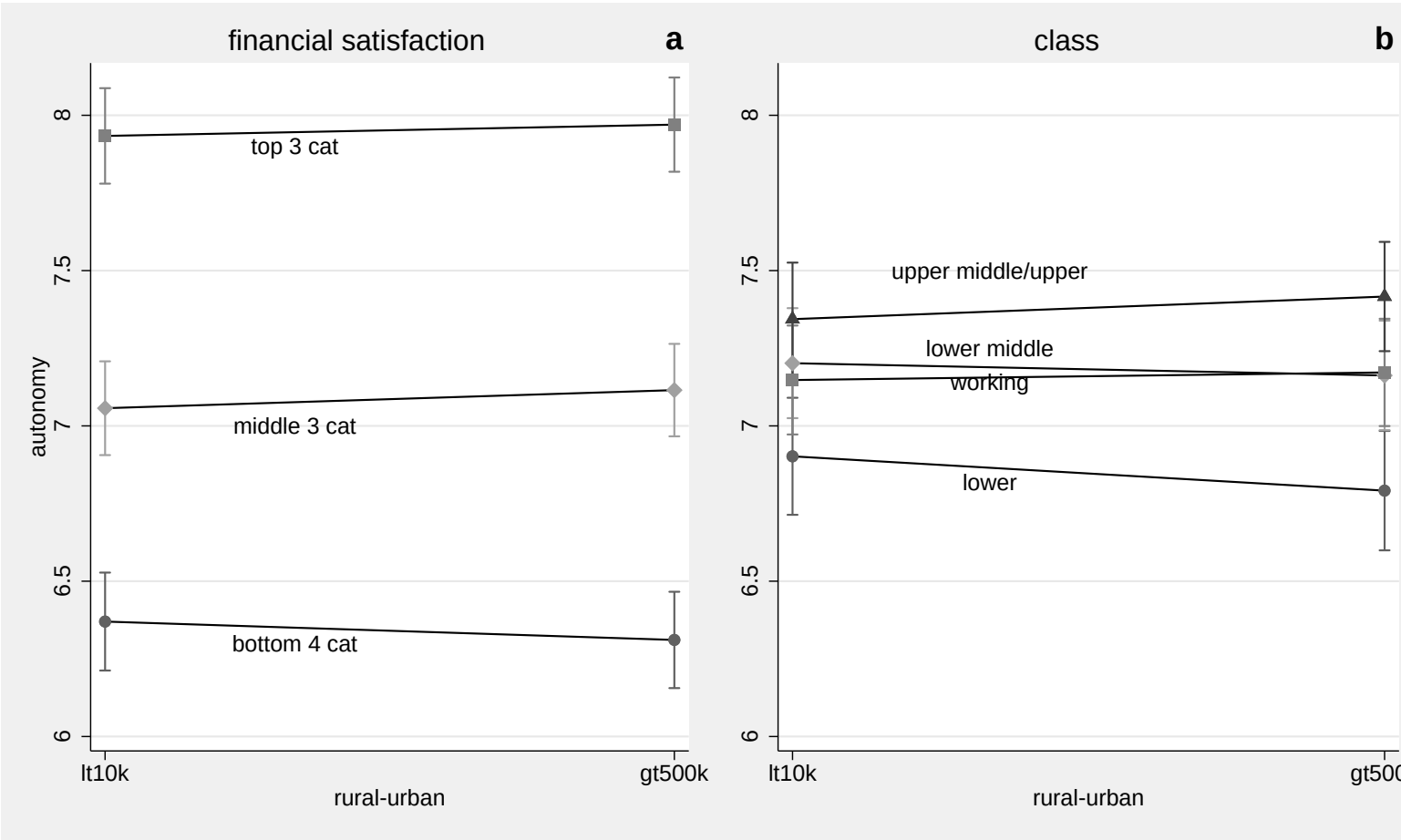


Figure 36: Adding education.

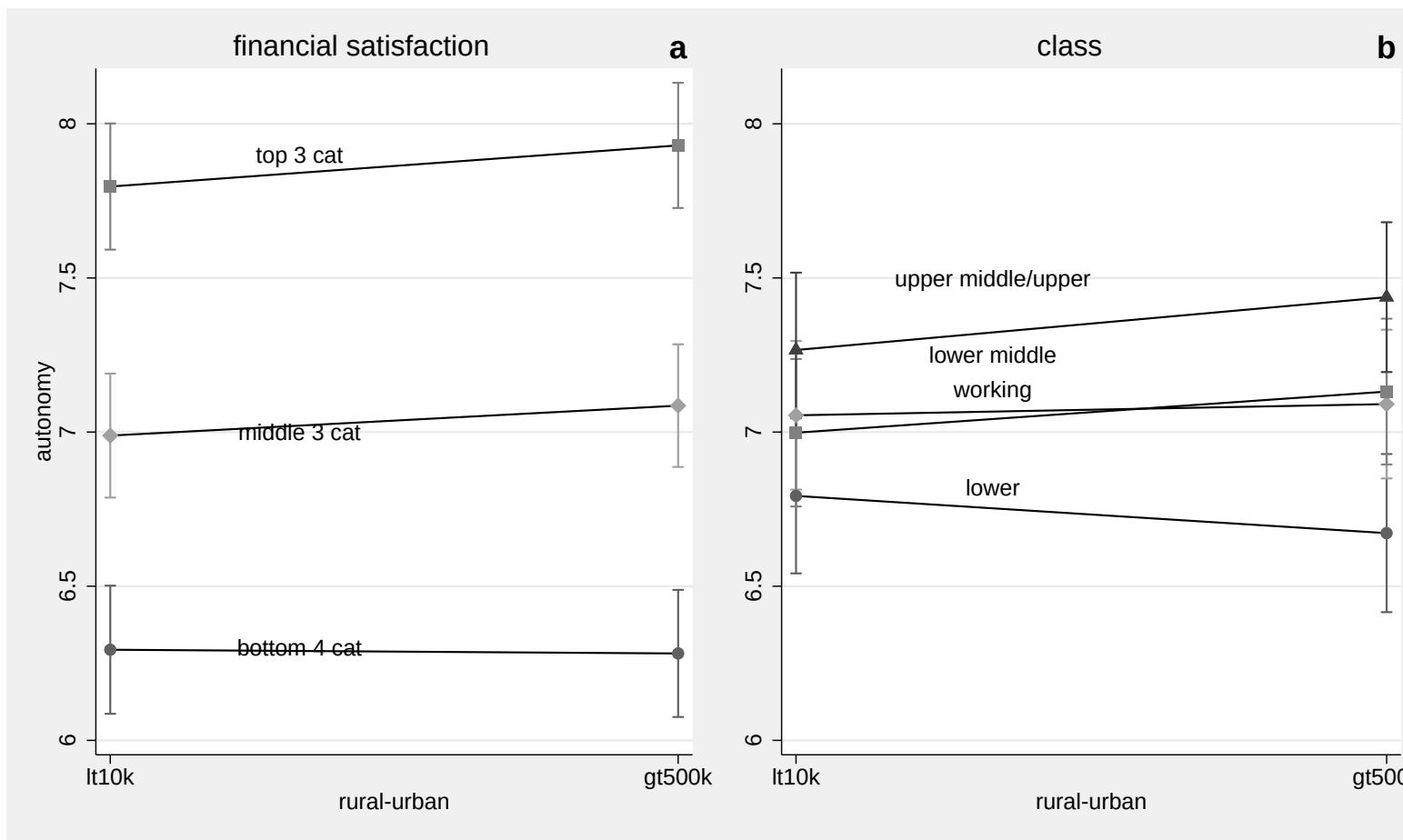


Figure 37: Only <2020.

### Initial interactions and alternative specifications

```
. mixed free c.town##c.satFin trust famUns criVic age age2 male mar i.emp he
> alth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184956.1
Iteration 1: Log likelihood = -184956.1
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
```

```
Number of obs   = 86,280
Number of groups = 64
Obs per group:
    min = 747
    avg = 1,348.1
    max = 4,018
Wald chi2(20)   = 11068.66
Prob > chi2     = 0.0000
```

Log likelihood = -184956.1

| free        | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |       |
|-------------|-------------|-----------|--------|------|----------------------|-------|
| town        | -0.03       | 0.01      | -3.99  | 0.00 | -0.05                | -0.02 |
| satFin      | 0.22        | 0.01      | 31.95  | 0.00 | 0.20                 | 0.23  |
| c.town#     |             |           |        |      |                      |       |
| c.satFin    | 0.01        | 0.00      | 5.90   | 0.00 | 0.00                 | 0.01  |
| trust       | 0.02        | 0.02      | 1.34   | 0.18 | -0.01                | 0.06  |
| famUns      | -0.11       | 0.01      | -12.26 | 0.00 | -0.13                | -0.09 |
| criVic      | -0.11       | 0.03      | -4.22  | 0.00 | -0.16                | -0.06 |
| age         | 0.02        | 0.00      | 5.99   | 0.00 | 0.01                 | 0.02  |
| age2        | -0.00       | 0.00      | -5.37  | 0.00 | -0.00                | -0.00 |
| male        | 0.01        | 0.02      | 0.67   | 0.50 | -0.02                | 0.04  |
| mar         | 0.00        | 0.02      | 0.03   | 0.98 | -0.03                | 0.03  |
| emp         |             |           |        |      |                      |       |
| Part time   | -0.08       | 0.03      | -2.82  | 0.00 | -0.13                | -0.02 |
| Self empl.. | -0.01       | 0.02      | -0.56  | 0.57 | -0.06                | 0.03  |
| Retired     | -0.04       | 0.03      | -1.32  | 0.19 | -0.10                | 0.02  |
| Housewife   | -0.20       | 0.03      | -8.00  | 0.00 | -0.25                | -0.15 |
| Students    | -0.14       | 0.04      | -3.79  | 0.00 | -0.21                | -0.07 |
| Unemployed  | -0.10       | 0.03      | -3.29  | 0.00 | -0.15                | -0.04 |
| Other       | -0.22       | 0.07      | -3.28  | 0.00 | -0.36                | -0.09 |

|         |      |      |       |      |      |      |
|---------|------|------|-------|------|------|------|
| health  | 0.33 | 0.01 | 37.13 | 0.00 | 0.31 | 0.35 |
| kids    | 0.02 | 0.01 | 2.67  | 0.01 | 0.00 | 0.03 |
| rel_imp | 0.04 | 0.01 | 4.11  | 0.00 | 0.02 | 0.05 |
| _cons   | 4.26 | 0.11 | 38.13 | 0.00 | 4.04 | 4.48 |

| Random-effects parameters | Estimate | Std. err. | [95% conf. interval] |      |
|---------------------------|----------|-----------|----------------------|------|
| cc: Identity              |          |           |                      |      |
| var(_cons)                | 0.28     | 0.05      | 0.19                 | 0.39 |
| var(Residual)             | 4.25     | 0.02      | 4.21                 | 4.29 |

LR test vs. linear model: chibar2(01) = 4378.78      Prob >= chibar2 = 0.0000

.19

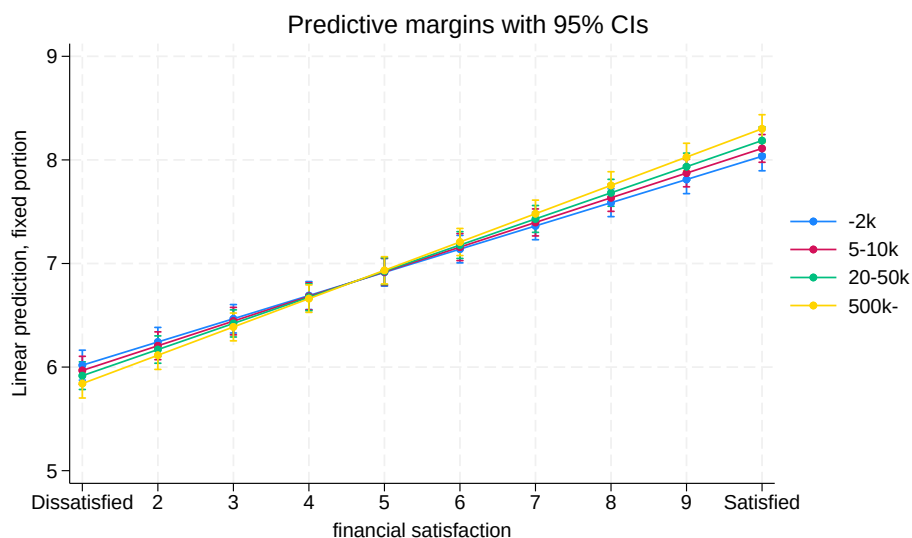


Figure 38: ini1

few things to note in the above—we have trust, famUns and criVic—which are rather predicted by urbanicity, not predictors, hence we may have overcontrol bias. Also here both town and satFin are forced to be linear, a tenuous assumption.

```

mixed free i.t4##i.sf3 trust famUns criVic age age2 male mar i.emp health
> kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -185241.39
Iteration 1: Log likelihood = -185241.39 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc

```

```

Number of obs   = 86,280
Number of groups = 64
Obs per group:
    min = 747
    avg = 1,348.1
    max = 4,018
Wald chi2(28)   = 10427.08
Prob > chi2     = 0.0000

```

Log likelihood = -185241.39

| free       | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |       |
|------------|-------------|-----------|--------|------|----------------------|-------|
| t4         |             |           |        |      |                      |       |
| 10-100k    | -0.14       | 0.04      | -3.39  | 0.00 | -0.21                | -0.06 |
| 100-500k   | -0.04       | 0.05      | -0.78  | 0.43 | -0.13                | 0.06  |
| gt500k     | -0.03       | 0.04      | -0.74  | 0.46 | -0.12                | 0.05  |
| sf3        |             |           |        |      |                      |       |
| 2          | 0.65        | 0.03      | 18.99  | 0.00 | 0.59                 | 0.72  |
| 3          | 1.50        | 0.04      | 40.61  | 0.00 | 1.42                 | 1.57  |
| t4#sf3     |             |           |        |      |                      |       |
| 10-100k#2  | 0.21        | 0.05      | 4.38   | 0.00 | 0.12                 | 0.31  |
| 10-100k#3  | 0.12        | 0.05      | 2.38   | 0.02 | 0.02                 | 0.22  |
| 100-500k#2 | 0.09        | 0.06      | 1.62   | 0.11 | -0.02                | 0.20  |
| 100-500k#3 | 0.07        | 0.06      | 1.17   | 0.24 | -0.05                | 0.19  |
| gt500k#2   | 0.12        | 0.05      | 2.43   | 0.01 | 0.02                 | 0.22  |
| gt500k#3   | 0.12        | 0.05      | 2.26   | 0.02 | 0.02                 | 0.23  |
| trust      | 0.03        | 0.02      | 1.80   | 0.07 | -0.00                | 0.07  |
| famUns     | -0.12       | 0.01      | -13.44 | 0.00 | -0.14                | -0.10 |
| criVic     | -0.12       | 0.03      | -4.75  | 0.00 | -0.17                | -0.07 |
| age        | 0.02        | 0.00      | 5.32   | 0.00 | 0.01                 | 0.02  |
| age2       | -0.00       | 0.00      | -4.66  | 0.00 | -0.00                | -0.00 |
| male       | 0.01        | 0.02      | 0.70   | 0.48 | -0.02                | 0.04  |
| mar        | 0.01        | 0.02      | 0.34   | 0.73 | -0.03                | 0.04  |

|             |       |      |       |      |       |       |
|-------------|-------|------|-------|------|-------|-------|
| emp         |       |      |       |      |       |       |
| Part time   | -0.08 | 0.03 | -3.06 | 0.00 | -0.14 | -0.03 |
| Self empl.. | -0.02 | 0.02 | -0.70 | 0.48 | -0.06 | 0.03  |
| Retired     | -0.05 | 0.03 | -1.66 | 0.10 | -0.11 | 0.01  |
| Housewife   | -0.22 | 0.03 | -8.41 | 0.00 | -0.27 | -0.17 |
| Students    | -0.14 | 0.04 | -3.88 | 0.00 | -0.21 | -0.07 |
| Unemployed  | -0.13 | 0.03 | -4.41 | 0.00 | -0.19 | -0.07 |
| Other       | -0.25 | 0.07 | -3.65 | 0.00 | -0.39 | -0.12 |
| health      | 0.35  | 0.01 | 39.72 | 0.00 | 0.34  | 0.37  |
| kids        | 0.01  | 0.01 | 2.26  | 0.02 | 0.00  | 0.03  |
| rel_imp     | 0.04  | 0.01 | 4.41  | 0.00 | 0.02  | 0.06  |
| _cons       | 4.82  | 0.11 | 45.42 | 0.00 | 4.61  | 5.03  |

| Random-effects parameters | Estimate | Std. err. | [95% conf. interval] |
|---------------------------|----------|-----------|----------------------|
| cc: Identity              |          |           |                      |
| var(_cons)                | 0.27     | 0.05      | 0.19 0.39            |
| var(Residual)             | 4.28     | 0.02      | 4.23 4.32            |

LR test vs. linear model: chibar2(01) = 4250.93      Prob >= chibar2 = 0.0000

.19

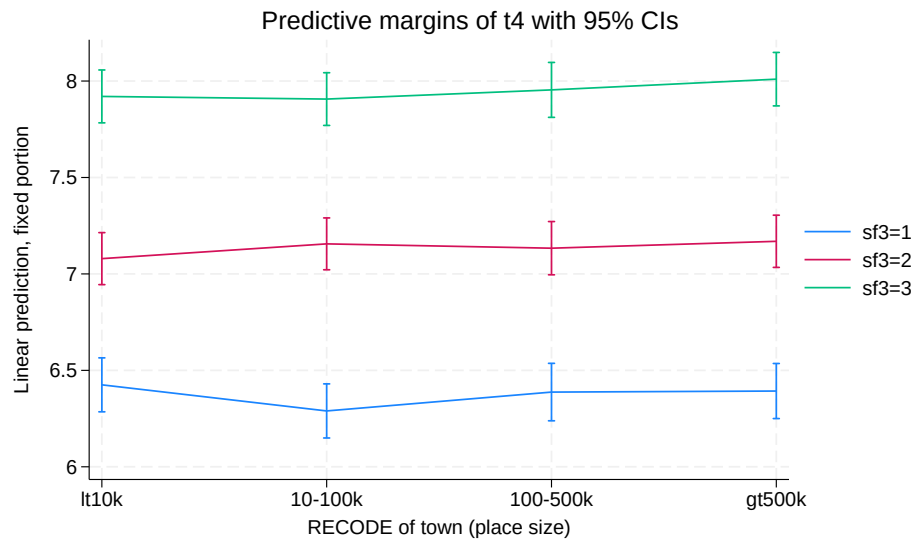


Figure 39: ini2

In the above a different take—financial satisfaction tertiles—for the 1st tertile the biggest drop is actually in the middle category at 10-100k.

```

mixed free c.town#c.inc trust famUns criVic age age2 male mar i.emp healt
> h kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184725.92
Iteration 1: Log likelihood = -184725.92 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs = 84,979
Number of groups = 64
Obs per group:
    min = 749
    avg = 1,327.8
    max = 4,018
Wald chi2(20) = 5460.49
Prob > chi2 = 0.0000
Log likelihood = -184725.92

```

| free         | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |
|--------------|-------------|-----------|--------|------|----------------------|
| town         | -0.00       | 0.01      | -0.43  | 0.67 | -0.02 0.01           |
| inc          | 0.11        | 0.01      | 13.01  | 0.00 | 0.09 0.12            |
| c.town#c.inc | 0.00        | 0.00      | 1.21   | 0.23 | -0.00 0.00           |
| trust        | 0.07        | 0.02      | 3.91   | 0.00 | 0.04 0.11            |
| famUns       | -0.17       | 0.01      | -18.82 | 0.00 | -0.19 -0.16          |
| criVic       | -0.16       | 0.03      | -6.05  | 0.00 | -0.21 -0.11          |
| age          | 0.01        | 0.00      | 2.60   | 0.01 | 0.00 0.01            |
| age2         | -0.00       | 0.00      | -1.15  | 0.25 | -0.00 0.00           |
| male         | -0.00       | 0.02      | -0.17  | 0.87 | -0.03 0.03           |
| mar          | 0.04        | 0.02      | 1.97   | 0.05 | 0.00 0.07            |
| emp          |             |           |        |      |                      |
| Part time    | -0.08       | 0.03      | -2.83  | 0.00 | -0.14 -0.02          |
| Self empl..  | -0.01       | 0.02      | -0.46  | 0.65 | -0.06 0.04           |

|            |       |      |       |      |       |       |
|------------|-------|------|-------|------|-------|-------|
| Retired    | 0.01  | 0.03 | 0.33  | 0.74 | -0.05 | 0.07  |
| Housewife  | -0.18 | 0.03 | -6.65 | 0.00 | -0.23 | -0.12 |
| Students   | -0.11 | 0.04 | -2.99 | 0.00 | -0.18 | -0.04 |
| Unemployed | -0.15 | 0.03 | -5.04 | 0.00 | -0.21 | -0.09 |
| Other      | -0.25 | 0.07 | -3.54 | 0.00 | -0.39 | -0.11 |
| health     | 0.46  | 0.01 | 51.21 | 0.00 | 0.45  | 0.48  |
| kids       | 0.01  | 0.01 | 1.93  | 0.05 | -0.00 | 0.03  |
| rel_imp    | 0.05  | 0.01 | 5.88  | 0.00 | 0.04  | 0.07  |
| _cons      | 4.75  | 0.12 | 39.88 | 0.00 | 4.51  | 4.98  |

---

| Random-effects parameters | Estimate | Std. err. | [95% conf. interval] |      |
|---------------------------|----------|-----------|----------------------|------|
| cc: Identity              |          |           |                      |      |
| var(_cons)                | 0.35     | 0.06      | 0.25                 | 0.50 |
| var(Residual)             | 4.51     | 0.02      | 4.47                 | 4.55 |

---

LR test vs. linear model: chibar2(01) = 5398.66      Prob >= chibar2 = 0.0000

.19

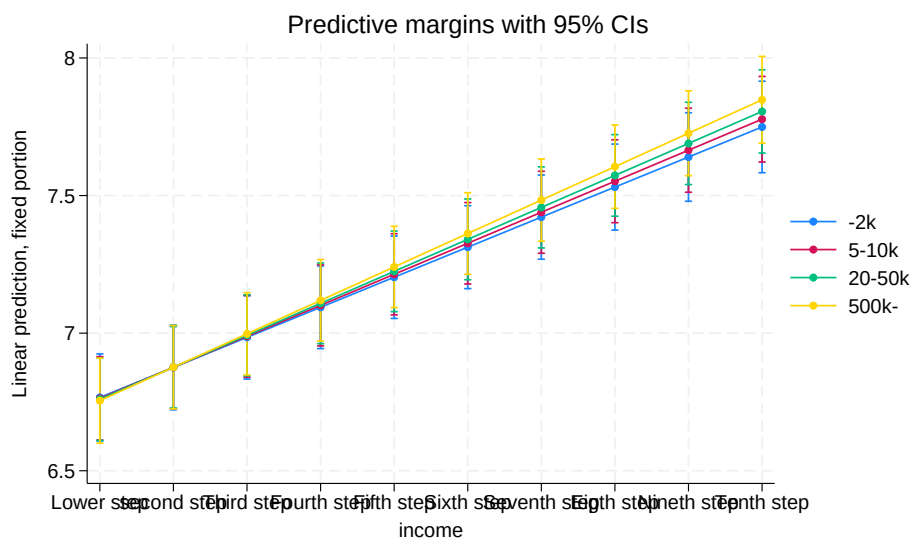


Figure 40: ini3

Here what is easy to see is that along the income spectrum the largest places > 500k change most; and smallest places < 2k change the least

```

mixed free c.town#c.class trust famUns criVic age age2 male mar i.emp hea
> lth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184877.86
Iteration 1: Log likelihood = -184877.86
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs   = 84,931
Number of groups = 64
Obs per group:
    min = 686
    avg = 1,327.0
    max = 4,018
Wald chi2(20)   = 4850.01
Prob > chi2     = 0.0000
Log likelihood = -184877.86

```

|             | free | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |       |
|-------------|------|-------------|-----------|--------|------|----------------------|-------|
| town        |      | -0.00       | 0.01      | -0.52  | 0.60 | -0.02                | 0.01  |
| class       |      | 0.14        | 0.02      | 7.62   | 0.00 | 0.10                 | 0.18  |
| c.town#     |      |             |           |        |      |                      |       |
| c.class     |      | 0.01        | 0.00      | 1.56   | 0.12 | -0.00                | 0.01  |
| trust       |      | 0.09        | 0.02      | 4.79   | 0.00 | 0.05                 | 0.13  |
| famUns      |      | -0.18       | 0.01      | -19.22 | 0.00 | -0.20                | -0.16 |
| criVic      |      | -0.17       | 0.03      | -6.34  | 0.00 | -0.22                | -0.11 |
| age         |      | 0.01        | 0.00      | 2.45   | 0.01 | 0.00                 | 0.01  |
| age2        |      | -0.00       | 0.00      | -1.24  | 0.22 | -0.00                | 0.00  |
| male        |      | 0.01        | 0.02      | 0.41   | 0.68 | -0.02                | 0.04  |
| mar         |      | 0.05        | 0.02      | 2.96   | 0.00 | 0.02                 | 0.09  |
| emp         |      |             |           |        |      |                      |       |
| Part time   |      | -0.10       | 0.03      | -3.45  | 0.00 | -0.16                | -0.04 |
| Self empl.. |      | -0.03       | 0.02      | -1.22  | 0.22 | -0.08                | 0.02  |
| Retired     |      | -0.02       | 0.03      | -0.64  | 0.52 | -0.08                | 0.04  |
| Housewife   |      | -0.21       | 0.03      | -7.84  | 0.00 | -0.26                | -0.16 |

|            |       |      |       |      |       |       |
|------------|-------|------|-------|------|-------|-------|
| Students   | -0.13 | 0.04 | -3.46 | 0.00 | -0.20 | -0.06 |
| Unemployed | -0.21 | 0.03 | -6.93 | 0.00 | -0.27 | -0.15 |
| Other      | -0.30 | 0.07 | -4.13 | 0.00 | -0.44 | -0.16 |
| health     | 0.48  | 0.01 | 52.29 | 0.00 | 0.46  | 0.49  |
| kids       | 0.01  | 0.01 | 1.81  | 0.07 | -0.00 | 0.02  |
| rel_imp    | 0.05  | 0.01 | 5.88  | 0.00 | 0.04  | 0.07  |
| _cons      | 4.86  | 0.12 | 39.41 | 0.00 | 4.61  | 5.10  |

---

| Random-effects parameters | Estimate | Std. err. | [95% conf. interval] |
|---------------------------|----------|-----------|----------------------|
| cc: Identity              |          |           |                      |
| var(_cons)                | 0.36     | 0.06      | 0.25 0.51            |
| var(Residual)             | 4.54     | 0.02      | 4.49 4.58            |

---

LR test vs. linear model:  $\chi^2(01) = 5460.71$       Prob >=  $\chi^2 = 0.0000$

.19

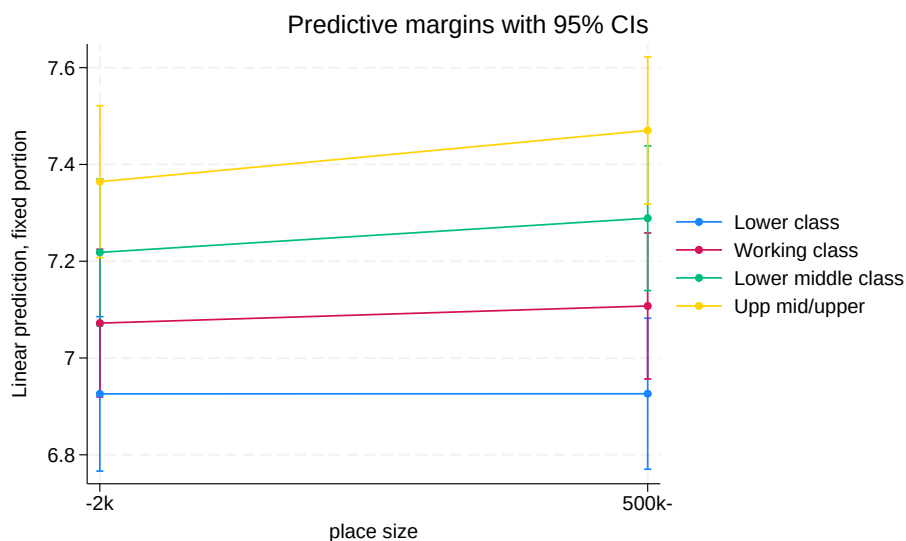


Figure 41: ini4

The above is a version of the main graph from the body of the paper, again here we are over-controlling with trust and the 2 crime variables, but it is not that different after all, similar patterns, just the main difference being that here the lower class is flat and in the body downward sloping and here the top class is quite steep and while in paper it is upward sloping it's little less so in the body of the paper

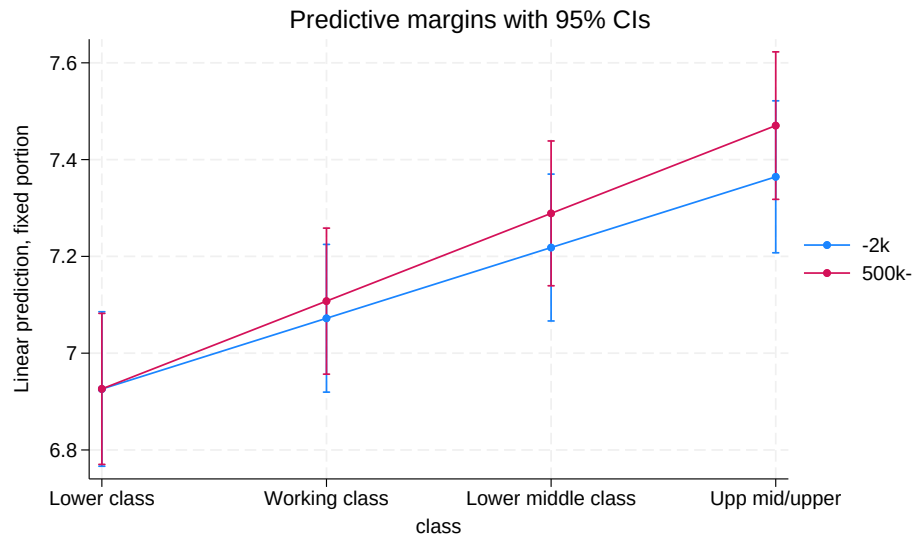


Figure 42: ini5

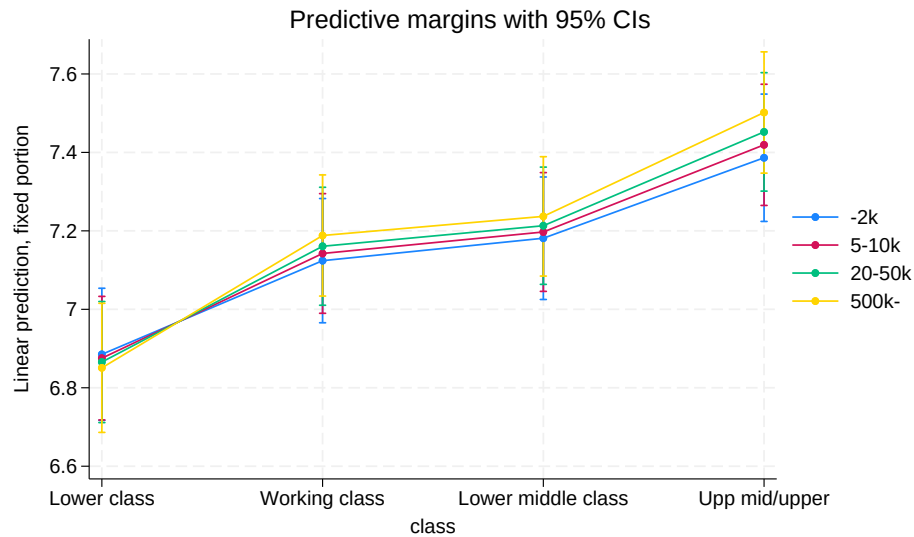
ini5 is from the same model, nicely shows that the higher the class the more difference between the smallest and the largest places

```

mixed free c.town##i.class trust famUns criVic age age2 male mar i.emp hea
> lth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184853.93
Iteration 1: Log likelihood = -184853.93 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs = 84,931
Number of groups = 64
Obs per group:
    min = 686
    avg = 1,327.0
    max = 4,018
Wald chi2(24) = 4900.64
Prob > chi2 = 0.0000
Log likelihood = -184853.93

```

|              | free | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |
|--------------|------|-------------|-----------|--------|------|----------------------|
| town         |      | -0.00       | 0.01      | -0.54  | 0.59 | -0.02 0.01           |
| class        |      |             |           |        |      |                      |
| Working c..  |      | 0.22        | 0.06      | 3.86   | 0.00 | 0.11 0.34            |
| Lower mid..  |      | 0.28        | 0.06      | 5.10   | 0.00 | 0.17 0.39            |
| Upp mid/u..  |      | 0.48        | 0.06      | 7.75   | 0.00 | 0.36 0.60            |
| class#c.town |      |             |           |        |      |                      |
| Working c..  |      | 0.01        | 0.01      | 1.34   | 0.18 | -0.01 0.03           |
| Lower mid..  |      | 0.01        | 0.01      | 1.28   | 0.20 | -0.01 0.03           |
| Upp mid/u..  |      | 0.02        | 0.01      | 1.95   | 0.05 | -0.00 0.04           |
| trust        |      | 0.09        | 0.02      | 4.82   | 0.00 | 0.05 0.13            |
| famUns       |      | -0.18       | 0.01      | -19.18 | 0.00 | -0.20 -0.16          |
| criVic       |      | -0.16       | 0.03      | -6.30  | 0.00 | -0.22 -0.11          |
| age          |      | 0.01        | 0.00      | 2.50   | 0.01 | 0.00 0.01            |
| age2         |      | -0.00       | 0.00      | -1.29  | 0.20 | -0.00 0.00           |
| male         |      | 0.01        | 0.02      | 0.35   | 0.72 | -0.03 0.04           |
| mar          |      | 0.05        | 0.02      | 2.87   | 0.00 | 0.02 0.09            |
| emp          |      |             |           |        |      |                      |
| Part time    |      | -0.10       | 0.03      | -3.39  | 0.00 | -0.15 -0.04          |
| Self empl..  |      | -0.03       | 0.02      | -1.10  | 0.27 | -0.07 0.02           |
| Retired      |      | -0.02       | 0.03      | -0.52  | 0.60 | -0.08 0.05           |
| Housewife    |      | -0.20       | 0.03      | -7.60  | 0.00 | -0.25 -0.15          |
| Students     |      | -0.13       | 0.04      | -3.34  | 0.00 | -0.20 -0.05          |
| Unemployed   |      | -0.20       | 0.03      | -6.63  | 0.00 | -0.26 -0.14          |
| Other        |      | -0.29       | 0.07      | -4.01  | 0.00 | -0.43 -0.15          |
| health       |      | 0.47        | 0.01      | 52.12  | 0.00 | 0.46 0.49            |
| kids         |      | 0.01        | 0.01      | 1.85   | 0.06 | -0.00 0.03           |
| rel_imp      |      | 0.05        | 0.01      | 5.85   | 0.00 | 0.04 0.07            |
| _cons        |      | 4.96        | 0.12      | 40.78  | 0.00 | 4.72 5.20            |



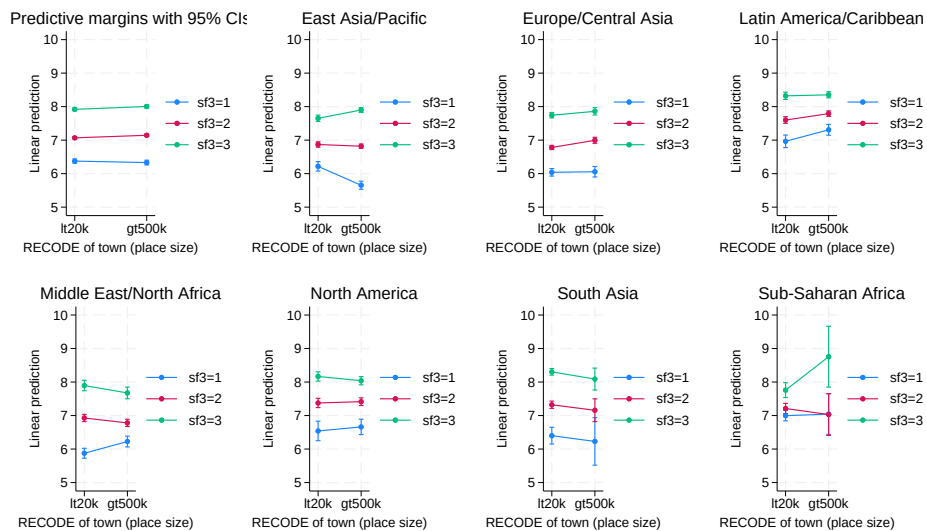
**Figure 43:** ini6

And then ini6: couple interesting things: first things flip from lower to working class, then working and lower middle class quite similar, and the differences get larger for upper middle and upper classes.

#### 2.4.3 By region and country income group: Interactions with financial satisfaction

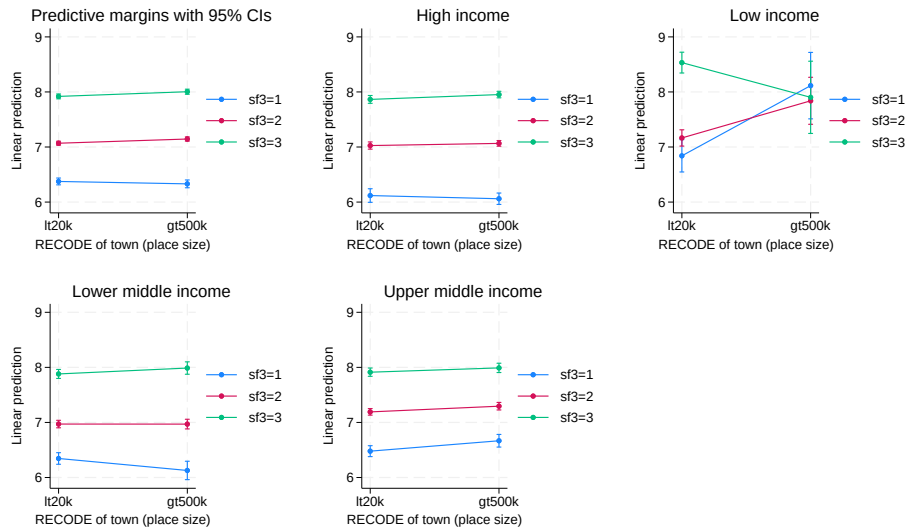
Here we repeat urbanicity interactions with financial satisfaction from the body of the paper but by region and income group; the first panel in each graph is repeated from the body of the paper.

In figure 44 we see that the pattern is much stronger for East Asia/Pacific, and opposite especially in Middle East/North Africa, but also in North America.



**Figure 44:** Urbanicity interactions with financial satisfaction by region.

In figure 45 high income has a similar pattern, and lower middle income panel has a strong pattern, and quite the opposite pattern for upper middle income.



**Figure 45:** Urbanicity interactions with financial satisfaction by income group. Note just 2 countries for low income group: ETH TJK.

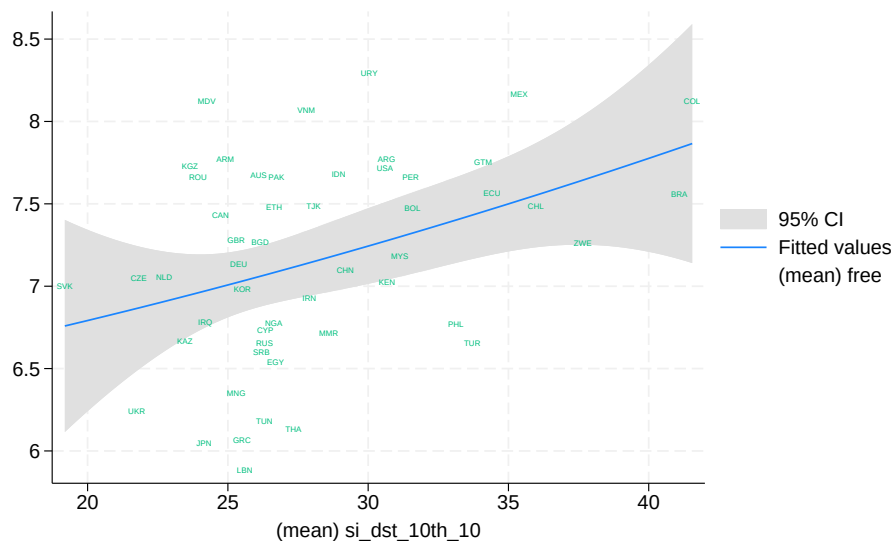
### 3 Auxiliary results and sidenote

#### 3.1 Some graphs

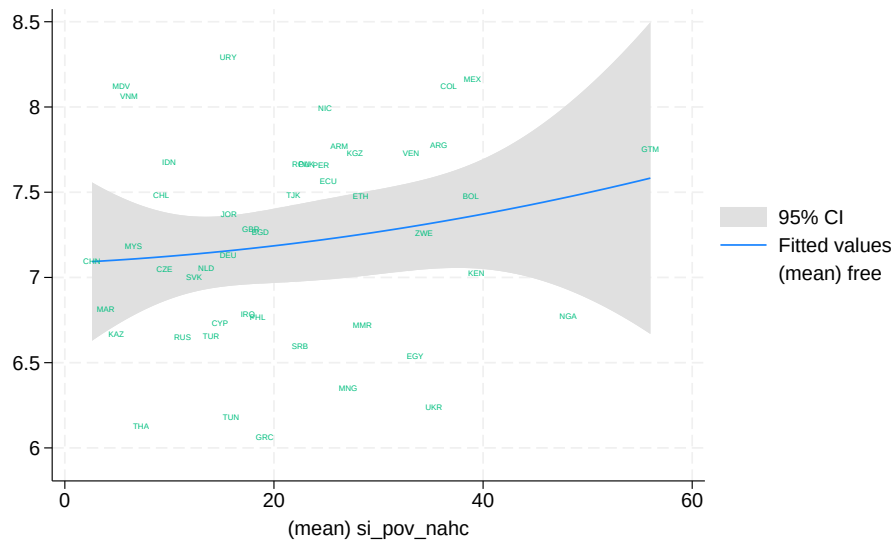
fig 46 and 47

weird to think that more ineq and pov would produce more freedom/autonomy; rather i guess lax capitalist friendly regulation would produce more ineq and pov, but should also produce less autonomy/freedom on avg, but it doesn't, there is positive relationship

or maybe the other way round: more freedom/autonomy more ineq and pov



**Figure 46:** more ineq (inc share held by top 10perc, more autonomy-conservatives would like it!! and pretty strong corr .4



**Figure 47:** likewise with poverty, but here corr just .2

## 3.2 Literature

Newman (2020) makes some useful points that may be worth citing in full:

Social theory typically conceives of urban growth as associated with political freedom. [...] But not even in the most likely of cases did the city makes urban peasants free. Focusing on Mexico City, I argue that because rural-to-urban migrants became urban denizens by establishing squatter settlements, they entered into dependent relations with political elites, which reinforced a system of political dependency, or clientelist politics. [...] Mexico is, perhaps, the most iconic case. A large cross section of the peasantry rose in arms against landowning oligarchs in the insurrectionary phase of the Mexican Revolution (1910-1917), ultimately precipitating the wholesale reconstruction of the political order. Thereafter, peasants continued to mobilize for radical change, constituting a major obstacle to postrevolutionary governing elites who sought to consolidate control. [...] When a large and growing portion of rural-to-urban migrants live in squatter settlements, and therefore have little recourse to legal protection in residential matters, they almost inevitably grow dependent on political elites—in an arrangement known as “clientelism.” Urban concentration gives rise to clientelism.

```
. tabstat ls, by(lwd7) stat(mean sd) format(%9.1f)
```

Summary for variables: ls  
Group variable: lwd7 (RECODE of lwd (do what you like))

| lwd7  | Mean | SD  |
|-------|------|-----|
| 0-4   | 5.6  | 2.7 |
| 2     | 7.6  | 2.3 |
| 3     | 7.6  | 1.8 |
| 4     | 8.0  | 1.5 |
| 5     | 7.8  | 1.9 |
| 6     | 8.2  | 1.5 |
| 7     | 8.9  | 1.5 |
| Total | 8.3  | 1.9 |

.9

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